

**InterACTS: A WEB-BASED MULTI-MODE FACE RECOGNITION
ATTENDANCE AND MONITORING SYSTEM FOR
EDUCATIONAL MANAGEMENT**

A THESIS

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DEDICATION

The author wholeheartedly dedicated this piece of work to the following who've been
the reasons to complete this report:

To their **family** and **loved ones**,

To their **classmates** and **friends**,

To their school and above all,

To Almighty God.

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ABSTRACT

This study presents the development and implementation of InterACTS: A Web-Based Multi-Modal Face Recognition Attendance and Monitoring System for Educational Management, designed to improve the efficiency, accuracy, and security of student attendance monitoring. The system addresses the limitations of traditional attendance methods, such as manual errors, time inefficiency, and lack of real-time monitoring. InterACTS uses facial recognition technology to automate attendance recording through multiple modes, including hallway monitoring, classroom attendance per subject, and event participation tracking. It also provides manual attendance options to support online classes and other learning environments. The system features real-time email notifications to inform parents or guardians about student attendance activities, improving communication and transparency. In addition, it includes modules for academic management, scheduling, teacher and student management, event handling, and device management, along with configurable attendance policies for institutional flexibility. The platform also offers analytics and reporting tools for monitoring attendance trends and generating reports. Furthermore, the event management module includes a payment verification process where students submit receipts for administrator approval before participation is confirmed. With role-based access for administrators, teachers, and students, InterACTS ensures organized system usage and data security. The findings of the study show that the system provides a reliable, user-friendly, and effective solution for modern educational attendance management.

List of Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
DBMS	Database Management System
CV	Computer Vision
CNN	Convolutional Neural Network
CRUD	Create, Read, Update, Delete
CSMS	Computer System Management System
DCNN	Deep Convolutional Neural Network
DepEd	Department of Education
GPS	Global Positioning System
GSM	Global System for Mobile Communication
HTTPS	Hypertext Transfer Protocol Secure
HOG	Histogram of Oriented Gradients
ISO	International Organization for Standardization
IoT	Internet of Things
JSON	JavaScript Object Notation

JWT	JSON Web Token
LBP	Local Binary Pattern
LAN	Local Area Network
NFR	Non-Functional Requirement
ML	Machine Learning
MVC	Model-View-Controller
OCR	Optical Character Recognition
PCA	Principal Component Analysis
RAM	Random Access Memory
REST	Representational State Transfer
RFID	Radio Frequency Identification
RGB	Red Green Blue
SRC	Sparse Representation-based Classification
SVDL	Sparse Variation Dictionary Learning
SDLC	Software Development Life Cycle
TCP/IP	Transmission Control Protocol / Internet Protocol

Definition Of Terms

AI	The simulation of human intelligence in machines used for face recognition, and data analysis.
Administrator	An authorized user responsible for managing system configuration, user accounts, facial data training, and reports.
Analytics	The process of analyzing attendance and system data to identify trends and patterns.
API	A set of rules that allows different software applications to communicate with each other.
Attendance Logging	The automated recording of student presence using face recognition technology.
Attendance Monitoring	The continuous tracking and supervision of student attendance status.
Authentication	The process of verifying a user's identity before system access is granted.
Biometric Technology	Technology that identifies individuals using unique physical characteristics such as facial features.
CNN	A deep learning algorithm used for image processing and face recognition.
Computer Vision	A field of artificial intelligence that enables systems to analyze images and video.

Dashboard	The main interface displaying attendance summaries, analytics, and notifications.
Database	A structured digital storage system for user data, facial records, and attendance logs.
Event Mode	A system feature for managing attendance and payment validation during school events.
Face Detection	The process of locating human faces in images or video frames.
Face Recognition	A biometric process that identifies individuals based on facial features.
Facial Data	Digitally captured facial images are used for training and recognition.
GSM	A mobile communication technology used for sending notifications.
ISO	An organization that develops international standards, including ISO 9126.
MBAMSUFTT	Mobile-Based Attendance Monitoring System Using Face Tagging Technology.
Monitoring Mode	A system function for real-time viewing of attendance activity.
MySQL	An open-source relational database management system.
Notification System	A feature that sends automated alerts and updates to users.

OpenCV	An open-source computer vision library used for face detection and recognition.
PHP	A server-side scripting language used for backend development
Role-Based Access Control	A security method that restricts access based on user roles.
System Accuracy	The ability of the system to correctly recognize users and record attendance.
Web-Based System	A system accessible through a web browser without installation

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CHAPTER 1

INTRODUCTION

In many schools and colleges, the traditional method of marking attendance is time-consuming and laborious because it deflects the staff from more critical tasks. A manual attendance system necessitates the user performing all these tasks themselves, including processing paper sheets and timecards, and the risk of human error because students only sometimes enter the correct data into attendance spreadsheets. If there were errors or issues with student attendance, the staff was flooded with questions and needed to adjust. As a result, many institutes have begun to use various other techniques for recording attendance.

Face recognition is a biometric technology that can identify or verify a person by analyzing their unique facial features. The process generally begins with detecting faces in images or video, often using tools like OpenCV and algorithms such as Haar Cascade. After detection, the system extracts important facial features such as the distance between the eyes, shape of the nose, and contours of the jawline and converts them into a numeric representation called an embedding. These embeddings are then compared with stored data to recognize or verify an individual. Face recognition technology enables accurate identification in real-time, reducing human error and improving efficiency. It is widely applied in various fields, including automated attendance systems, security and surveillance, access control, and smartphone authentication. Advances in artificial intelligence, particularly deep learning and neural networks, have significantly increased the accuracy and robustness of face recognition systems, even under varying

lighting conditions, facial expressions, or partial occlusions such as masks or glasses. As a result, face recognition has become a reliable and essential tool in modern security and monitoring applications.

It provides numerous benefits for schools by automating and improving the accuracy of attendance logging compared to manual roll calls or paper-based systems, thereby reducing human errors. It allows for real-time monitoring, enabling schools to track which students are present, absent, late, or leaving early. This technology also enhances security, as parents and school authorities are informed of students' presence on campus, helping prevent unauthorized access and ensuring student safety. Additionally, face recognition systems can generate analytics and reports on attendance, participation, and engagement, allowing teachers and administrators to identify patterns of absenteeism or tardiness. It also provides convenience for students, who can log attendance automatically, and for parents, who receive real-time notifications about their child's time-in, time-out, and whereabouts. Furthermore, schools can integrate face recognition with event participation or payment validation to monitor attendance and participation more effectively.

Project Context

The project includes the development of InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for educational management, accessible online through multiple platforms using web browsers such as Google Chrome and Microsoft Edge. The system supports three distinct modes: Attendance Mode for real-time face recognition and automated attendance logging, Monitoring

Mode for tracking student performance and generating analytics, and Events Mode for validating student payments and managing school activities. It is designed to deliver optimal performance across devices, including laptops, personal computers, tablets, iPads, and smartphones. To ensure both institutional and user security and privacy, the web application is structured into four user roles the administrator, Student, Teacher, and Parent each with unique access levels and responsibilities.

InterACTS provides a solution designed to address specific challenges in educational management. Through its modules, students register with their personal information and facial data, enabling the system to automatically record attendance and generate unique login credentials for each student. This eliminates manual attendance errors and streamlines the process, while allowing students to access their records and monitor their own attendance. The system is also capable of managing student information, validating attendance and event payments, posting announcements for upcoming events or exams, and generating analytical reports, ensuring accurate and transparent record-keeping. Users receive real-time notifications regarding student attendance and participation in school activities, keeping them well-informed. The system can track attendance per subject and per grading period, adjust schedules in case of conflicts, and provide detailed analytics to evaluate student performance. This allows teachers to offer timely support, identify patterns in attendance or academic performance, and maintain accurate and up-to-date records. Students can access their personal analytics, gaining insights into their progress, attendance trends, and areas for improvement, promoting self-awareness and responsibility.

In InterACTS, student payments for school activities are validated using AI-powered receipt detection and ensuring the authenticity of uploaded transaction records. The standalone face recognition scanner allows verified students to participate in events without presenting physical receipts, improving convenience, reducing manual verification, and ensuring security. Parents are automatically notified once payments are confirmed, fostering transparency and accountability. Administrators and registrars can generate reports on student participation, monitor payment statuses, and ensure organized and secure management of school events. By integrating these functionalities, InterACTS provides a secure, efficient, and reliable platform that automates attendance, monitors academic performance, validates event payments, facilitates communication between school and home, and supports data-driven decision-making for students, parents, teachers, and administrators alike. It ensures a modern, intelligent, and fully integrated educational environment where attendance, monitoring, and event management are accurate, convenient, and transparent

Purpose and Description of the Study

The purpose of this study is to present InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management, which aims to improve school attendance management, maintain secure access control, send automated parent notifications, and utilize real-time analytics to support data-driven decision-making. The integration of an events mode further enhances the system's capability to organize and monitor special occasions, such as convocations and school activities, through targeted attendance tracking and controlled system access. To support

these functions, the system incorporates ten modules working together as a unified platform.

The user management regulates user roles and permissions, while the authentication and security, Module ensures that only authorized individuals can access sensitive features. The train Data Module processes facial images to build an accurate recognition database, which works alongside the Real-Time Face Recognition Module responsible for identifying individuals as they appear on camera. These recognition results are then utilized by the Monitoring Module for real-time activity viewing and by the Attendance logging module for automated time-in and time-out recording. The central interface module provides a centralized view of ongoing operations, whereas the analytics and reports generate summaries and trends that guide decision-making. To maintain accuracy, the verification validates uploaded receipts and ensures correct event payment records, and the notification module distributes timely alerts to parents and staff regarding attendance updates or system events. The events ode complements all these modules by enabling specialized monitoring and access control for programs and school-wide activities. Together, these components create a secure, efficient, and data-driven management system for educational institutions.

Administrators serve as the primary managers of the system, overseeing all aspects of user management, attendance monitoring, and system configuration. They can create and manage user accounts, including registering teachers, and students. When students are added, the system automatically generates their accounts, allowing them to log in securely without manual setup. Administrators are also responsible for training

and updating facial data in the database to ensure accurate recognition for attendance purposes. They can monitor real-time attendance activities, review and modify attendance records when necessary, and ensure that data integrity is maintained throughout the system. The administrators can also configure system settings, including notifications and events mode, to optimize operations for school programs and special occasions. Additionally, they can generate detailed analytics and reports to support data-driven decisions and communicate directly with parents and teachers through the notification module, ensuring timely updates on attendance, absences, or irregularities. This comprehensive management capability allows administrators to maintain smooth and secure operation of the entire system while providing reliable information to all stakeholders.

Students interface provides personalized access to academic and payment-related information. Students can view their subjects, and attendance analytics, including their number of absences and presents. Their attendance data is securely stored and accessible for review at any time. When a student reaches an absence rate of 20%, the system automatically notifies the Student and Parents ensuring they are promptly informed of their child's attendance status. They can upload receipts for event payments by taking a photo, entering the transaction number, and specifying the event name. Once submitted, the registrar receives the information for confirmation. To improve efficiency during school events, standalone cameras are deployed to automatically mark students as paid and present upon recognition, eliminating the need for long queues and manual validation.

Teachers have the authority to access and use a monitoring-based interface that focused on class attendance management. Upon logging in, teachers can access their assigned subjects, schedules, classrooms, and student lists. During each session, they can initiate facial recognition attendance through the system's camera interface, ensuring that only enrolled students are marked present. The system also performs real-time verification of recognition results to maintain accuracy. Teachers can manually edit attendance records if necessary and view analytics that display individual student attendance trends over time. When a student accumulates excessive absences or shows irregular attendance patterns, the system automatically generates alerts to notify the teacher and, if needed, directs the student to consult with the registrar. This integrated approach allows teachers to efficiently manage attendance, monitor student participation, and take timely actions when issues happen.

Parents receive notifications through the system's Notification API to stay informed about their child's attendance. They can see the student's time-in and time-out records, providing security and ensuring that they know when their child has left the school premises. This allows parents to monitor their child's attendance in real-time and be aware of their whereabouts during school hours. Additionally, parents will be notified through the Notification API when the student has made payments for events if the administrator validate the receipt.

Statement of the Problem

The main problem addressed by this study is the inefficiency and inaccuracy of traditional attendance and monitoring methods used in schools, which often result in fraudulent reporting, manual errors, and lack of transparency between students, teachers, parents, and administrators. This aims to enhance and to help teachers, administrators, and school staff manage student attendance and participation more effectively, while also supporting better tracking of events and classroom activities. To address these issues, the study aims to develop a InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational. The system is composed of ten integrated modules, each serving a specific function to ensure comprehensive management of school operations. Each module is designed to perform distinct functions, from securing user access and managing profiles, to capturing and recognizing facial data, logging attendance, validating payments, sending notifications, generating reports, and providing a centralized interface for all users.

Specifically, this study seeks to:

1. Design and Develop InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management.
 - 1.1 Register and manage user accounts: students, teachers, and parents, while assigning appropriate roles and permissions for secure, role-based access.
 - 1.2 Generate automatically unique student IDs and login credentials to allow immediate access.

1.3 Capture and train facial images to ensure a reliable database for accurate recognition.

1.4 Handle notifications and validates uploaded receipt transactions efficiently.

1.5 Provide an admin-friendly interface for inputting event details, announcements, and school activities, allowing efficient management and easy updates.

1.6 Generate analytical reports on attendance, event participation, and payment status.

1.7 Monitor overall school activities, attendance trends, and event statistics.

1.8 Record and track student attendance per subject in real-time using face recognition

1.9 Monitor attendance per grading period (Prelim, Midterm, Pre-Final, and Final).

1.10 Edit and manage class records, grades, and participation logs to ensure accuracy and up-to-date student information.

1.11 Register for event and submits payments online securely.

1.12 Receive automatic notifications regarding attendance, events, and payments.

2. The acceptability of the "InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management" is determined using ISO 9126 in terms of:

2.1 Functionality

2.2 Reliability

2.3 Usability

2.4 Efficiency

2.5 Maintainability

2.6 Portability

3. What implementation plan can be proposed for the adaptation of the InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management?

Significance of the Study

The study of InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management emphasizes the growing importance of adopting intelligent and automated solutions in school operations. With its integration of Real time face recognition and monitoring the students. The system aims to enhance efficiency, strengthen accountability, and support smarter decision-making within the educational environment. Through these capabilities, the system provides meaningful advantages to its different stakeholders, each benefiting in ways that improve their tasks, responsibilities, and overall engagement with the institution.

Administrators will significantly help by providing a centralized and automated platform for managing student attendance. Through real-time data and accurate records generated by the face recognition system, administrators can easily monitor attendance patterns, prepare reports, and make informed decisions that support effective school

management. The system adds value by improving data organization, enhancing record accuracy, and strengthening the overall efficiency of administrative processes.

Teachers will benefit from the system through faster and more convenient attendance monitoring. With instant recognition and automatic recording, teachers can save time and direct more attention to teaching and classroom activities. The system also provides teachers with accessible attendance summaries that help them track student participation and identify any patterns that may require intervention or follow-up. This contributes to improved classroom management and better support for student performance.

Students will benefit from a more efficient and fair attendance process. The face recognition system provides quick identity verification and immediate recording, allowing students to enter their classes without delays. This reduces errors and ensures that attendance is accurately documented. The system also encourages discipline, punctuality, and accountability, helping students develop positive academic habits and experience a smoother daily routine.

Parents will experience improved communication and involvement through the system's attendance visibility features. By receiving timely updates or having access to their child's attendance records, parents can maintain awareness of their child's punctuality and presence in school. This helps them guide their children more effectively and encourages responsible habits. The system promotes stronger collaboration between home and school by making accurate attendance information readily available.

School Management will benefit also because this study lies in providing an updated and improved School Management system for the institution. The new development introduces features such as multi-mode face recognition for attendance, automated notifications, monitoring, and streamlined verification, which enhance efficiency, accuracy, and user experience. This ensures that administrators, teachers, students, and parents can manage and access school-related information more effectively, addressing gaps in the previous system and supporting a more organized, secure, and data-driven educational environment.

This study serves as a reference for future researchers who aim to develop or improve school management systems. The findings, methodologies, and system design presented in this research can guide them in creating more efficient, secure, and user-friendly applications. Future researchers may explore additional features, integration with other educational tools, or enhancements in facial recognition accuracy and data analytics to further optimize school management processes.

Scope and Limitations of the Study

The scope of the project includes the development of InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management, which is accessible online through multiple platforms using web browsers like Google Chrome and Microsoft Edge. The system is designed to deliver maximum performance across these devices, including laptops, personal computers, tablets, iPads, and smartphones. To ensure both institutional and users security and privacy the web

application is designed to four distinct roles: Administrator, Student, Teacher and Parents each unique access levels and responsibilities.

The Administrator has full control over the platform, managing the registration of students and teachers by collecting their details. They are also responsible for training the system's AI models to recognize student faces in real-time, ensuring accurate attendance logging and monitoring. This system utilizes artificial intelligence and computer vision technologies, including tools for face detection and recognition, to automatically identify students. The platform integrates an API-based notification system to ensure timely alerts and updates, as well as an AI Detection Receipt for validation of attendance. Additionally, administrators handle notifications, ensuring that all necessary activities or uploaded information are delivered promptly. They also have access to system analytics, allowing them to monitor user engagement, track attendance, and generate comprehensive reports for events and other activities, enabling accurate records and efficient platform management.

Teachers have access to monitor student with attendance per subject in real-time and review attendance logs. They are authorized to update or delete records and adjust the schedule if there are conflicts or changes needed for a particular subject. Teachers also receive notifications through the system whenever there are announcements from the Administrator, and the platform can automatically notify them about students with frequent absences. Additionally, they can view analytics and generate reports to track student attendance and performance.

Students are the main core users of the system and have the privilege to use real-time face recognition for attendance. They can upload receipts to their accounts and enter transaction numbers for validation. It also allows verified students to participate in events without presenting physical receipts, improving convenience, reducing manual verification, and ensuring security. Students also have access to view their personal analytics, such as attendance records and performance, and can view announcements from the Administrator regarding any school activities. If a student has frequented absences, the system will notify them, and they are required to consult with the school based on the school's policies

Parents receive notifications through the system's Notification API to stay informed about their child's attendance. They can see the student's time-in and time-out records, providing security and ensuring that they know when their child has left the school premises. This allows parents to monitor their child's attendance in real-time and be aware of their whereabouts during school hours. Additionally, parents will be notified through the Notification API when the student has made payments for events and administrator validate it.

The limitations of the study primarily focus on the technical and functional constraints of the face recognition attendance and monitoring system. Firstly, as a web-based application, the system relies heavily on stable internet and network connectivity, and any disruption or slow connection may cause delays, errors in recording attendance, or temporary inability to access certain system features. Secondly, scalability presents potential challenges, as an increasing number of users and larger datasets can strain the

system's resources, which may affect overall performance, especially during peak usage times. Thirdly, the accuracy of face recognition is influenced by environmental and physical factors. Poor lighting conditions, low-resolution cameras, or obstructions can reduce recognition rates, while changes in students' appearances, such as hairstyles, glasses, facial hair, or even facial expressions, may prevent accurate identification. Additionally, proper user cooperation is required; students must face the camera correctly and remain relatively still during scanning, otherwise recognition errors may occur.

Moreover, while the system aims to ensure efficiency and convenience, security and data privacy remain critical limitations. Sensitive information, including data and personal details, must be protected from unauthorized access, system breaches, or misuse. Finally, user familiarity with the system can impact its effectiveness. Students, teachers, or staff who are not adequately trained may encounter difficulties navigating the system or fully utilizing its features, potentially reducing the overall efficiency of the attendance monitoring process.

These limitations highlight areas where the system's functionality may be constrained, particularly in high demand situations or when personalized responses are required.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents the review of related literature and studies that serve as the foundation of the proposed system. It includes both local and foreign sources obtained from books, journals, research projects, and online articles that are relevant to the development of the InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management. These sources provide essential knowledge, theories, and technological concepts that guided the researchers in identifying the strengths and weaknesses of existing systems. The review also helps in determining how the proposed system can improve current processes in terms of automation, accuracy, and monitoring efficiency.

Foreign Literature

Chen et al. (2021) emphasize face recognition problems widespread in the field of computer vision. In recent years, Deep Convolutional Neural Networks (DCNNs) have significantly improved the performance of face recognition systems through their layered end-to-end learning frameworks and deep feature extraction techniques from local to global representations. These networks, such as AlexNet, VGG, InceptionNet, ResNet, and DenseNet, have become the foundation of modern face recognition. Among them, AlexNet, proposed by Krizhevsky et al., is considered the pioneering architecture that won the ImageNet Large-Scale Challenge. They emphasized that effective feature representation plays a crucial role in improving face recognition accuracy. However, due to high intra-class variation and inter-class similarity, learning discriminative features

remains challenging. Although the softmax function with cross-entropy loss is widely used in DCNN training, it often fails to achieve sufficient feature discrimination. To address this issue, various loss functions such as center loss, pairwise loss, and triplet loss were introduced to minimize intra-class variance and maximize inter-class variance. Despite their improvements, these methods still have limitations; for instance, the center loss neglects inter-class separability, while pairwise and triplet losses require time-consuming sample selection. To overcome these challenges, recent studies have shifted from Euclidean to angular space approaches. Large-Margin Softmax Loss and A-Softmax Loss impose angular margin constraints to achieve higher inter-class separability. Furthermore, the Additive Margin Softmax (AM-Softmax) loss simplifies optimization by integrating angular margins additively, resulting in better performance and easier implementation.

Lu et al. (2021) define face recognition as an active topic and attracted extensive attention due to its wide potential applications in many areas. There are multiple modalities of face data that can be used in face recognition, such as near infrared images, depth images, and Red Green Blue (RGB) images. Compared with near infrared and depth images, RGB images include more information and have broader application scenarios. In the past decades, many RGB-based face recognition methods have been proposed, and great progress has been made, especially with the development of deep learning. However, there are still many problems to be solved. Face recognition with a single sample per person (SSPP) is one of the most important issues. In SSPP, there is only one training sample per person but various testing samples with appearances

different from training samples. This situation could appear in many actual scenarios such as criminal tracing, ID card identification, and video surveillance. In SSPP, the limited training samples provide insufficient information of intra-class variations, which significantly decreases the performance of most existing face recognition methods. Many methods have been suggested to solve the SSPP problem, which can be roughly divided into three categories: robust feature extraction, generic learning, and synthetic face generation. Algorithms in the first category extract features that are robust to various variations. Some of them extract more discriminative features from single samples based on variants of improved principal component analysis (PCA). Others focus on capturing multiple face representations, mostly by dividing face images into a set of patches and applying various feature extraction techniques to get face representations. For instance, a novel discriminative multi-manifold analysis (DMMA) method was proposed to learn features from patches by constructing a manifold from each individual and formulating face recognition as a manifold–manifold matching problem. Another approach represented human faces using Histogram of Oriented Gradients (HOG), which captures edge or gradient structure and is invariant to local geometric and photometric transformations. The Local Binary Pattern (LBP) texture feature extractor has also been explored for face recognition due to its computational efficiency and invariance to gray-level changes. With the development of deep learning, there are many other methods that utilize deep learning to extract more robust features, such as VGGNet, GoogleNet, FaceNet, ResNet, and SENet. Generic learning methods attempt to utilize a generic dataset, in which each person has more than one training sample, to enhance the

generalization ability of the model. The idea of sharing information across datasets has been widely used and has achieved promising results. Sparse-representation-based classification (SRC) is often used for face recognition, but its performance depends on adequate samples for each subject. An improved technique, called sparse variation dictionary learning (SVDL), was introduced to learn projections from both gallery and generic sets and rebuild a sparse dictionary to perform single-sample-per-person face recognition.

Bramesh et al. (2023) found that the student attendance tracking is very important for improving academic outcomes, promoting student engagement, and enhancing overall school and or college operations. However, most of the existing systems are using single-authentication mechanism for example Biometric or GPS for attendance tracking. This paper presents a dual authentication mechanism by integrating facial recognition and geo-location verification to address the limitations of traditional and or single-factor attendance mechanisms. The proposed system uses computer vision techniques to detect and recognise faces in real time and also uses GPS-based distance estimates from pre-defined classroom locations to verify the students' physical presence in real time. Also, the proposed system has been investigated in real time by considering one classroom of 10 students in a college campus and four sessions per day where each session is of one hour. Finally, the proposed mechanism is a reliable and flexible solution for current educational settings because of its modular construction, which enables future scalability, including interaction with institutional systems and extra biometric features. Attendance tracking is crucial in academic institutions for promoting student success,

ensuring accountability, and supporting data-driven decision-making. It fosters a disciplined learning environment, helps to identify at-risk students for early intervention, and also aids compliance with regulatory standards. However, the traditional attendance tracking mechanisms like paper-based sign-in sheets and roll calls, were time-consuming and also prone to errors like proxy attendance and manual recording errors. On the other hand, with the rise of automation, academic institutions are moving towards systems that increase efficiency and accuracy by minimizing manual errors, saving faculty time, and offering real-time monitoring and analytics. Hence, there is need for the development of more dependable and effective attendance tracking systems. This paper has addressed this by proposing a smart attendance system, which integrates Global Positioning System (GPS) for student location verification and facial recognition for student recognition.

According to Abdalkarim et al. (2022), every institution that depends on people must account for its employees as a first step in the modern day. As a result, creating and maintaining a suitable management system costs different organizations a substantial sum of money. In many countries, government organizations and educational institutions keep track of attendance using paper-based methods. For example, to monitor each student's attendance, it takes time to call out their names at the beginning of the class. False signatures, missing names on spreadsheets, manual data entry, and the possibility of proxy attendance are further problems. Such techniques have several issues that have increased over time. Therefore, to track attendance effectively, it is crucial to replace these outdated practices with modern systems. As a result, extensive research has been conducted in this area using current technologies. Automatic recognition of individuals

based on distinguishing characteristics such as QR codes, IDs and passwords, facial recognition, and fingerprint recognition has become an area of interest for researchers. This paper presents a literature overview of recent works on automated and smart attendance tracking systems. In terms of technology, application domain, and key findings, the critical assessment emphasizes the growing body of research in this field. Keeping track of employees or students in a class is one of the most time-consuming activities in any school, institution, or educational setting. Taking attendance, for instance, consumes both the teacher's time and the lecture period. If the instructor skips this procedure, the school and community will be unaware of whether the students are participating in the classes. Various human and automated tracking approaches and techniques have been developed to ensure that users' and employees' attendance is checked and recorded regularly. It is commendable that several studies have been conducted to address this issue. Researchers have explored various technologies available today, including biometric-related systems, which use data about a person for identification. Biometric systems require precise information on distinctive biological features to function correctly. Passing this data into algorithms for specific outputs, typically linked to a user's or individual's identity, is fundamental to how biometric systems operate.

Selwyn et al. (2021), emphasize that facial recognition technology is now being introduced across various aspects of public life, including its integration into compulsory schooling to address issues such as campus security, automated registration, and student emotion detection. According to the authors, these technologies have largely been seen

as routine additions to school systems with already extensive cultures of monitoring and surveillance. In recent years, automated facial recognition systems have been implemented across a range of social realms. While these technologies are most frequently associated with strengthening public safety, they are also used in applications such as verifying the identity of bank users and “smart billboards” that display advertisements based on the moods of passers-by. According to Andrejevic and Selwyn (2020), facial recognition technologies are beginning to be implemented in schools for campus security systems, automated rollcalls, and student emotion and attention monitoring. In countries like the US, UK, and Australia, these technologies have so far prompted little controversy or pushback because schools in these countries have long utilized video surveillance and other forms of technology-based monitoring. However, the authors problematize the possible consequences of facial recognition technology in schools, including how it might alter schooling along divisive, authoritarian, and oppressive lines. Furthermore, according to Carney that the facial recognition systems can be used to target political dissidents and restrict access to services, while Anscombe noted that facial detection systems are used to track de-identified individuals across spaces to infer characteristics like gender, age, and mood.

Local Literature

Bayocot and Romanillos (2024) indicated that in the department of education, monitoring student attendance is a critical aspect in schooling. Traditionally, attendance tracking has been a manual and time-consuming process, often prone to errors and inefficiencies. However, the rapid advancement of technology has provided an

opportunity for educational institutions to revolutionize the way they manage attendance through computer-based attendance systems. Student attendance serves as an effective predictor of future academic achievement as well as of high school graduation. An analysis of attendance data indicates a direct link between absenteeism and school achievement, with chronically absent students scoring lower on achievement tests and dropping out of high school at higher rates than peers with better attendance. it is conducted a longitudinal investigation into the relationship between student attendance and their academic performance. The study has collected data over an extended period to examine how attendance patterns influenced academic achievement. students' attendance could impact on academic success therefore students' attendance monitoring should be done in educational institutions. Computer-based attendance systems have gained significant attention and adoption in recent years. By implementing such systems, educational institutions aim to achieve a variety of goals, including improved accuracy in attendance recording, time savings for both educators and students, enhanced data analysis, and a more robust foundation for fostering student engagement and accountability. Biometric attendance systems are technology solutions that use unique physical or behavioral characteristics of individuals, such as fingerprints, facial recognition, or iris scans, to verify and record their presence at a particular location or event. These systems have gained prominence in various sectors, including education, businesses, and government institutions, due to their potential to provide efficient and secure attendance monitoring. Otherwise, RFID technology has become increasingly

common in schools and workplaces. In addition, Machine learning algorithms have shown promise in automating attendance tracking.

Cabrillas et al. (2021) found that the regular school attendance is crucial to the development and education of learners. Attendance rate is important because students are more likely to succeed in academics when they attend school consistently attendance management of students through the conventional method had been a challenge in the recent years. Moreover, the process of checking large number of student's attendance by means of calling their names one by one will notify the teacher if the student is present or not then recording the attendance result in a form of paper and pen manually that may lead to inconsistency in data entry and generate errors that will give unreliable attendance result many systems have been developed to solve problems in checking and monitoring students' attendance in most accurate and reliable way. Nowadays, the use of Radio Frequency Identification (RFID) system in attendance monitoring is currently being practiced by most schools (public or private) under the Department of Education. The RFID attendance system has the ability of uniquely identifying each student based on their RFID tag type of ID card that makes the process of taking the attendance easier, faster and secure as compared to conventional method However, this system used to take student's attendance as they enter and exit the school premise only, so teachers in every subject still need to check and monitor students' attendance manually to see if there is no one doing cutting classes. Attendance marking in a classroom in every subject is not only burdensome but also a time-consuming task. It is extremely difficult for every teacher to manually identify the students who skip their subject on regular basis. As a result, school

administrators and teachers still need to consume time and effort in doing attendance summary report religiously. Moreover, this system can easily be compromised through the help of fellow students tapping RFID tag on behalf of their friends. Due to a usual large number of students entering and exiting the school premise as they are tapping their RFID card, there is always a possibility of proxy attendance. Hence, modern methods like biometrics have been proposed as a solution to the problems faced by traditional methods. Biometric authentication is the process that utilizes unique human traits to identify a person in an accurate manner. It requires any device that has digital photographic technology to generate and obtain the images to create and record the biometric facial pattern of the person that needs to be identified. Face recognition uses unique mathematical and dynamic patterns to recognize and identify a person in the safest and most effective one.

Domingo and Ladia (2024) emphasize that face recognition is a computer application capable of detecting, tracking, identifying, or verifying human faces from an image or video captured using a digital camera. Although significant progress has been achieved in the field of face detection and recognition for security, identification, and attendance purposes, issues still hinder the progress toward achieving human-level accuracy. A face recognition system is an application of computer vision and image processing that performs two major tasks: identifying and verifying a person from an image or a video database. Furthermore, using biometric applications for person authentication has gained considerable attention among computer vision researchers. Biometric applications play a vital role in reliable person authentication, especially in

recording attendance for employees in large organizations. Among the different biometric systems such as fingerprint recognition, face detection is considered the most optimal solution for enrollment and attendance purposes the attendance system is essential in schools and colleges since the manual attendance process has many challenges, including reduced accuracy and time inefficiency. Implementing a face recognition technique increases accuracy and minimizes the time required for recording attendance Efficient attendance management also leads organizations to improve overall corporate performance and accomplish specific goals. Maintaining accurate attendance records helps ensure working discipline and productivity Moreover, manual attendance systems are prone to errors and fraudulent timekeeping, which may affect institutional productivity. The attendance monitoring system serves as an essential component of organizational information systems

A study by Latonero (2024) mentioned that the newly installed facial recognition attendance and security system at San Beda University replaced the traditional ID card scans with contactless facial scans, enabling real-time monitoring of attendance, minimizing human error, and enhancing overall campus security. This transition to biometric technology was part of the institution's modernization efforts to streamline daily operations while ensuring data accuracy and accountability among students and faculty. The system's advanced features include anti-spoofing technology to prevent fraudulent check-ins, data encryption to protect sensitive biometric information, and a user-friendly administrative dashboard that allows instant report generation and tracking. Furthermore, the implementation demonstrated how automation can reduce manual

workload for staff and improve institutional efficiency. Latonero noted that early challenges emerged, particularly in network calibration and biometric registration, where students had to undergo multiple scans for accurate enrollment. Despite these initial setbacks, the system proved to be effective in ensuring security, transparency, and convenience in attendance monitoring, setting a model for other educational institutions in the Philippines to follow.

Grefaldo and Bausa (2025) revealed that in recent years, technology has significantly transformed the landscape of educational institutions, particularly in the area of monitoring and records management. In the Philippines, many schools are gradually adopting solutions for student monitoring and attendance systems; however, the challenges of large student populations and reliance on manual methods persist. According to Reyes and Gonzales (2023), outdated systems continue to affect record accuracy and administrative efficiency. This study aims to develop technology-based administrative processes, including attendance tracking and records management, to address these issues. Furthermore, the Department of Education (DepEd, 2022) has expressed its support for integrating digital tools in schools to promote efficiency and transparency. Despite this progress, several public high schools still rely on traditional manual methods for recording attendance, which remain time-consuming and prone to errors. According to Dela Cruz (2023), while some urban schools have started using biometric systems, many rural schools face challenges in implementing such technologies due to limited resources, highlighting the need for accessible and scalable solutions. Attendance monitoring systems utilizing biometric technologies, such as face

recognition, have gained popularity worldwide due to their efficiency, accuracy, and non-intrusiveness.

Foreign Studies

Jadhav et al. (2025) developed an automated attendance system utilizing face recognition technology to enhance the efficiency and accuracy of attendance tracking in educational institutions. The integration of face recognition technology into attendance systems offers a promising alternative to traditional methods. Leveraging computer vision and machine learning algorithms, these systems can identify and verify individuals based on their facial features, allowing attendance to be recorded automatically and accurately. Traditional methods such as manual roll calls or sign-in sheets are not only time-consuming but also prone to errors and fraudulent practices such as proxy attendance. These conventional approaches disrupt classroom activities and place additional administrative burdens on educators, while lacking real-time tracking and immediate communication with parents or guardians. Recent advancements have seen the incorporation of additional features to improve functionality and reliability. For instance, integrating attendance systems with class timetables ensures that attendance is recorded only during scheduled sessions, preventing students from entering or leaving classes outside the allotted time. Another significant feature is the use of automated SMS notifications to inform parents when their child is absent. This fosters better communication between educational institutions and families, ensuring timely intervention and promoting accountability. Nguyen-Tat et al. (2024) introduced an attendance management system utilizing Haar Cascade algorithms combined with

OpenCV2 on embedded devices such as the NVIDIA Jetson Nano. Their system emphasizes cost-effectiveness and resource efficiency, making it suitable for institutions with limited computational capabilities. The design not only automates attendance recording but also addresses common challenges such as the need for low-power devices and affordability, which are often critical in large-scale implementations. Arya et al proposed a smart attendance system employing Convolutional Neural Networks (CNN) for real-time face recognition. By utilizing deep learning techniques, their system overcomes limitations associated with traditional methods, including sensitivity to lighting conditions and variations in student posture or orientation. The CNN-based approach significantly enhances recognition accuracy, enabling institutions to reliably monitor attendance without manual intervention.

A study by Rao (2022) emphasize that the AttenFace system uses a snapshot-based face recognition technique to automatically monitor student attendance and reduce proxy attendance is a mandatory part of every class in colleges. Often, there is a minimum attendance requirement for courses taken by students. The simplest methods of taking attendance include rollcall or manually signing on a document. These methods are tedious, time-consuming, and do not take advantage of technology in any way. Attendance will have to be manually entered from the attendance sheet into the database. Further, proxy attendance is easy, wherein a student gives attendance to another student, either by forging a signature or calling out a name during rollcall. A current solution to make attendance easier, and gaining popularity, is biometric attendance. The biometric machine can be connected to the database and update attendance automatically. The

problem of proxy is solved due to the uniqueness of thumbprints. However, this method still wastes time because students need to queue up for biometric attendance. Passing the biometric machine around in class solves this issue, but it can be disturbing. All the aforementioned methods of taking attendance share a common problem: there is no way to ensure that students sit in class for the entire duration. A student can leave class immediately after attendance or enter class just before attendance. AttenFace's novel snapshot technique of face recognition solves all these problems, including the issue of proxy attendance, which is common in colleges. Students no longer have to manually give attendance, as their presence is automatically recognized by a camera. Since the camera records at all times, it is easy to capture how long a student remains in class. Final attendance can be given only if the student remains in class above a certain threshold of time, which can be decided by the professor teaching the class. Further, the system includes an easy-to-use portal for students to check their attendance for any class and course, and for professors to override default attendance rules for a particular class or student if necessary.

Chakraborty et al. (2021) reported that the student attendance is taken manually using attendance sheets issued by department heads as part of the legislation in most learning institutions. In these sheets, students sign up for future study, which is then filled in or signed on to a computer manually. This approach is slow, time-consuming and inconsistent as some students often sign for their missing colleagues. It also makes it difficult to track the attendance of individual students in a large classroom setting. In our work, we propose the design and implementation of a face detection and recognition

system to automatically detect students attending a lecture in the classroom and recognize their attendance by recognizing their faces. While other methods of biometric authentication may be more reliable, students usually have to queue for a long time when they enter the classroom. Because of its non-intrusive nature and familiarity, face recognition is chosen because people recognize other people primarily based on their facial characteristics. The biometric (facial) system consists of an enrollment process in which the specific features of a person's face are registered in a database and then the recognition and authentication processes. The detected face (acquired from the webcam) in a picture is matched with the previously stored faces captured in this process at the time of attendance. Many universities still use conventional methods for student attendance. Because this approach is used, many students support their peers by signing in their attendance if they are absent from the school. So, while using this process, the faculty analyzes and manually manages attendance records to know the number of present and absent students. If the attendance sheet is lost, the faculty will have to attend again and in this case students who are absent will have the chance to make their present in a new sheet. The practice will also affect students as time is spent on signing, checking and sending the attendance sheet manually, in addition to being inconvenient for the lecturer. Therefore, it is necessary to develop a computerized system that can handle and allow the lecturers to easily attend and sustain the attendance. This program can be easily accessed by the faculty. The program must take careful monitoring and control of student attendance data in order to minimize the faculty's manual review of student attendance. The program analyzes all the data automatically as it is transmitted by the faculty. Instead

of traditional methods, by using our face detection and recognition system, you can get a fast and effective method to collect student attendance reliably while providing safe, stable and robust system record storage where allowed.

Prakash and Reddy (2023) emphasize that the face recognition-based attendance systems automate the attendance process by identifying individuals through facial features, improving accuracy and efficiency. Attendance systems leverage computer vision technology to automate the process of taking attendance by identifying and verifying an individual's identity based on their facial features. These systems typically involve capturing an image of the person's face and comparing it to a database of pre-stored images to determine if a match is found. Upon a successful match, the individual is marked as present for that session, and the attendance record is updated in real-time. face recognition-based attendance system include increased accuracy, efficiency, and security. By automating the attendance marking process, the system reduces human errors associated with manual entry and eliminates the possibility of proxy attendance. Additionally, the real-time updating of attendance records ensures timely and accurate tracking of student presence. A typical face recognition-based attendance system operates by capturing a live video feed from a camera and identifying individuals by matching their facial features with pre-stored facial data. Technologies such as OpenCV and the face recognition library in Python are commonly utilized to detect and recognize faces in real-time. Initially, the system requires the collection of facial data of individuals and their respective names, which are stored in a database for reference. During operation, the live video feed is processed, and individuals are identified by comparing

their facial features with the pre-stored data. If an individual's facial features match with the data in the database, their attendance is automatically marked, and the timestamp is recorded.

Siyan Zhu (2023) discusses the thesis on smart campus with classmates, that smart campus contains many components be founded, the part about monitoring tracking management and face recognition is very interesting. With the development of related technologies, the current monitoring system should be able to combine Internet of Thing (IoT) technology to do more things in various fields and enrich its functions. Related application fields are not limited to smart campuses, and more practical applications are tried in various public security, hospitals, smart transportation and other fields. For example, in terms of intrusion detection and alarm, the traditional method is generally composed of detectors (or sensors), control parts, sound and light alarm circuits, power supplies and related auxiliary circuits. In practical applications, there are many types of detectors. Generally, infrared, ultrasonic, microwave, photoelectric switch, vibration and other sensors are used for one or more combined detections. There are various types of equipment and rich system components. To a certain extent, these systems have problems such as poor stability, environmental restrictions on installation, and high false alarm rate. With the development of the Internet of Things industry and technology, many related problems can be solved based on the Internet of Things technology, so that related industries have broader prospects and application space. This thesis analyzes the possible and current uses of surveillance tracking management and face recognition in various scenarios and their possible impact on the future, whether it is currently possible to use

this device and possible solutions for the current industry. In general, surveillance tracking management and face recognition is attracting attention as a way to solve various safety and efficiency problems. Many companies have already begun to try related businesses. There are still many problems to be solved, such as legal issues, application environment and technical immaturity.

Local Studies

A study by Bautista (2021) emphasizes that the Face Recognition with GSM Technology system developed at ACLC College of San Pablo is designed to monitor both student and employee attendance by recording their entry and exit times. This study aims to develop a Facial Recognition System equipped with GSM technology to monitor all students, faculty, administrative staff, and visitors entering and exiting the school premises, thereby enhancing the security of everyone within the school. The system will be able to record and recognize individuals as they come in and out of the school. Using GSM technology, it will send real-time notifications to guardians, parents, or other concerned individuals whenever their child or ward enters or leaves the school vicinity. Attendance plays a crucial role in a student's academic performance, as a single absence can significantly affect learning outcomes. Factors such as lack of interest in classes, laziness, preference for leisure activities like visiting computer shops, or peer influence often contribute to absenteeism. Traditional manual attendance monitoring is time-consuming, delaying the reporting of absences to parents or guardians, which can worsen the problem as students may miss multiple classes before their absence is noticed. With today's technological advancements, sustainable solutions have emerged that improve

both educational delivery and learning effectiveness. ACLC College of Gapan, a pioneer in adopting technological innovations in the city, currently has 360 students, 11 administrative staff, and 26 employees. Despite 20 years of existence, the school still relies on traditional methods for monitoring attendance. ACLC College Monitoring System using Facial Recognition and GSM Technology aims to provide parents and administrators with timely and accurate information regarding the attendance of students and faculty. GSM technology offers a fast and reliable way to inform parents, guardians, and administrators about attendance anytime and anywhere.

Aquino et al. (2023) develop an attendance system that simplifies and improves the accuracy of tracking employees' or students' attendance. The primary goal of the project is to develop a system that automatically records attendance using facial recognition. The results indicate that the system is operational, accurate, and provides significant benefits to both students and faculty members. Currently, various attendance and monitoring solutions are used in the industry. Even though these solutions are mostly automated, they all have certain limitations and flaws. To address these issues, a new deep learning-based facial recognition attendance system has been proposed in this research. The study details the process of constructing a facial recognition component by integrating advanced methodologies and deep learning improvements. It was discovered that, even with fewer face images, the proposed data augmentation approach achieved high accuracy, reaching 95.02%. The study also explores different solutions for face identification, an area that remains largely underexplored. According to Sawhney et al., the primary objective of their study is to develop a smart attendance management system

based on facial recognition, which addresses the shortcomings of current automated systems used in modern environments. While authenticating an individual for attendance is a long-established process, previous advancements have often failed to keep pace with the fast-evolving, technology-driven lifestyle. Traditional attendance methods, such as fingerprint scanning, require queuing and consume a considerable amount of time. In cases of finger injuries, fingerprint accuracy may be compromised, and fingerprints can even be forged. Similarly, ID card scanning cannot always verify the cardholder's identity, resulting in fraudulent attendance behavior. In contrast, a facial recognition system can identify a person from a distance without physical contact or interaction. Additionally, the system enhances security and crime deterrence since captured images can be stored in a secure repository and used later for identification purposes.

According to Sebastian et al. (2024), they emphasize that the Design and Development of an IoT-Based Automated Attendance System Using Facial Recognition demonstrated that a contactless and portable attendance system can effectively replace traditional log sheets and fingerprint-based methods. The system was developed using Python and OpenCV for facial detection and recognition, while a mini-PC paired with a webcam served as the primary hardware controller. They also created a mobile application through Android Studio to provide an accessible user interface for monitoring attendance records. Their study showed that the system achieved satisfactory overall performance after being tested with 100 participants, fulfilling the main objective of creating an efficient and automated attendance mechanism. They also highlighted that the entire prototype cost only 14150, proving that facial-recognition systems can be

implemented even with budget limitations. The authors concluded that IoT-enabled facial-recognition attendance systems can significantly enhance efficiency, accuracy, and hygiene in academic environments. Automatic attendance management systems have become essential, as this step not only enhances efficiency but also boosts effectiveness. The existing system is fraught with uncertainty, leading to inaccuracies and inefficiencies in attendance tracking and record maintenance. Numerous challenges emerge when institutions are unable to enforce regulations within the current system. Therefore, through technological means, the project aimed to address the shortcomings present in the traditional process. By automating most tasks, it seeks to elevate attendance tracking to a significantly improved level. Through the utilization of the Python programming language, the development of a facial recognition program and an automated attendance system became achievable. The system facilitates the automatic organization of attendance records into an Excel spreadsheet, delivering neatly arranged and structured information for easy viewing and management. The development of this system helps faculty workers organize the attendance log more efficiently. The mobile application also assists both students and instructors who need information regarding whether someone specific is currently present or absent for the day. Administrators are able to conveniently access all attendance records through the software interface, which retrieves information directly from the database and simplifies the process compared to traditional methods. Staff employees typically record attendance for both in-class and absences using the traditional method.

A study by Cañete (2021) indicated that the Real-time Face recognition used the Descriptive research method to develop a prototype model of a system to achieve the objectives of the study. Prototype Model is used to achieve the System Development Method in which the requirements are gathered, build a quick design, build a prototype Tested by the user through user evaluation and then reworked of refining the prototype as needed until an acceptable prototype will be implemented and maintain from which the complete system can now be used. Daily attendance of a student is a major factor in every educational institution, manual checking of attendance sheets is time consuming, tedious and laborious for a teacher that cannot memorize all students who are attending the class especially during pandemic which face to face classes are prohibited. Truancy and Cutting Classes are common problems of every school which student usually attending the class to get their self- present during attendance but eventually will skipping the class. This paper aimed to develop a system software prototype that will help for the automated student attendances that can be used anywhere. There are (4) processes in which the student must do to take attendance before taking a class. First, fill in the information given including name, surname and student ID, etc. Second, the image of a student will be captured like a selfie by the camera. Third, the Student needs to save the profile, and the system will ask for a password. The system will recognize the face that is registered in name with the help of the dataset and will automatically list their names in the system, Face recognition evaluates student faces to recognize. Lastly, the registered name in images recorded in registration will automatically be saved in an excel file which the final step to take attendances in the record. To develop a useful and

applicable Real-time Face recognition system there are several parts need to take in hand. The system will only recognize the faces that are registered in the system. The teacher controls or holds of taking the attendance of the student. The purpose of this study is to develop a system using OpenCV that automatically takes attendance using face recognition in class and converted it into an excel file for a record. Specifically, an OpenCV Real-time face recognition system can be used to save time, effort, misspellings of names and to prevent fake attendances.

In the context of Cabrillas et al. (2021), the manual checking of attendance may lead to inconsistency of data inputs and may generate unreliable attendance result. Hence, Radio Frequency Identification (RFID) system has been developed to solve this problem, but it allows only checking student's attendance as they enter and exit the school premise only. In consequence, teachers in every subject still need to check and monitor students' attendance manually. Nevertheless, due to a usual large number of students entering and existing the school premise as they are tapping their RFID card, there is always a possibility of proxy attendance. Thus, Mobile-Based Attendance Monitoring System Using Face Tagging Technology (MBAMSUFTT) was developed to provide an attendance monitoring system through biometric authentication such as face recognition. The system serves as a tool for teachers to check and monitor student's attendance in most reliable and accurate way using their smart phones. The MBAMSUFTT generates attendance report intended for close monitoring and printing of student's attendance result. But the reliability of the attendance result (output) of the system depends on the quality of picture (input) sent by the user. Camera specification, ambiance lighting

condition, and proper position of students while taking photo is exclusively required. quality characteristics by the IT experts, and its functionality, performance efficiency, and usability by the

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the technical foundation of the proposed system. It consists of four major sections: Research Approach, Methodology, Requirements Analysis, and System Requirements. These sections establish the conceptual and technical basis for the planned design and development of the InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management.

Research Approach

The study employed a developmental research approach, focusing on the design, development, and evaluation of the InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management. The system was developed following a defined software engineering process, while its acceptability will be assessed using the ISO 9126 Software Quality Standard. ISO 9126 covers six core software quality attributes: functionality, reliability, usability, efficiency, maintainability, and portability. An evaluation instrument aligned with these ISO dimensions was used to determine the system's acceptability based on user responses.

Methodology

The Computer System Management System (CSMS) Waterfall Model was adopted in this study. The Waterfall approach is preferred because of its structured and sequential process, ensuring that each phase is completed before proceeding to the next.

The development process was consisted of the following phases: System Planning, System Design, System Implementation, System Testing, System Deployment, and System Maintenance. Each phase was carefully executed and documented to ensure accuracy, reliability, and consistency in the system's overall performance. Stakeholder feedback from students, teachers, parents, and administrative personnel were gathered at key stages of development to ensure that the system remains relevant, user-centered, and aligned with institutional goals.

The waterfall process, originally called the waterfall model, was among the earliest formal software engineering lifecycle processes used in software development. It is considered one of the simplest and most structured ways of organizing lifecycle activities into a clear and systematic process. The model was named “waterfall” because it follows an ordered and essentially linear sequence of phases, where each stage flows into the next like cascading tiers of a waterfall.

This methodology emphasizes a structured progression through phases such as requirements analysis, system design, implementation, testing, deployment, and maintenance. In this process, each phase must be completed before moving on to the next, allowing developers to focus on one stage at a time while ensuring proper planning, thorough documentation, and careful evaluation throughout the software development cycle, reducing the risk of overlooked details and errors.

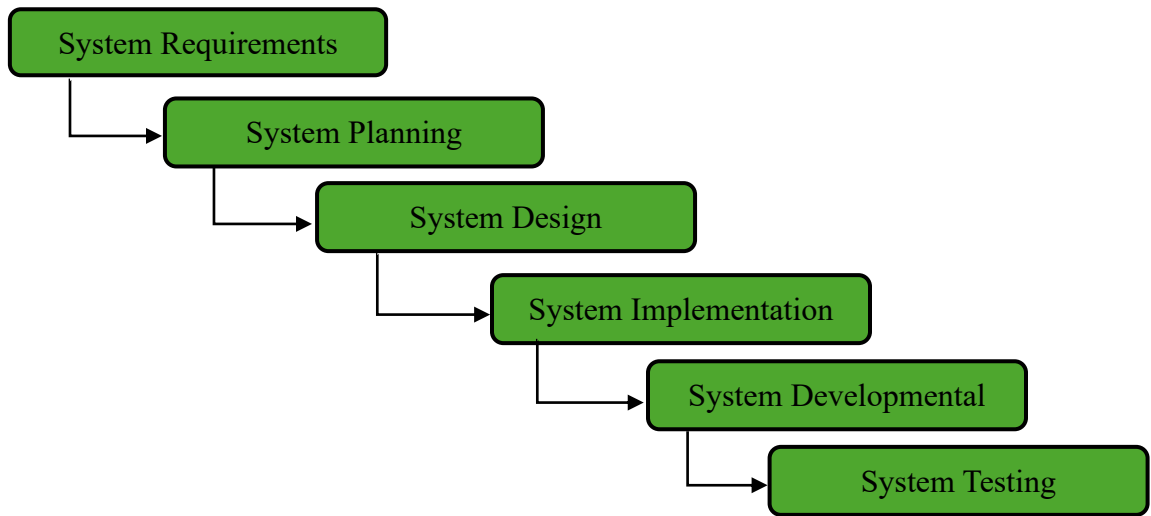


Figure A. Waterfall Model Diagram

Figure A presents the Waterfall Model Diagram, which illustrates the step-by-step and sequential progression of the system development process. Each phase ranging from planning to deployment must be fully completed before moving to the next, ensuring clarity, structure, and minimal overlap of tasks. This model provides a clear visual guide to how the project was organized and executed, emphasizing a disciplined and methodical workflow throughout the study.

System Requirements

The system requirements identify the specific needs and expectations of the users to ensure the proposed system functions effectively. It covers both functional requirements, which define the features and operations the system must perform, and non-functional requirements, which describe the quality, performance, and constraints of the system. Information for these requirements were gathered through interviews, observation, and documentation analysis.

Table 1*Data Gathering Tools*

Tool / Techniques	Purpose	Target Users
Interview	To gather detailed information about user needs and expectations for the system	Teachers, Students, Parents, Admin
Observation	To observe current attendance and monitoring processes	Classrooms, Offices
Documentation Analysis	To review existing records, reports, and forms for workflow insights	Attendance Logs, Reports, Forms
Survey / Questionnaire	To collect additional feedback from users	Teachers, Students, Parents, Admin

Table 1 illustrates the data gathering tools of the InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management. To collect the necessary data for the development of the InterACTS face recognition and monitoring system, several data gathering tools were systematically utilized. Interviews were conducted with teachers, administrators, students, and parents to gain a comprehensive understanding of the existing attendance and monitoring processes, identify specific challenges and pain points, and determine the desired features and functional requirements for the new system. These interviews provided qualitative insights into user experiences, expectations, and suggestions, which informed the design and functionality of the system. Observation was performed in classrooms,

administrative offices, and other relevant educational settings to examine the actual execution of attendance tracking and monitoring. This method enabled the research team to identify gaps, delays, repetitive tasks, and operational inefficiencies, as well as to observe how technology and human factors interact in real-time attendance scenarios. Documentation analysis involved a thorough review of existing attendance logs, daily reports, forms, and other related records to understand the current workflow, data recording methods, and management practices. This review highlighted inconsistencies, redundancies, and areas where automation could significantly improve accuracy, organization, and accessibility of attendance data. Together, these data gathering tools provided a multi-mode perspective that combined user input, real-world practices, and historical documentation, ensuring that the InterACTS system was designed based on actual needs, practical considerations, and evidence-based insights. The integration of these tools allowed the research team to discuss the findings, validate information from different sources, and develop a system that addresses both functional requirements and operational challenges, ultimately contributing to a more efficient, reliable, and user-friendly attendance monitoring solution.

Table 2*Functional Requirements*

Modules	Function
1.1 Authentication & Security	Provides secure login, role-based access control, and protection of sensitive data to ensure that only authorized users can access the system.
1.2 User Management	Allows creation, updating, and management of accounts for teachers, students, parents, and administrators, including user roles and permissions.
1.3 Train Data Face Module	Captures and trains the system with facial data to enhance the accuracy and reliability of face recognition for attendance purposes
1.4 Real Time Face Recognition	Detects and recognizes faces of users in real time to automate attendance and monitoring, reducing manual errors and saving time.
1.5 Monitoring Module	Sends notifications, alerts, and updates to users about attendance, schedule changes, or important announcements.
1.6 Attendance Logging	Automatically records and stores attendance data, ensuring accurate tracking and easy retrieval of records.
1.7 Analytics and Reports	Generates detailed reports and analytics for administrators, teachers, and parents to evaluate attendance patterns, trends, and system usage.
1.8 Verification	Verifies the identity of students and staff to prevent fraud or errors in attendance records.
1.9 Notification	Sends notifications, alerts, and updates to users about attendance, schedule changes, or important announcements.
1.10 Central Interface	Displays a comprehensive summary and overview of attendance, monitoring, notifications, and system statistics for quick decision-making.

The Table 2 provides a functional requirement of InterACTS, focusing on delivering a comprehensive and efficient system for attendance and monitoring in an educational setting. The system ensures secure access through authentication and role-based controls, protecting sensitive data while allowing only authorized users to perform specific actions. User management capabilities allow administrators to create, update, and maintain accounts for teachers, students, parents, and staff, assigning appropriate roles and permissions to ensure accountability and proper access. The face recognition modules, including data training and real-time recognition, automate attendance recording by accurately detecting and identifying users, thereby reducing manual errors and saving valuable time for both faculty and administrative staff. Monitoring and attendance logging features provide real-time tracking of activities, ensuring that attendance records are reliably stored and easily retrievable when needed. Analytics and reporting tools allow administrators and teachers to evaluate attendance patterns, identify irregularities, and make data-driven decisions to improve student engagement and operational efficiency. Additional modules, such as verification, notifications, and the dashboard, support system integrity, enhance communication, and facilitate quick decision-making by presenting an organized overview of attendance, alerts, and key performance metrics. Furthermore, the integration of these functional components enables the system to adapt to diverse educational workflows, supports scalability for different class sizes, and ensures interoperability with existing institutional infrastructure.

Table 3*Non - Functional Requirements*

Requirement Type	Function
2.1 Performance Requirements	NFR-01: InterACTS must recognize a student's face and record attendance within 2–3 seconds to ensure fast and efficient processing. NFR-02: The system must handle at least 80–100 simultaneous login and attendance requests without slowing down. NFR-03: Pages such as login, dashboard, and attendance records must load within 3–5 seconds under standard internet speed. NFR-04: The system must support large databases of students and attendance logs without performance degradation. NFR-05: The system must maintain stable performance even during long hours of continuous school operations
2.2 Usability Requirements	NFR-06: The interface must be simple and easy to navigate for students, teachers, and administrators. NFR-07: Key functions such as scanning attendance, viewing records, and generating reports must be accessible within three clicks. NFR-08: The system must show clear on-screen instructions during face scanning to minimize user errors. NFR-09: Buttons, labels, and icons must be consistent and easily understandable for first-time users. NFR-10: The system must be usable by individuals with basic computer literacy.
2.3 Security Requirements	NFR-11: All facial data, attendance records, and user information must be encrypted to ensure data protection. NFR-12: Only authorized users such as administrators and faculty members may access restricted system modules. NFR-13: The system must automatically log out users after 10 minutes of inactivity to prevent unauthorized access. NFR-14: User passwords must follow minimum security standards and be stored in encrypted form. NFR-15: The system must restrict face recognition features to logged-in and authorized users only.
2.4 Reliability Requirements	NFR-16: The system must remain operational at least 95% of the time during school hours. NFR-17: Attendance data must remain intact and accurate even during unexpected shutdowns or system failures. NFR-18: The system must perform automatic data backups at least once per day. NFR-19: The system must quickly recover from minor errors and interruptions without affecting stored data. NFR-20: The face recognition function must work consistently even with variations in lighting..
2.5 Compatibility Requirements	NFR-21: The system must be fully functional on major web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. NFR-22: The system must adjust correctly to different Gadgets User.

Table 3, the Non-Functional Requirements of InterACTS, describes the quality standards that guide the system's overall performance and user experience. It ensures that the system performs its tasks accurately, including managing users, recognizing faces in real time, logging attendance, monitoring activities, sending notifications, generating reports, verifying data, and displaying the dashboard. The system is designed to operate consistently and dependably, maintain a user-friendly interface for easy navigation, work efficiently even with multiple users accessing it simultaneously, allow easy updates and maintenance, and function seamlessly across different devices such as PCs, laptops, tablets, and mobile phones. By adhering to these non-functional requirements, InterACTS provides a reliable, efficient, and accessible platform that supports smooth day-to-day school operations and meets the needs of all users.

System Planning

It involves organizing and outlining the structure and flow of the InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management. It includes planning the modules, user interactions, and data flow to ensure that all requirements are properly addressed. Tools such as context diagrams, data flow diagrams, and use case diagrams are used to visualize the system's functionality and interactions, serving as a foundation for the design phase.

Gantt Chart

As part of the system requirements, the researchers used a Gantt Chart to organize and schedule the activities involved in developing InterACTS, the web-based multi-modal face recognition attendance and monitoring system. The chart provided a clear timeline for completing each task, helping the researchers manage time effectively and track the progress of the system's development in a systematic manner. It also facilitated proper coordination among the researchers, ensuring that each phase of the project was executed efficiently and that deadlines were met. By visualizing the sequence of activities and their respective durations, the Gantt Chart contributed to a more structured approach, allowing the team to identify potential delays early and adjust plans, accordingly, thereby improving overall productivity and ensuring the timely completion of the system.

Activities	START	END	DAYS	December				January				February			
Weeks				1	2	3	4	1	2	3	4	1	2	3	4
System Planning															
1.1 Manuscript Revision	Sept 21	Oct 10	20												
1.2 Building a Database and Flow Charts	Oct 21	Nov 21	32												
1.3 Continuation of System	Oct 21	Nov 21	32												
1.4 Consultation to our Adviser	Oct 21	Oct 21	32												

Figure B. September-November Gantt Chart of InterACTS

Figure B presents a Gantt Chart of the planning phase, illustrating the timeline for the InterACTS project over a period of two and a half months, from September to November. The Planning Phase is divided into four key activities: topic setup, information gathering, identifying system users and requirements, and determining the project's scope and functionalities, which encompassed the capabilities of the system. This comprehensive planning phase was essential in laying a strong foundation for the development of the InterACTS system, ensuring that all critical aspects of the project were carefully considered and addressed. Through systematic execution of topic setup, information gathering, user and system identification, and scope and functionality determination, the project team aimed to create a clear roadmap for the subsequent development and implementation stages, promoting efficient workflow.

Activities	START	END	DAYS	December				January				February			
Weeks				1	2	3	4	1	2	3	4	1	2	3	4
System Planning															
1.1 Manuscript Revision	Dec 1	Dec 12	12												
1.2 Building a Database and Flow Charts	Dec 1	Feb 28	90												
1.3 Continuation of System	Dec 1	Feb 28	90												
1.4 Consultation to our Adviser	Jan 23	Feb 28	37												

Figure C. December-February Gantt Chart of InterACTS

Figure C presents the Gantt Chart of the development phase, illustrating the timeline of the InterACTS project from December to February. This phase spans approximately three months and focuses on the continuous improvement and development of the system. The Development Phase consists of four major activities: manuscript revision, building of the database and flowcharts, continuation of system development, and consultation with the project adviser. Manuscript revision was conducted to refine the documentation and ensure alignment with the system's objectives and requirements. Simultaneously, the construction of the database and flowcharts was carried out to establish a structured foundation for data management and system processes. The continuation of system development involved the actual implementation and enhancement of system features based on the defined scope and functionalities. Moreover, regular consultation with the project adviser played a crucial role in providing guidance, validation, and necessary adjustments throughout the development process. This phase was essential in ensuring that both the system and its documentation were accurately developed, well-structured, and aligned with academic and technical standards, ultimately contributing to the successful progression of the InterACTS project.

Activities	START	END	DAYS	March				April				May			
Weeks				1	2	3	4	1	2	3	4	1	2	3	4
System Planning															
1.1 Gathering Information data	March 31	April 10	11												
1.2 Continuation of Manuscript	March 18	May 8	52												
1.3 Continuation of System	March 1	April 18	49												
1.4 Manuscript Bookbind	May 1	May 10	10												

Figure D. March - May Gantt Chart of InterACTS

Figure D presents the Gantt Chart of the finalization phase, illustrating the timeline of the InterACTS project from March to May. This phase covers the concluding activities necessary to complete both the system and its corresponding documentation. The Finalization Phase consists of four key activities: gathering of information data, continuation of manuscript, continuation of system development, and manuscript bookbinding. The gathering of information data was conducted to supplement and validate the remaining details required for both the system and the manuscript. The continuation of the manuscript focused on refining, organizing, and completing all chapters to ensure clarity, coherence, and compliance with academic standards. At the

same time, the continuation of system development involved final adjustments, debugging, and enhancement of system functionalities to ensure optimal performance. Lastly, manuscript bookbinding marked the completion of the documentation process, preparing the study for final submission and evaluation. This phase was crucial in ensuring that the entire InterACTS project was polished, complete, and ready for presentation, demonstrating the culmination of all efforts from planning to development and final implementation.

Furthermore, the finalization phase also ensured that all system components were properly integrated and functioning according to the intended objectives of the study. Necessary revisions and improvements were implemented based on feedback, observations, and testing results to enhance the reliability, accuracy, and usability of the system. This phase also emphasized the importance of quality assurance by reviewing both technical and documentation aspects of the project to minimize errors and inconsistencies. Through careful evaluation and completion of all remaining tasks, the researchers were able to finalize a system that is organized, functional, and aligned with the objectives and requirements of the study.

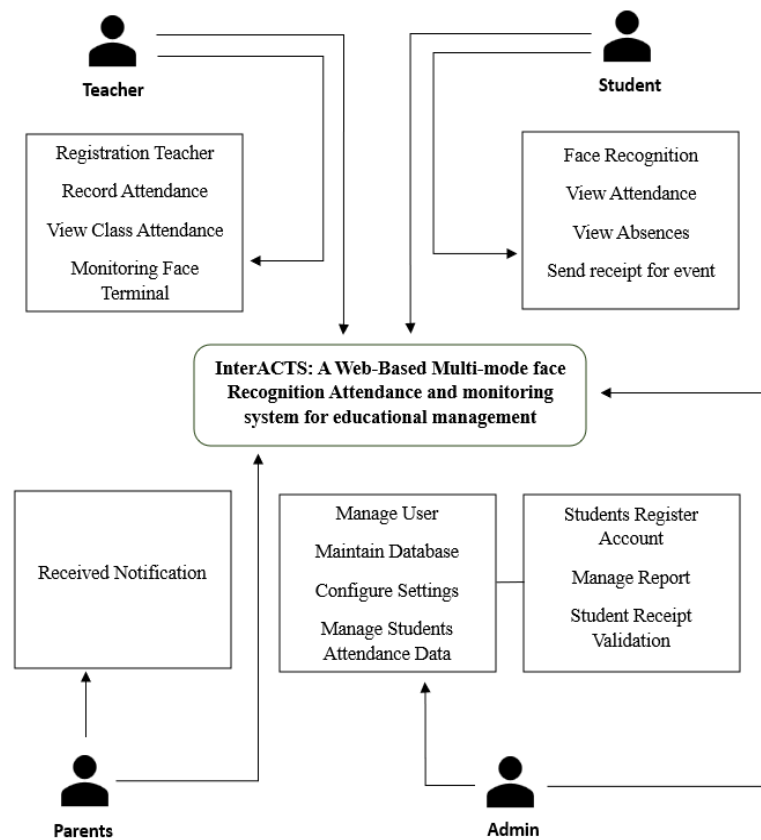


Figure E. Use Case Diagram

The use case diagram of InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management illustrates the interaction between the system and its primary users: teachers, students, parents, and administrators. The diagram highlights the system's role as the central hub connecting all users, while each user interacts with specific functionalities tailored to their roles provides a high-level overview of the system's functional requirements, illustrating how different users engage with InterACTS to streamline attendance monitoring and educational management.

Table 4*Project Scope*

Category	Description
Project Title:	InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management.
Objective:	To provide an automated, reliable, and efficient system for attendance tracking and monitoring using face recognition technology, benefiting teachers, students, parents, and administrators.
Scope: (Inclusions)	Includes user management, face data training, real-time face recognition, attendance logging, monitoring, notifications, analytics and reports, verification, authentication & security, and dashboard functionalities.
System Boundaries:	Excludes non-educational usage, hardware procurement, integration with external apps, and manual attendance methods outside the system.
End Users:	Teachers, students, parents, and school administrators who will interact with or benefit from the system.
Deliverables:	Fully functional web-based face recognition attendance and monitoring system, complete with user manuals, training guides for teachers and students, and administrators.
Assumptions:	Users have basic computer literacy, access to compatible devices (PC, laptop, tablet, mobile) and reliable internet, and willingness to adopt and use the system. It is assumed that the institution will support system implementation and provide necessary resources for testing and deployment
Constraints:	Development is limited by budget, time restrictions, server capacity, and dependency on stable internet connectivity. Other constraints include limited access to historical data, user unfamiliarity with face recognition technology, and the need to comply with data privacy regulations.

Table 4 presents the project scope of the InterACTS system: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational

Management. The table outlines the objectives, inclusions, exclusions, end users, deliverables, assumptions, and constraints of the system. It defines the boundaries of the project by specifying the features to be included, such as user management, face recognition, attendance logging, monitoring, notifications, analytics, verification, authentication, and dashboard functionalities, as well as what is excluded, such as non-educational usage, hardware procurement, and integration with external applications. The table also identifies the target users, expected deliverables, assumptions regarding user capabilities and institutional support, and constraints related to budget, time, infrastructure, an compliance with data privacy regulations.

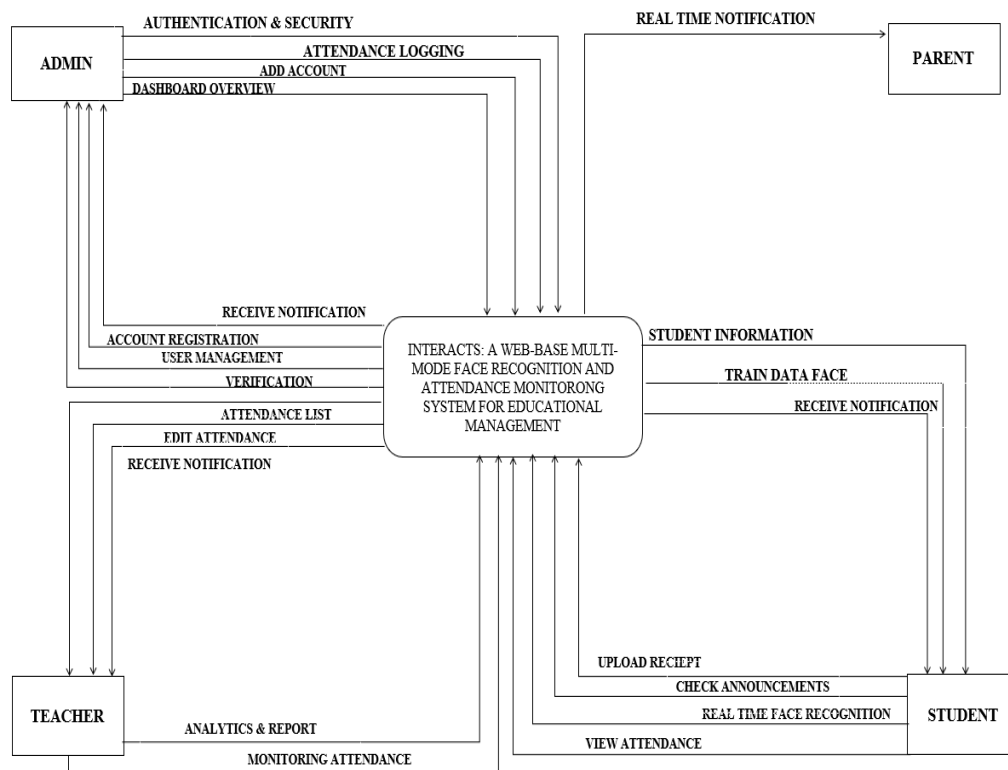


Figure F: Context Diagram

The Figure D illustrates the context diagram that provides a high-level overview of the entire Face Recognition and Monitoring System by illustrating how it interacts with its external users and entities. It shows the system as a single central process and highlights the flow of information going in and out. The purpose of the context diagram is to present a simplified view of the system's boundaries such as what enters, what leaves, and who is involved without showing internal processes or detailed steps. In this diagram, the main users such as the Admin, Teachers, Students, and Parents are represented as external entities. Each entity exchanges specific data with the system. Teachers and admin provide and receive attendance information, students provide facial data during the recognition process, and parents receive notifications or monitoring updates.

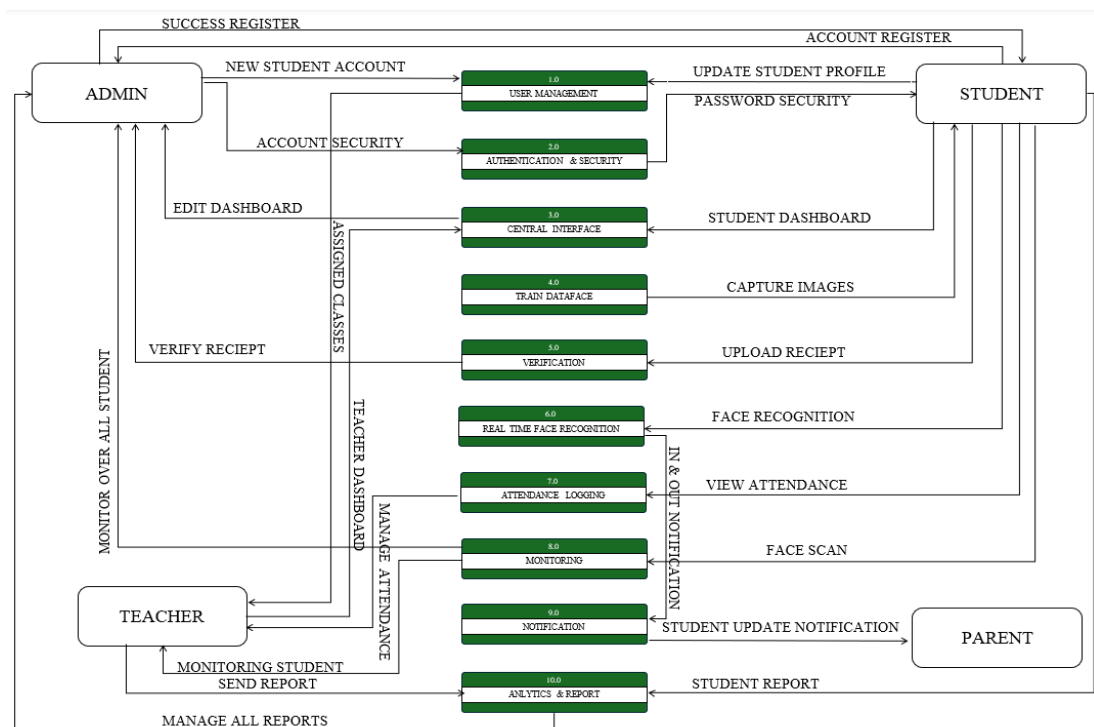


Figure G: Data Flow Diagram Level 1

Figure E illustrates the level 1 data flow diagram of InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management shows the main processes of the system and how data flows between them. It illustrates the interactions between users, teachers, students, parents, and administrators and the system's core modules, including user management, face data training, real-time face recognition, attendance logging, monitoring, analytics, notifications, verification, authentication & security, and the dashboard. The diagram highlights how data is input, processed, and stored in the database, providing a clear overview of the system's primary operations and functional workflow

System Design

The system design phase focuses on planning and creating the architecture and structure of InterACTS, the web-based multi-modal face recognition attendance and monitoring system. This phase involves translating the requirements gathered during the analysis stage into detailed specifications, diagrams, and models that guide the development of the system. It includes designing the system architecture, user interfaces, data flow, and the interactions between different components, ensuring that the system will be efficient, reliable, and user-friendly once implemented. The design provides a blueprint for developers and serves as a reference for future updates or enhancements.

System Architecture Diagram

The structure of InterACTS shows the key components of the system and how they interact. It highlights the relationships between users, devices, the web application server, software modules, and the database, providing a clear visual representation of the

system's design and data flow. This diagram serves as a guide for understanding how the system operates and supports its development and implementation. It also explains the integration of core modules such as user management, real-time face recognition, attendance logging, monitoring, analytics, notifications, verification, authentication, and dashboard functionalities. The diagram emphasizes data integrity, security, and efficient communication between components, ensuring that the system can reliably manage attendance processes and deliver accurate information to teachers, students, parents, and administrators. Additionally, it provides guidance for developers in planning system deployment, addressing potential technical issues, maintaining scalability for future updates, and ensuring that all components work cohesively to provide a seamless and efficient user experience.

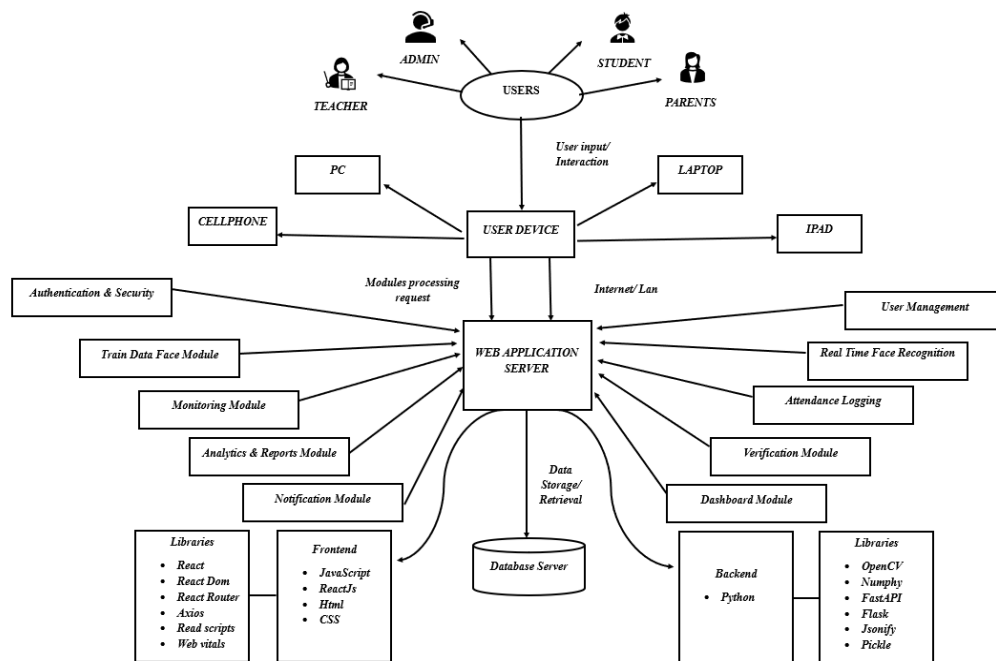


Figure H. System Architecture Diagram

Figure F illustrates the system architecture diagram of InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management and shows the main components and their interactions. Users, including teachers, students, parents, and administrators, access the system through a web interface on PCs, laptops, tablets, and mobile devices. Data flows between the user interface, application modules, face recognition engine, and database, ensuring that attendance and monitoring processes are automated and efficient. Modules such as user management, real-time face recognition, attendance logging, monitoring, analytics, notifications, verification, authentication and security, and dashboard work together to provide a seamless and reliable experience. The architecture also ensures data integrity and privacy while supporting quick access to reports and analytics. Furthermore, the system is designed to be scalable, allowing additional users and modules to be integrated with minimal disruption. Redundancy and backup mechanisms are implemented to prevent data loss, and security protocols are in place to protect sensitive information. Overall, the architecture provides a robust, efficient, and secure environment that supports the educational management processes effectively.

Entity-Relationship Diagram

This figure illustrates the Entity-Relationship Diagram (ERD) for InterACTS, showing the structure of the system's data and the logical connections between its main components. It provides a conceptual view of how information is organized and flows within the system, serving as a guide for future database design and development. Although the actual database is not yet finalized, the ERD helps ensure that all essential

components and their interactions are considered, allowing the system to manage attendance, face recognition data, and notifications efficiently. This conceptual representation also aids the researchers and developers in planning the system architecture and understanding how data will be stored, accessed, and processed once the database is implemented. Moreover, the ERD highlights the relationships between entities such as students, teachers, administrators, and attendance records, ensuring data integrity and minimizing redundancy. It also supports the identification of primary and foreign keys, which is essential for maintaining relational

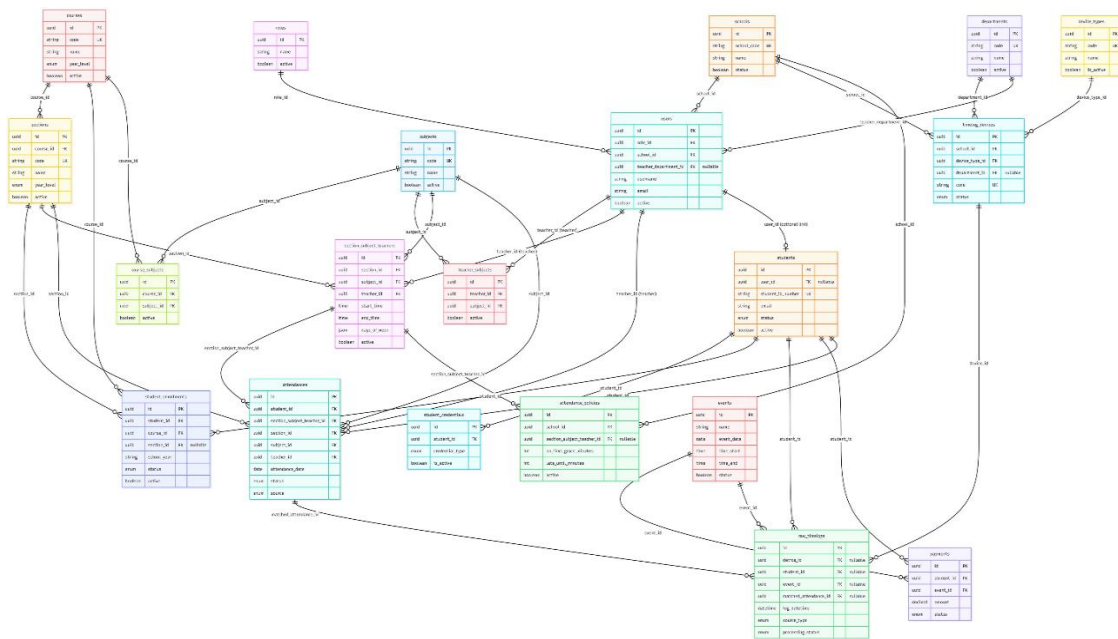


Figure I. Entity-Relationship Diagram of InterACTS

User Interface Mockups or Wireframes

This section presents the User Interface (UI) mockups and wireframes of InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management. It illustrates the layout, structure, and visual design

of the application, providing a clear representation of how users will interact with its features before actual development. These mockups serve as a visual guide in understanding the system flow, functionality, and overall user experience, ensuring that the design aligns with the intended requirements of the system. In addition, it helps the developers and stakeholders visualize the proposed interface, identify possible improvements early in the design stage, and maintain consistency across all system modules.

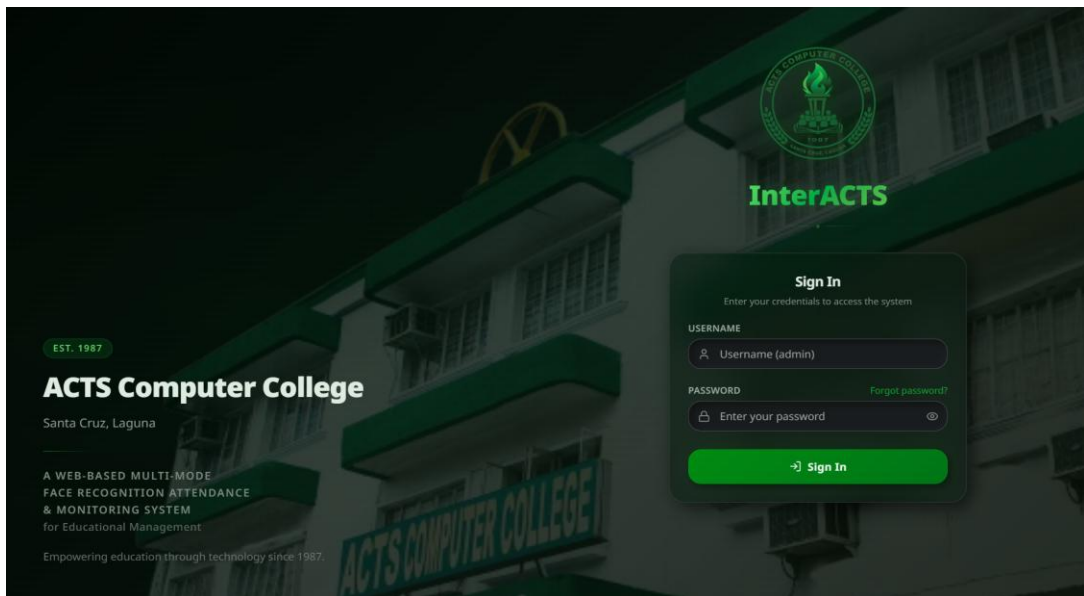


Figure J: User Log in Page of InterACTS

Figure G shows the Log In page of InterACTS, where users can securely access the system by entering their credentials. It highlights the layout and design of the login interface, including input fields for username and password, and the login button, ensuring ease of use, proper navigation, and secure access for authorized users. The design also emphasizes clarity and simplicity, allowing users to quickly identify required

actions, while maintaining a professional and user-friendly interface consistent with the overall theme of the system.

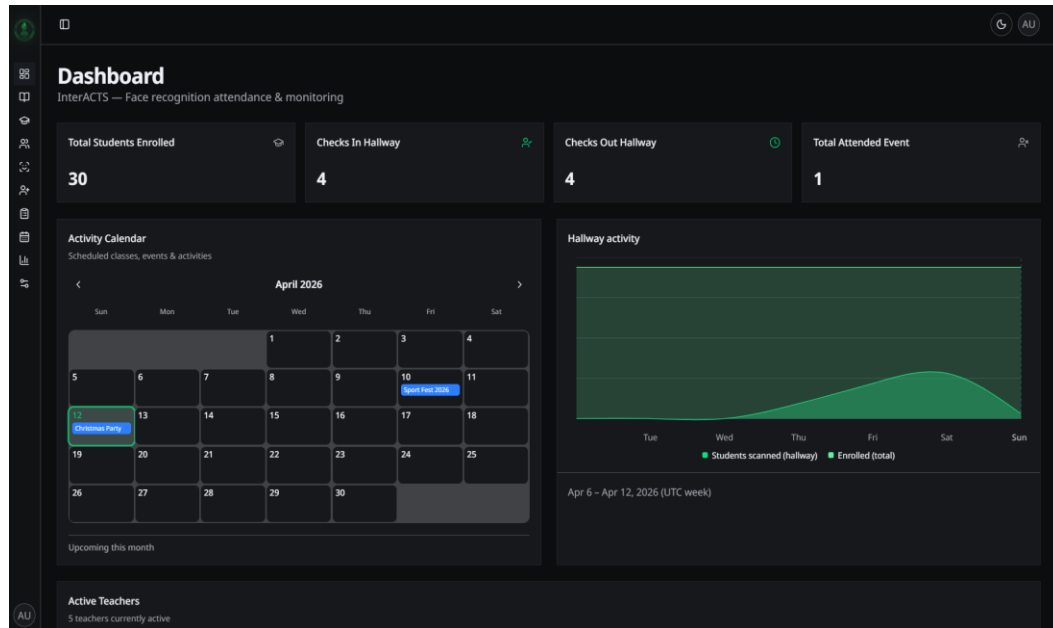


Figure K. Admin Dashboard of InterACTS

The Admin Dashboard of InterACTS provides a comprehensive overview of key system analytics and activities. It displays essential information such as the total number of enrolled students, as well as the number of students who have checked in and checked out within the hallway monitoring system. It also shows the total number of events attended by students. In addition, the dashboard includes a calendar feature for tracking scheduled activities and events, along with a list of active teachers who are currently registered in the system, serving as a centralized panel for monitoring student participation and system usage in real time.

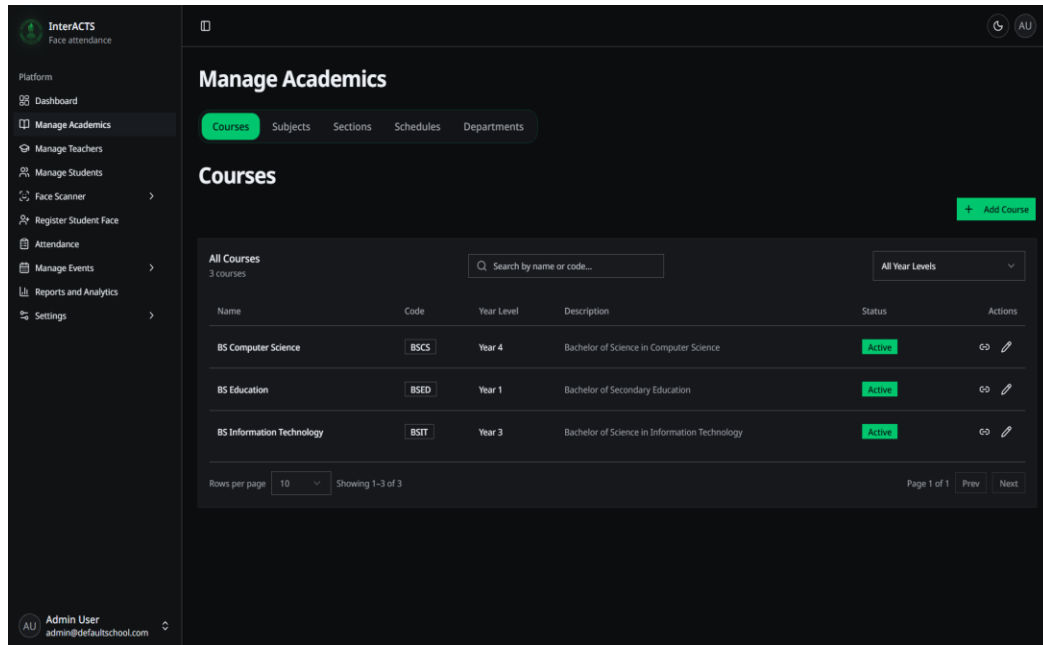


Figure L. Manage Academics of InterACTS

The Manage Academics of InterACTS allows the administrator to manage and organize academic-related data within the system. It enables the creation of courses and subjects, as well as the assignment of teachers to specific subjects. The module also supports adding sections and assigning subjects to each section. In addition, it allows the scheduling of classes and the management of departments, ensuring that all academic structures are properly organized and systematically arranged within the system.

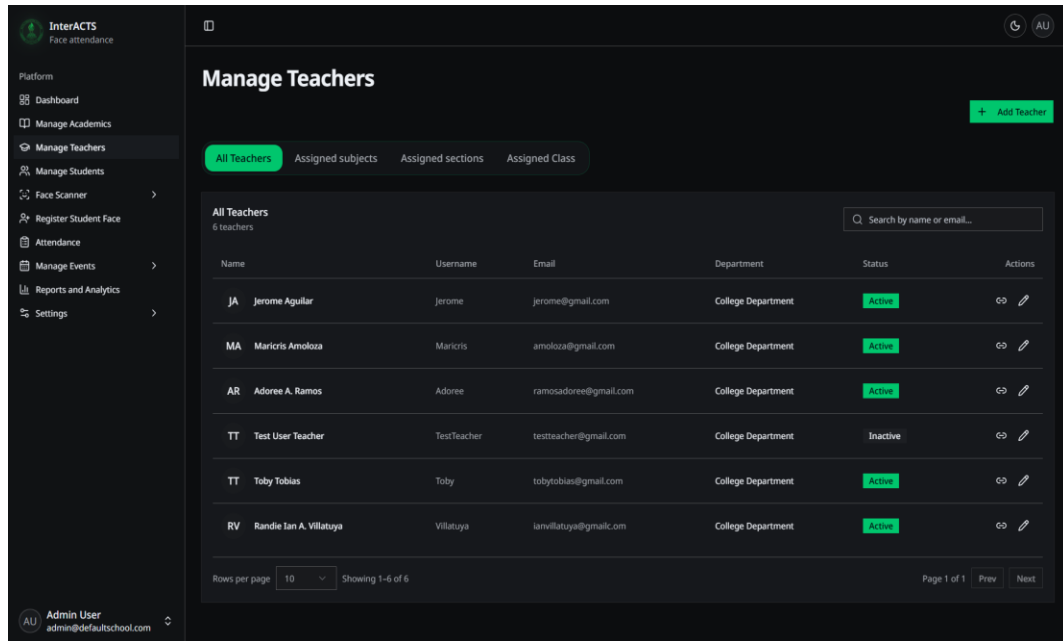


Figure M. Manage Teachers of InterACTS

The Manage Teachers of InterACTS allows the administrator to add and manage teacher accounts within the system. It provides functionality for registering new teachers and maintaining their information in an organized manner. The module also displays all registered teachers together with their assigned subjects, assigned sections, and assigned classes, allowing for easier monitoring of academic responsibilities. In addition, it helps ensure proper distribution of teaching loads and supports efficient management of faculty assignments within the system.

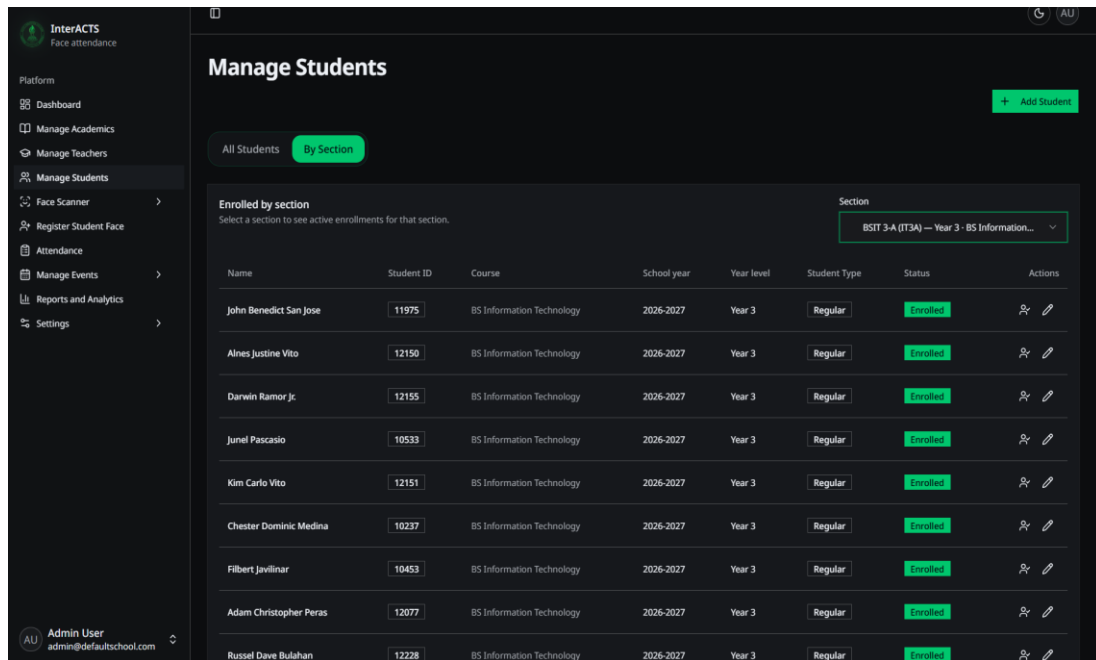


Figure N. Manage Students of InterACTS

The Manage Students of InterACTS allows the administrator to view and manage all created student records within the system. It provides functionality to update student status, such as enrolled, graduated, or dropped, depending on their current academic condition. The module also allows editing of student information when necessary. In addition, students are organized and displayed by section, making it easier to monitor and manage student distribution within each class.

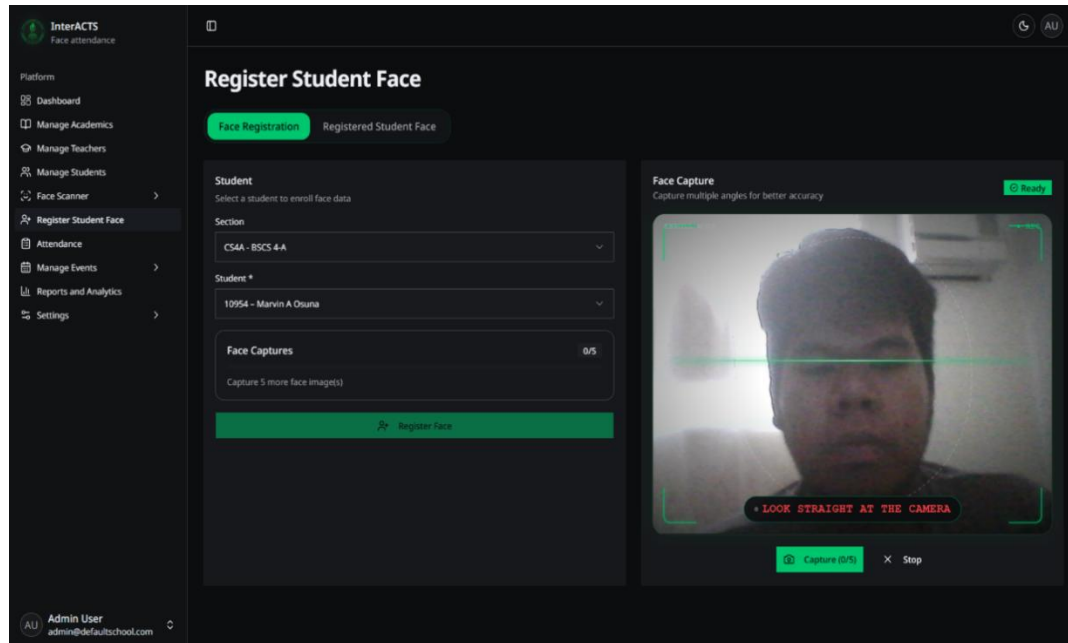


Figure O. Register Student Face of InterACTS

Figure O illustrates the Register Student Face module of the InterACTS system, where the administrator manages the enrollment of student facial data for recognition purposes. In this module, the administrator selects the appropriate section and chooses the student to be registered. The system then allows the capturing and training of the student's facial images in different angles, including front, left, right, upward view, and facial expressions such as smiling. This ensures a more accurate and reliable facial recognition process during attendance monitoring. The system also supports face retraining when necessary to improve recognition accuracy and update existing facial data.

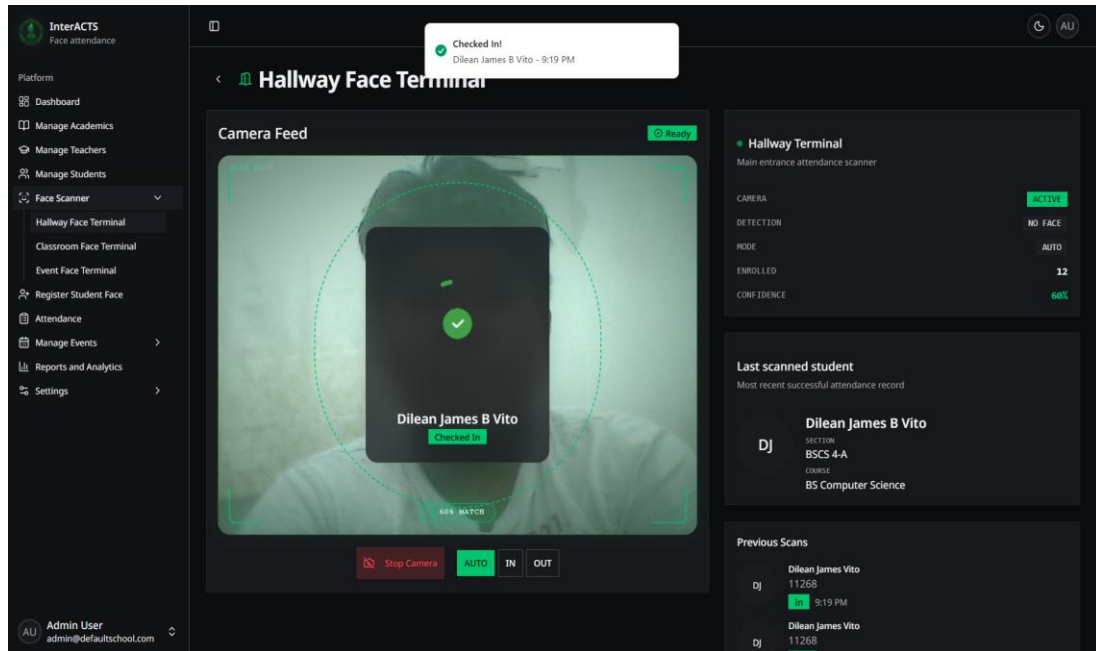


Figure P. Hallway Face Terminal of InterACTS

The Hallway Face Terminal of InterACTS serves as the main attendance scanning point for students after face training and registration. It allows students to perform check-in and check-out within the hallway using facial recognition. Each time a student checks out, the system automatically records the attendance activity and triggers a notification sent to the parents or guardians through email. This feature ensures real-time monitoring of student movement and provides timely updates to guardians regarding their attendance status.

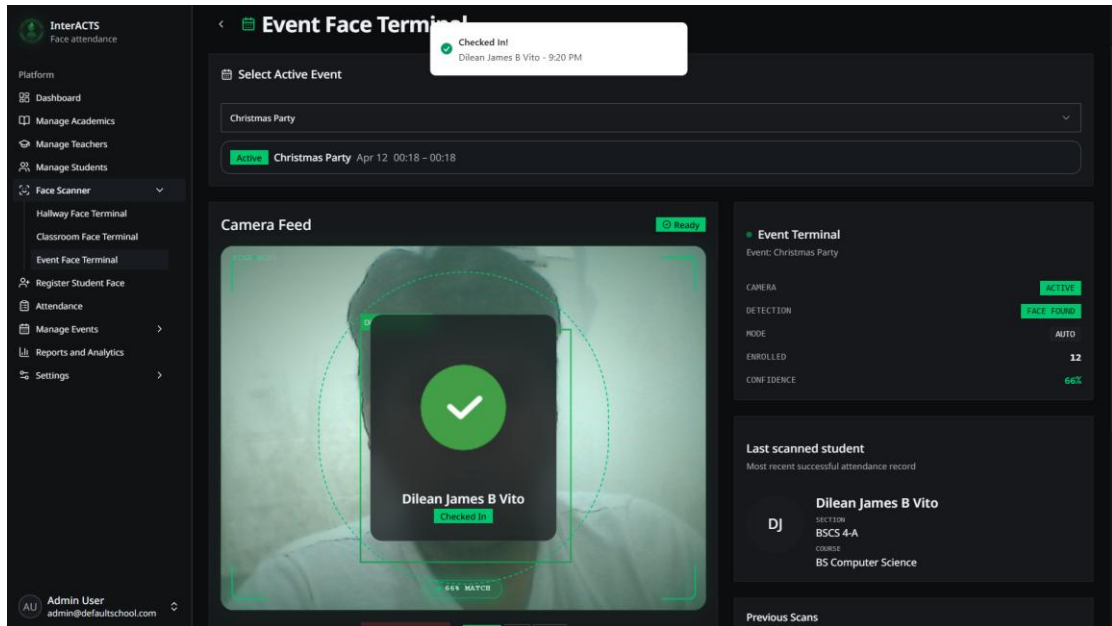


Figure Q. Events Face Terminal of InterACTS

The Events Face Terminal of InterACTS is designed for attendance processing during school events. In this mode, students are required to present or submit a valid receipt as proof of payment before they can be recognized by the system. If a student is not verified or marked as unpaid, their attendance will not be recorded, and they will not be recognized by the system during the event scanning process. This feature ensures that only authorized and validated participants are allowed to enter and have their attendance recorded in event-related activities.

Attendance Log

Attendance Records
119 of 119 records

Search by name... All Types All Devices 10/04/2026 11/04/2026 Clear

Student	Type	Source Type	Date	Time	Confidence
JC Jeric Rodelas	Check Out	WEB_PORTAL	Apr 11, 2026	11:56:16 AM	56%
JC Jeric Rodelas	Check In	WEB_PORTAL	Apr 11, 2026	11:56:03 AM	60%
Dilean James Vito	Check In	WEB_PORTAL	Apr 11, 2026	11:44:49 AM	54%
Dilean James Vito	Check Out	WEB_PORTAL	Apr 11, 2026	11:44:34 AM	55%
John Mark Manangkil	Check In	WEB_PORTAL	Apr 11, 2026	11:44:13 AM	55%
John Denver Esperanzate	Check In	WEB_PORTAL	Apr 11, 2026	11:44:02 AM	54%
JC Jeric Rodelas	Check In	WEB_PORTAL	Apr 11, 2026	11:43:51 AM	60%
Maricon Dela Cruz	Check In	WEB_PORTAL	Apr 11, 2026	11:43:42 AM	56%
John Mark Manangkil	Check Out	WEB_PORTAL	Apr 11, 2026	11:43:07 AM	66%
Maricon Dela Cruz	Check Out	WEB_PORTAL	Apr 11, 2026	11:42:58 AM	56%

Figure R. Attendance Log of InterACTS

The Attendance Log of InterACTS records and displays all student check-in and check-out activities for both hallway monitoring and event attendance. It includes important details such as the date and time of each transaction, allowing for accurate and precise tracking of student movements and participation. The log serves as a centralized record for monitoring attendance history, ensuring transparency and accountability within the system. In addition, the module supports data export in CSV format, which allows for easy data processing, reporting, and backup of attendance records for administrative use.

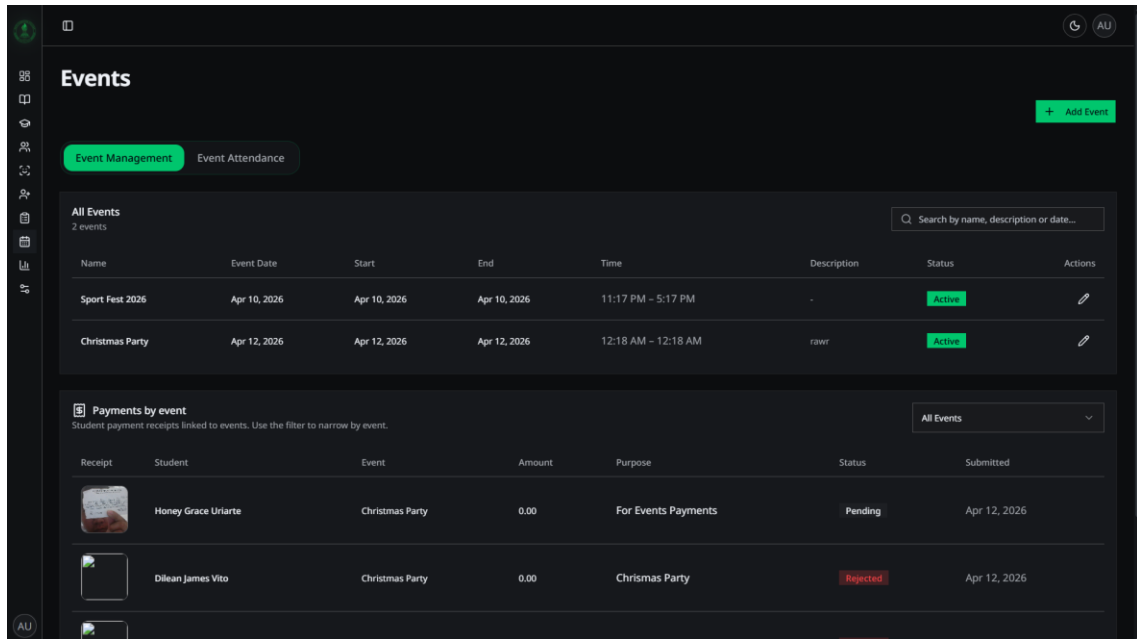


Figure S. Manage Events of InterACTS

The Manage Events of InterACTS allows the administrator to create and manage school events within the system. It enables the setup of event details such as date and time, payment schedules, payment deadlines, and the start and end time of each event. This module also supports monitoring of student submissions, including uploaded receipt images used as proof of payment. Through this feature, administrators can efficiently organize events, validate payments, and ensure proper management of event participation.

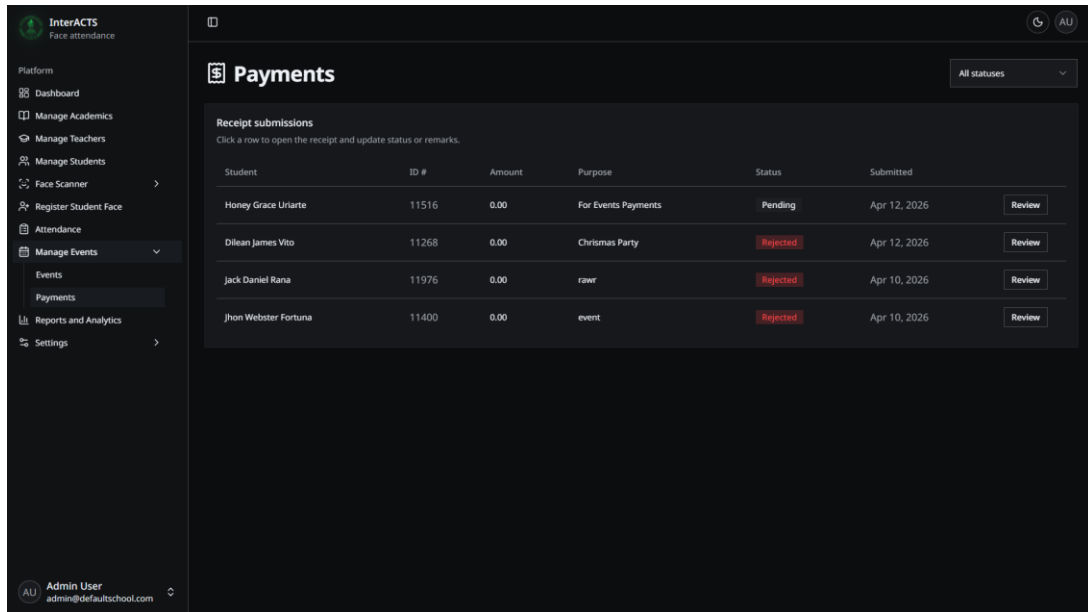


Figure T. Manage Events Payment of InterACTS

The Manage Events Payment of InterACTS handles the processing and validation of student payments for events. It serves as the main section where students submitted receipt images are received and reviewed. The module displays important payment details such as the amount paid and the corresponding event or purpose of the payment. The administrator is responsible for validating each submission and updating its status as pending, approved, or rejected based on the verification results. This ensures that only verified payments are accepted for event participation and provides proper control over financial-related transactions within the system.

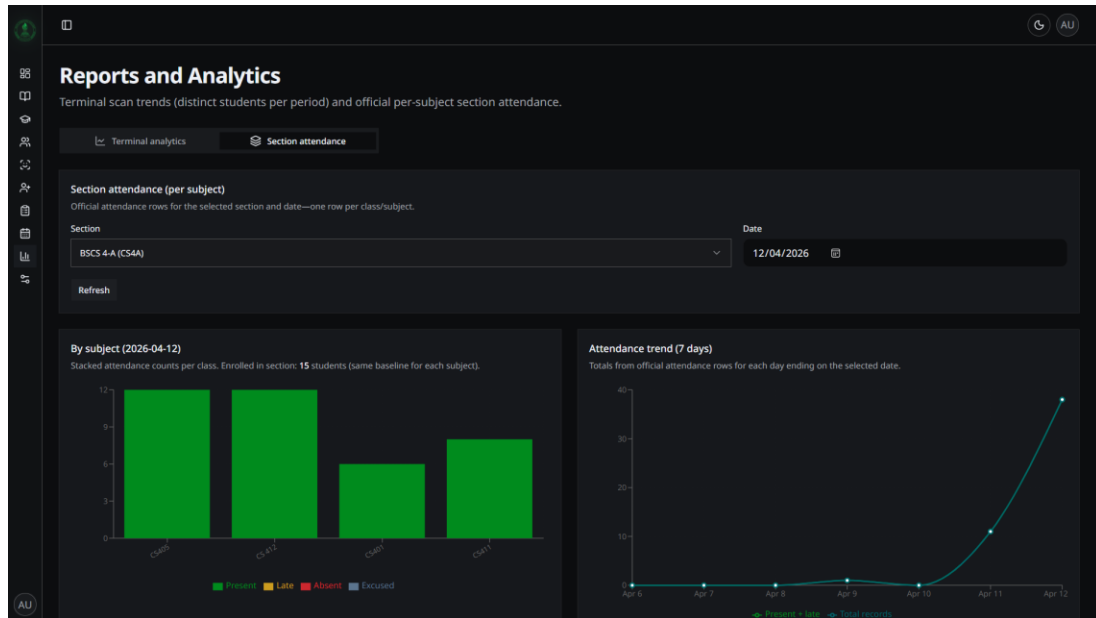
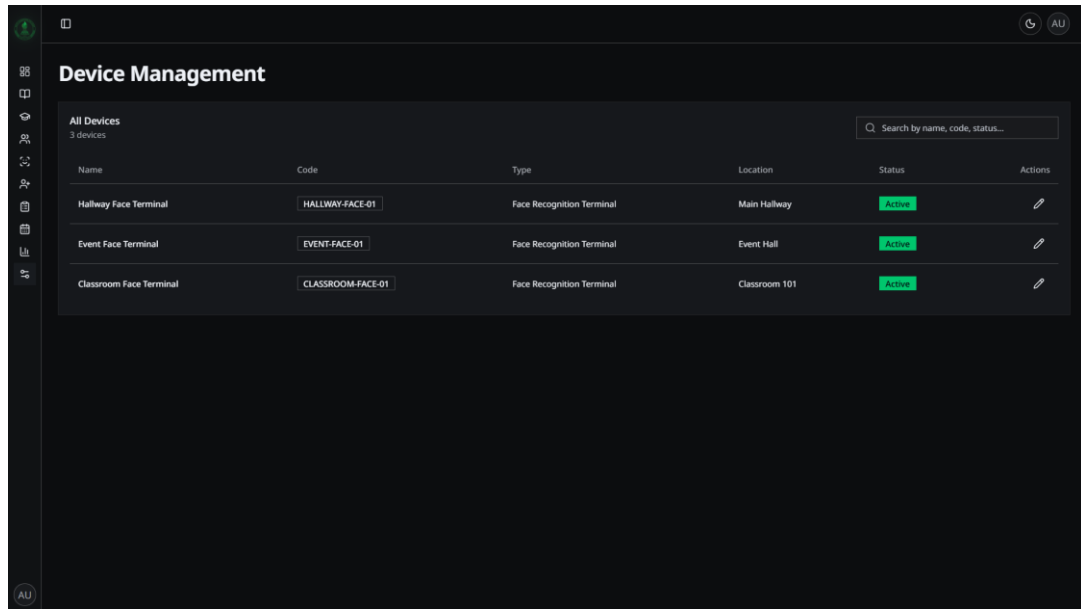


Figure U. Reports and Analytics of InterACTS

The Reports and Analytics of InterACTS provides a comprehensive overview of system-generated data that supports monitoring and evaluation of student activities within the platform. It includes detailed information such as the total number of registered and trained students, peak distinct student records, and the overall number of enrolled students. It also tracks daily system usage, including how many students have scanned, checked in, and checked out within a specific day, allowing administrators to monitor real-time participation and attendance trends. Furthermore, the module offers separate analytics for hallway, classroom, and event-based attendance, enabling data segmentation for better analysis and comparison. These analytics can be viewed in different time ranges such as daily, weekly, or monthly, depending on reporting needs. It also provides attendance summaries per section, showing the number of students who attended in each group for better academic monitoring. In addition, the module supports

CSV export functionality, allowing administrators to download and utilize the data for reporting, documentation, and further analysis.



Name	Code	Type	Location	Status	Actions
Hallway Face Terminal	HALLWAY-FACE-01	Face Recognition Terminal	Main Hallway	Active	
Event Face Terminal	EVENT-FACE-01	Face Recognition Terminal	Event Hall	Active	
Classroom Face Terminal	CLASSROOM-FACE-01	Face Recognition Terminal	Classroom 101	Active	

Figure V. Device Management of InterACTS

The Device Management module of InterACTS allows the administrator to monitor and control all connected system devices, particularly the cameras used for face recognition and scanning. It provides functionality to check whether each device is properly working and actively transmitting data. Through this module, devices can be set to active, inactive, offline, or placed under maintenance depending on their condition or the requirements of the school. This feature ensures that all hardware components are properly managed, maintained, and monitored to guarantee the continuous and reliable operation of the system.

Attendance policy
School-wide defaults for late / absent rules, how early students may scan before class start, and how long after class end a check-out scan is accepted. Teachers can override per class.

School default
Applies when a class has no per-assignment policy. Used by the classroom face terminal for eligibility and status.

On-time grace (minutes after start): 10

Early arrival allowance (minutes before start): 0

Late window (minutes after start): 30

Late check-out grace (minutes after class end): 20

Absent after late window
If off, scans stay "late" instead of absent after the late window. ☒

Save school policy

Figure W: Attendance Policy (Admin's View) of InterACTS

The Attendance Policy module of InterACTS defines the rules and guidelines for student attendance recording within the system. It includes configurations such as on-time marking, grace period for early arrival, and late arrival window with a defined limit such as a 30-minute late check-in allowance. It also defines check-out rules, including grace minutes after class end time before a student is considered officially checked out. This module ensures consistent and fair attendance computation by standardizing time-based rules for all students across different classes and activities.

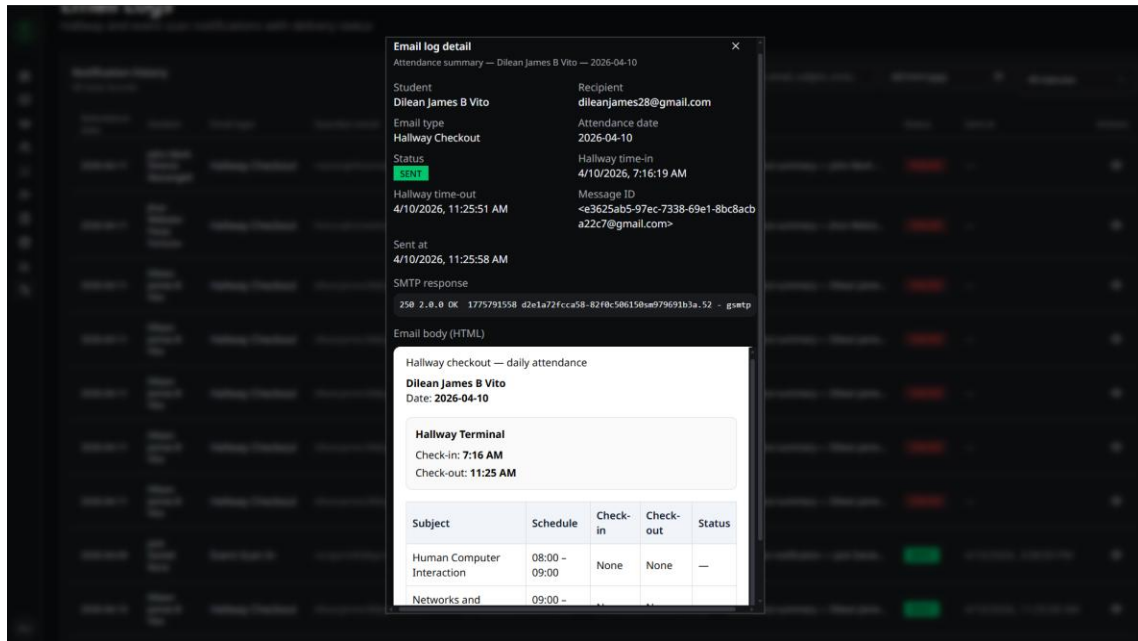


Figure X: Email Logs of InterACTS

The Email Logs module of InterACTS records all system-generated email notifications sent to parents or guardians through Gmail. It includes notifications related to student check-in and check-out activities, allowing parents to be informed about their child’s presence and movement within the school. This module also provides an overview of the student’s attendance schedule and records of their participation in school activities. Through this feature, the system ensures transparent communication between the school and parents regarding student attendance and school presence.

Name	Username	Role	Email	Status	Actions
Jhon Webster Perez Fortuna	11400	Student	fortunajhonwebster@gmail.com	Active	
Jack Daniel Rana	11976	Student	ranajack369@gmail.com	Active	
Dehn Mark Oliveros	10752	Student	oliverosdehnmark@gmail.com	Active	
John Denver Abanilo Esperanzate	11364	Student	denver@gmail.com	Active	
Maricon Sagal Dela Cruz	11164	Student	maricon@gmail.com	Active	
Patricia Alaisa	11072	Student	patricia@gmail.com	Active	
Isra-Enos Suazo	12345	Student	enos@gmail.com	Active	
Adrian Aquino	11016	Student	adrian@gmail.com	Active	
Jerome Aguilar	Jerome	Teacher	jerome@gmail.com	Active	

Figure Y. Users of InterACTS

The Users module of InterACTS provides a centralized list of all registered users within the system. It displays different user accounts, including their basic information and roles, allowing the administrator to easily manage and monitor system access. Through this module, users can be organized and reviewed to ensure proper account management and security within the system. It serves as the main reference for all individuals who have access to the InterACTS platform.

Assigned Class
View your handled classes and students from regular enrollment plus irregular overrides.

Assigned Class
Select a class tab to view merged student rosters for your teaching context.

BSCS 4-A BSIT 3-A

BSIT 3-A (IT3A) 15 students

Student	Student ID	Status
AC Adam Christopher Peras	12077	Enrolled
AJ Aines Justine Vito	12150	Enrolled
CD Chester Dominic Medina	10237	Enrolled
DR Darwin Ramor Jr	12155	Enrolled
DM Dehn Mark Oliveros	10752	Enrolled
FJ Filbert Javilinar	10453	Enrolled
HG Honey Grace Uriarte	11516	Enrolled
JD Jack Daniel Rana	11976	Enrolled
JB John Benedict San Jose	11975	Enrolled
JR John Rey Bahan	14025	Enrolled

Figure Z. Assigned Class of Students (Teacher’s View) of InterACTS

This figure illustrates the Assigned Class of Students (Teacher’s View) in InterACTS, where teachers can view the classes assigned to them per course. It displays the list of students under each assigned class, allowing teachers to easily identify and manage their respective groups. The interface organizes student information based on course and section, making it more convenient for teachers to monitor class composition and student distribution. It also supports efficient class handling by providing a clear and structured view of assigned teaching responsibilities within the system.

The screenshot displays the 'Attendance' module in InterACTS. It includes a sidebar with navigation icons, a top header with the title 'Attendance', and a main content area. The main area features a 'Class attendance' section with filters for 'From' (12/04/2026), 'To' (12/04/2026), and 'Status filter' (All statuses). A summary bar shows 'Total records 8', 'Present 8', 'Late 0', 'Absent 0', and 'Excused 0'. Below this, there are tabs for different subjects: 'BSCS ... Human Computer Inter...', 'BSCS ... Networks and Communi...', and 'BSIT ... Human Computer Inter...'. A search bar is present with the text 'Search by name or student ID...'. The main table lists student attendance records with columns for Student, Student ID, Status, Time in, Time out, First scan, Last scan, and Logs. The table contains 8 rows of data, with students marked as 'PRESENT' or 'ABSENT'.

Student	Student ID	Status	Time in	Time out	First scan	Last scan	Logs
AA Adrian Aquino	11016	PRESENT	9:00 AM	10:30 AM	9:00 AM	10:30 AM	Mark Absent IE Logs
AC Anjanette Castor	09720	PRESENT	9:00 AM	10:30 AM	9:00 AM	10:30 AM	Mark Absent IE Logs
DJ Dilean James B Vito	11268	ABSENT	—	—	—	—	Mark Present IE Logs
IS Isra-Enos Suazo	12345	PRESENT	9:00 AM	10:30 AM	9:00 AM	10:30 AM	Mark Absent IE Logs
JJ JC Jeric Marginio Rodelas	09840	PRESENT	9:00 AM	10:30 AM	9:00 AM	10:30 AM	Mark Absent IE Logs
JW Jhon Webster Perez Fortuna	11400	PRESENT	9:00 AM	10:30 AM	9:00 AM	10:30 AM	Mark Absent IE Logs
JD John Denver Abanion Esperanzate	11364	ABSENT	—	—	—	—	Mark Present IE Logs
JM John Mark Silverio Manangkil	11225	PRESENT	9:00 AM	10:30 AM	9:00 AM	10:30 AM	Mark Absent IE Logs

Figure AA. Attendance of Students (Teacher's View) of InterACTS

This figure illustrates the Attendance of Students (Teacher's View) in InterACTS, where teachers can manage and monitor student attendance for each subject. It allows teachers to view attendance records generated through face recognition, including the specific time when students were recognized by the system. In addition, the module supports manual attendance input, which is useful in cases such as online classes where face recognition may not be applicable. Attendance data is organized per subject, enabling teachers to efficiently track and manage student participation across their assigned classes.

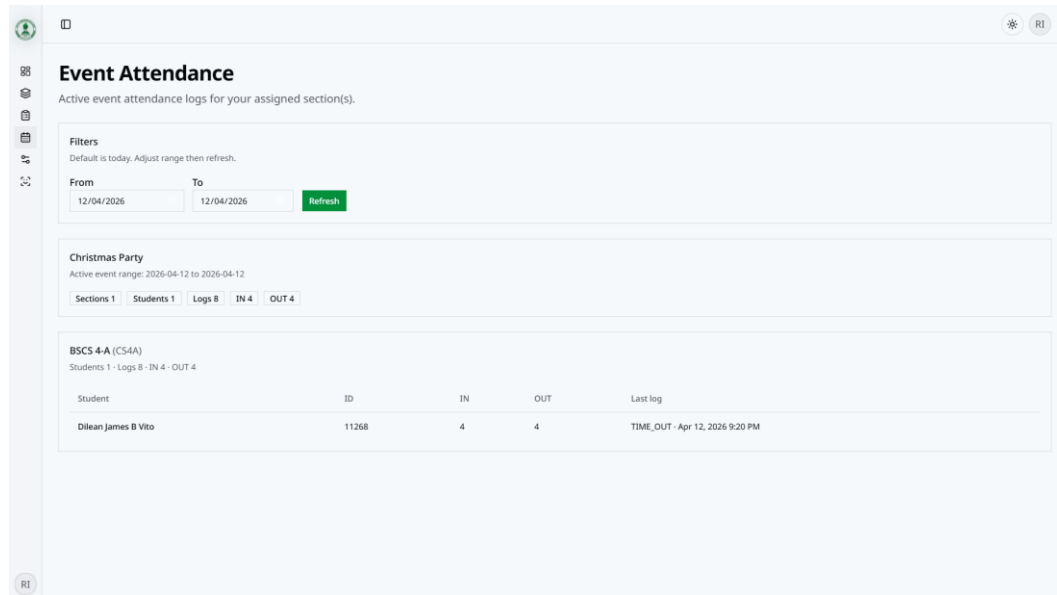


Figure AB. Event Attendance (Teacher's View) in InterACTS

This figure illustrates the Event Attendance (Teacher's View) in InterACTS, where teachers can monitor the attendance of students in school events. It displays the list of students who attended events under the courses or classes assigned to the teacher. The interface allows teachers to easily track participation and verify which students have successfully recorded their attendance during events. This feature supports efficient monitoring of event participation and ensures that attendance records are properly organized per course and class.

Class attendance policies

Set rules for your assigned sections and subjects. Scans outside the allowed window are blocked at the classroom terminal.

Per-class overrides

Values below apply to this class only. School defaults are shown for reference.

School defaults: grace 10m · late window 30m · early allowance 0m · late check-out 20m

3SCS 4-A · Human Computer Interactor · SCS 4-A · Networks and Communication · SIT 3-A · Human Computer Interaction

Custom override

On-time grace (minutes): 10

Late window (minutes): 20

Early arrival allowance (minutes): 10

Late check-out grace (minutes after class end): 20

Allows face check-out for this many minutes after the scheduled end; also used when the next class starts immediately after.

Absent after late window: ☒

Save for this class

Figure AC. Class Attendance Policies of InterACTS

This figure illustrates the Class Attendance Policies of InterACTS, where attendance rules are configured per class or course. It allows the setting of parameters such as early arrival allowance, on-time grace period, and late arrival limits, including the maximum time window for students to scan using face recognition. These policies help standardize attendance recording based on specific class requirements. The interface provides teachers with clear visibility and control over attendance rules, ensuring accurate and consistent monitoring of student attendance per course.

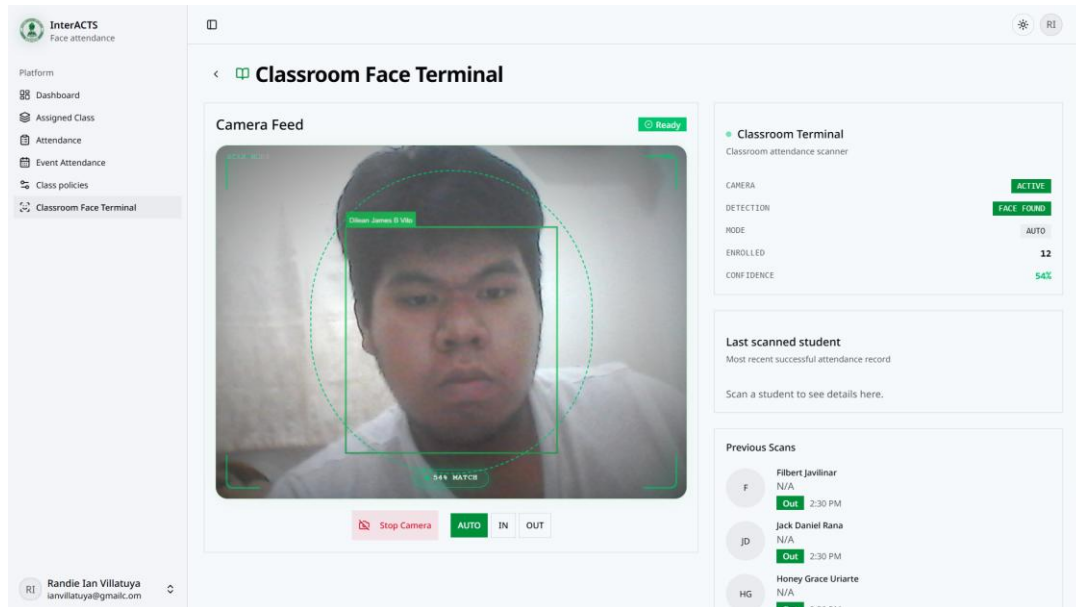


Figure AD: Classroom Face Terminal of InterACTS

This figure illustrates the Classroom Face Terminal of InterACTS, which is one of the cores and most complex functionalities of the system under its multi-mode operation. It is designed to handle subject-based face recognition within the classroom setting. In this module, the teacher initiates the scanning process, allowing students to check in at the start of the class and check out at the end of the session using facial recognition. Attendance is recorded per subject, ensuring accurate tracking of student participation in each class. The system requires a high level of accuracy to prevent errors such as overlapping records or incorrect attendance logging, which could affect the reliability of the overall system. This ensures that attendance data remains precise, consistent, and aligned with scheduled class sessions.

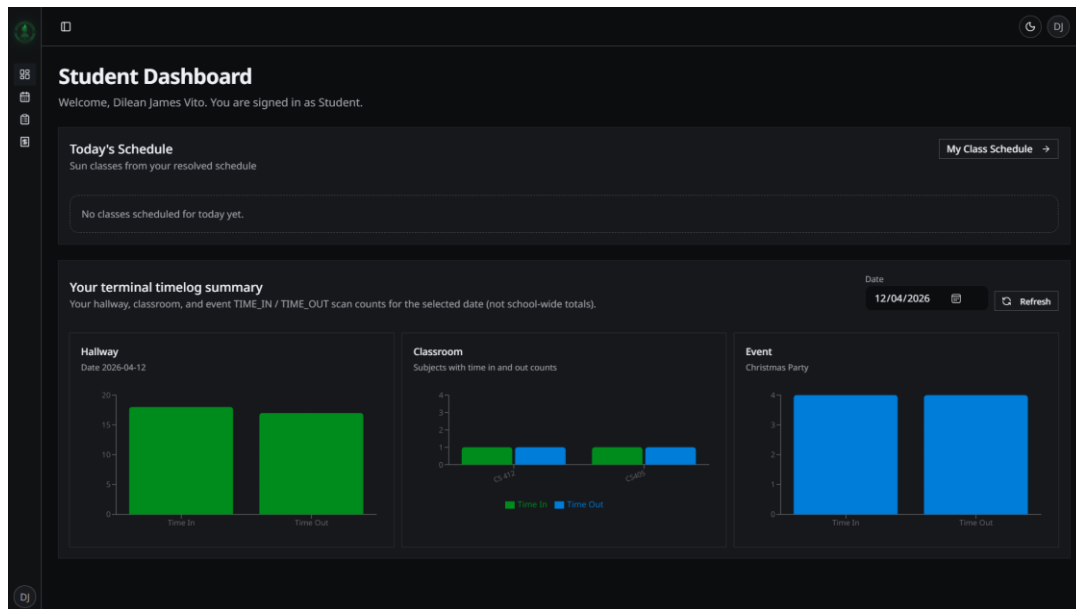


Figure AE. Student's Dashboard of InterACTS

This figure illustrates the Student's Dashboard of InterACTS, which provides an overview of the student's daily academic activities and attendance-related data. It displays the student's scheduled classes for the day, allowing them to stay informed about their subjects and timetable. The dashboard also presents analytics related to classroom, hallway, and event attendance, particularly based on the student's face recognition records. Through this interface, students can monitor their own attendance history and participation across different system modules, ensuring awareness and transparency of their recorded activities.

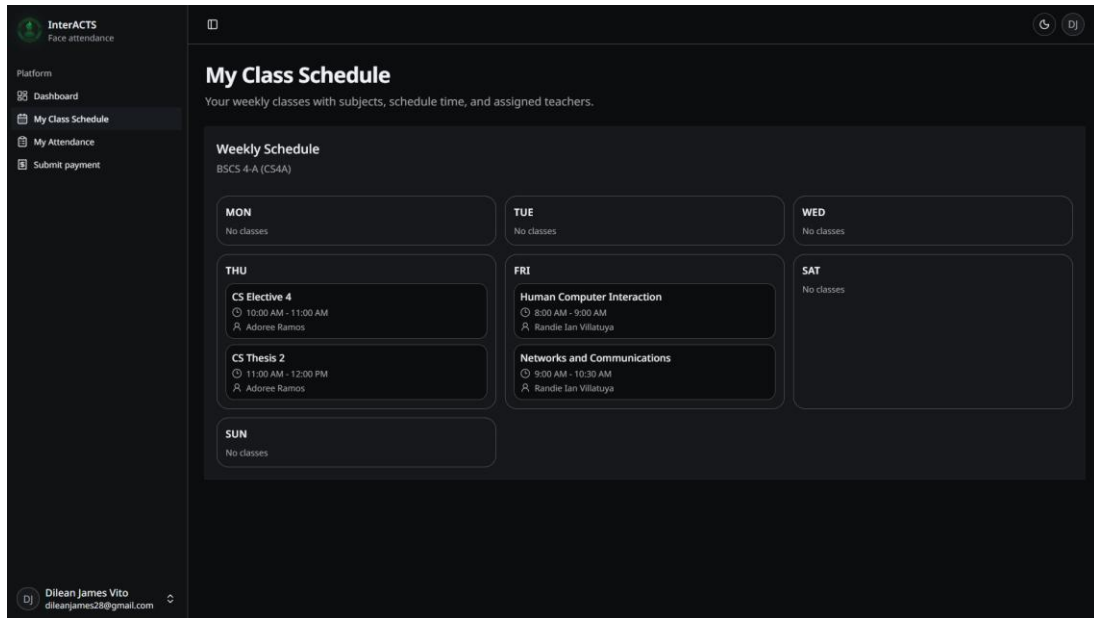


Figure AF. Class Schedule (Student's View) of InterACTS

This figure illustrates the Class Schedule (Student's View) of InterACTS, where students can view their complete class schedule for the entire week. It displays all assigned subjects along with their corresponding days and time slots, allowing students to clearly track their academic timetable. The interface provides an organized and structured view of scheduled classes, helping students manage their time effectively and stay informed about their daily and weekly academic responsibilities.

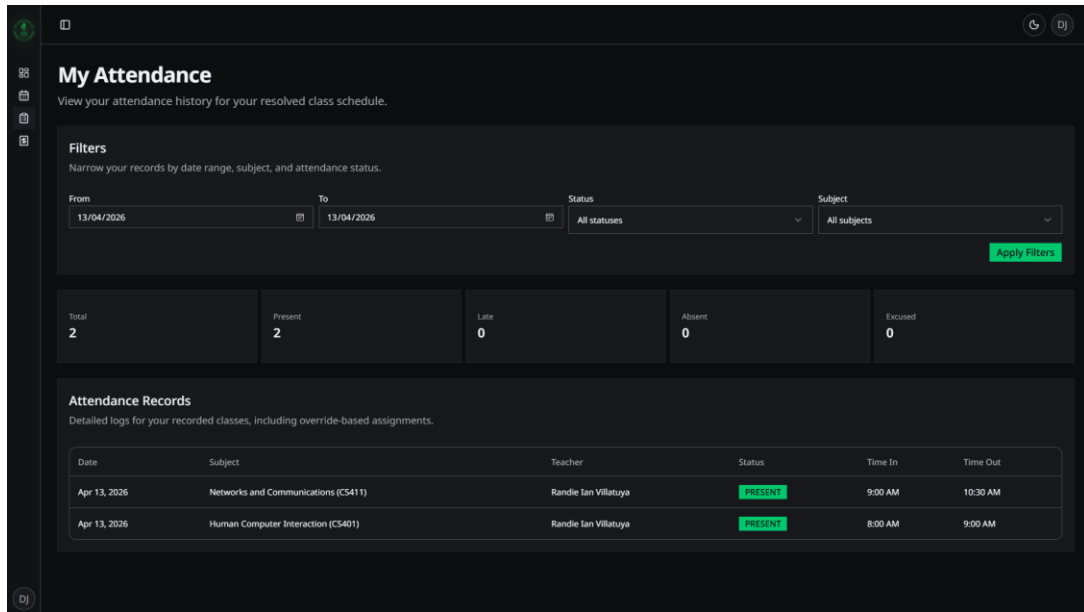


Figure AG. Student Attendance Record (Student’s View) of InterACTS

This figure illustrates the Student Attendance Record (Student’s View) of InterACTS, where students can view their attendance status per day, including whether they are marked as present, late, excused, or absent. The interface allows students to monitor their attendance history across different subjects they are enrolled in. In addition, it provides filtering options so students can view attendance records per subject or view all subjects combined, making it easier to track and review their overall academic attendance performance.

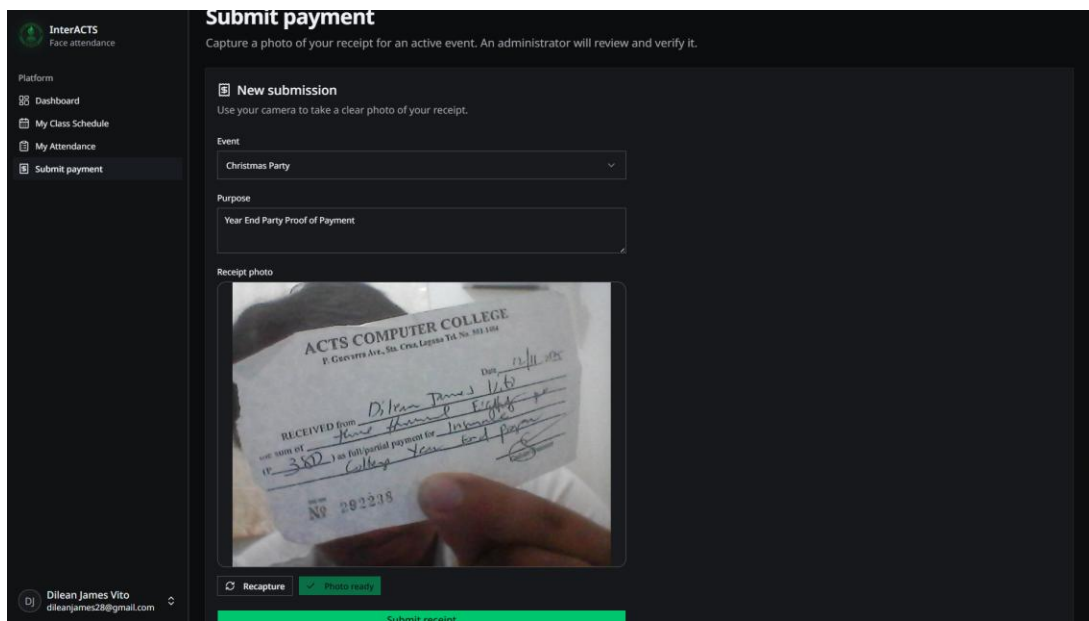


Figure AH. Submit Events Payment (Student’s View) of InterACTS

This figure illustrates the Submit Events Payment (Student’s View) of InterACTS, where students can submit their event payment proof by uploading a picture of their receipt. The interface allows students to select the specific event they are paying for and submit the corresponding payment details for validation. It also provides real-time status updates, showing whether the submitted receipt has been approved, rejected, or is still pending review by the administrator. This feature ensures proper tracking and verification of event payments within the system.

System Implementation

The section describes the development, testing, operation, support, and security of the system.

Table 5

Implementation Plan Table for InterACTS

STRATEGY	ACTIVITIES	PERSONAL INVOLVE	DURATION
Provide a letter for the user of the InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management.	System user for Implementation	Developers, Admin, students, parents and the IT Experts.	One (1) day
Coordination and assessment with the Admin, Students, Parents and IT experts for the Development of InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management.	System Implementation	Developers, Admin, students, parents and the IT Experts	Five (5) days

Table 5 outlines the implementation plan for InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational

Management. It presents the strategies, activities, personnel involved, and their corresponding duration. The plan covers important steps such as securing user approval for system implementation, setting up the required software, hardware, and databases, and coordinating closely with developers, admin, students, teachers, parents, and IT experts. In addition, it includes system configuration, face data enrollment, testing and validation of the face recognition accuracy, user training, and final system deployment. These activities ensure that the system is properly implemented, functional, and aligned with the institution's operational and security requirements.

System Development

Software Specification

The development of InterACTS required a Microsoft Windows 10 Enterprise 64-bit Operating System, XAMMP and Visual Studio Code.

Hardware Specification

In term of Hardware specification, the development used Ryzen 5 7000 with AMD Radeon (TM) Graphics 400MHz, 16GB memory (RAM) and 512GB storage (ROM).

Program Specification

The researchers developed InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management, using Artificial Intelligence (AI) technology to improve and simplify student communication and engagement processes. This system automates attendance tracking and monitors

student movement within the campus through face recognition, reducing manual work and errors.

Programming Environment

The researchers employed Visual Studio Code to develop both frontend and backend components of the web-based system.

Frontend

The frontend development used the React Framework, along with HTML, CSS, JavaScript to build the user interface. React is an open-source JavaScript library capable of handling the views layer for web and mobile applications. Additionally, Ajax was employed to facilitate asynchronous communication with the backend server, enabling dynamic updates to specific parts of the user interface without requiring a full page reload.

Backend

The backend development of the InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management, utilized a stack comprising Python, Database, and MongoDB currently. This combination ensures efficient data handling, secure storage, scalability, and high system performance, making it suitable for attendance and monitoring systems in educational institutions.

System Testing

After a comprehensive testing phase to ensure optimal performance, InterACTS is segmented into Ten modules: Authentication & Security, User Management, Train Data Face Module, Real Time Face Recognition, Monitoring Module, Attendance Logging,

Analytics and Reports, Verification, Notification, and Central Interface. The InterACTS system integrates multiple modules to provide secure, accurate, and efficient attendance and campus monitoring. It manages user access, accounts, and roles, processes and recognizes facial data in real time, tracks student movement, and automatically logs attendance. In addition, the system includes an email notification feature through Gmail, which automatically sends attendance updates and important alerts to users, ensuring timely communication between the system and stakeholders.

CHAPTER 4

RESULTS AND DISCUSSION

In this chapter, the evaluation of InterACTS is presented from various perspectives, focusing on its functionality, reliability, usability, efficiency, maintainability, and portability in accordance with the ISO 9126 quality standard. The system's performance was assessed based on feedback gathered from Admin, Students, Parents, Teachers, and IT experts, ensuring a comprehensive evaluation of its overall capability and compliance with internationally recognized software quality standards.

The results indicate that InterACTS obtained high ratings across all evaluated criteria. In terms of functionality, the system is rated highly in accuracy, compliance, security, and reliability, showing that it effectively meets user requirements and institutional policies. Reliability results further demonstrate strong fault tolerance and recoverability, indicating that the system can maintain stable operations even under different or unexpected conditions.

Usability is also rated positively, indicating that the system is easy to understand, learn, and operate, allowing smooth user interaction. Efficiency results show good resource utilization, although some respondents raised concerns regarding time behavior, which may need minor optimization. For maintainability, the system is generally rated positively in terms of stability, changeability, and testability, while analyzability received lower feedback, suggesting an area for improvement. Lastly, portability results show that the system is easy to install and adaptable to different environments while complying with institutional standards.

The evaluation using the Likert scale reveals that InterACTS is acceptable and performs well across all ISO 9126 quality attributes, indicating that the system is reliable, efficient, and suitable for implementation in an educational environment.

Table 6

Likert's Scale

Point of Scale	Range of Weighted Mean	Adjectival Rating (AR)
5	4.21 – 5.00	Highly Acceptable (HA)
4	3.41 – 4.20	Very Acceptable (VA)
3	2.61 – 3.40	Acceptable (A)
2	1.81 – 2.60	Disagree (D)
1	1.00 – 1.80	Strongly Disagree (SD)

Table 6 presents the Likert Scale, associating points on the scale with ranges of weighted mean scores and their corresponding adjectival ratings. A weighted mean ranging from 4.21 to 5.00 corresponds to Highly Acceptable (HA), which indicates that the respondents perceived the system as highly effective and very satisfactory in meeting their needs. A range of 3.41 to 4.20 is interpreted as Very Acceptable (VA), signifying that the system performs well and meets most of the users' expectations. A weighted mean of 2.61 to 3.40 is described as Acceptable (A), suggesting that the system is generally satisfactory but may still require improvements in certain areas. A range of 1.81 to 2.60 corresponds to Disagree (D), indicating that the respondents are not satisfied with the system's performance. And the weighted mean between 1.00 and 1.80 is interpreted as Strongly Disagree (SD), which reflects strong dissatisfaction and poor acceptance of the system.

Table 7*Level of Acceptability of InterACTS as Perceived by the Admin and IT Experts*

Criteria	Sub-Characteristic	IT Experts		Admin	
		Rate	AR	Rate	AR
Functionality	Sustainability – The system's features effectively support attendance tracking across hallway, monitoring, and event modes.	4.75	HA	5.00	HA
	Accuracy – The system accurately identifies individuals and records attendance using face recognition	4.50	HA	5.00	HA
	Compliance – The system ensures that attendance data is properly recorded and managed in an organized and systematic manner.	4.40	HA	5.00	HA
	Security – The system is secure by allowing access to its administration, whereas the function of confidentiality of the data.	4.50	HA	5.00	HA
Average Weighted Mean		4.54	HA	5.00	HA
Reliability	Fault Tolerance – The system continues to operate and record attendance even when minor errors occur during scanning.	4.50	HA	5.00	HA
	Recoverability – The system can quickly restore its operation and data after system interruptions or failures.	4.50	HA	5.00	HA
Average Weighted Mean		4.50	HA	5.00	HA
Usability	Understandability – The system interface is clear and easy to understand across all modes (hallway, monitoring, events).	4.75	HA	4.50	HA
	Learnability – Users can easily learn how to use the system with minimal guidance.	5.00	HA	5.00	HA
	Operability – Users and administrators can efficiently operate the system for attendance tracking and monitoring	5.00	HA	5.00	HA
Average Weighted Mean		4.92	HA	4.83	HA
Efficiency	Time Behavior – The system quickly processes face recognition and attendance recording with minimal delay.	4.50	HA	4.50	HA
	Resource Behavior – The system efficiently utilizes hardware resources such as CPU, memory, and network during operation.	4.50	HA	5.00	HA
Average Weighted Mean		4.50	HA	4.75	HA
Maintainability	Analyzability – The system allows easy detection and identification of errors during operation.	5.00	HA	4.50	HA
	Changeability – The system can be modified or updated easily to improve features or performance.	4.50	HA	5.00	HA
	Stability – The system maintains consistent performance even after updates or modifications.	4.75	HA	4.50	HA
	Testability – The system can be easily tested to ensure that updates and changes function correctly	5.00	HA	5.00	HA
Average Weighted Mean		4.81	HA	4.75	HA
Portability	Installability – The system can be easily installed and set up in different monitoring locations (e.g., hallways, entrances, event areas).	5.00	HA	4.50	HA
	Conformance – The system can be deployed and used in various environments within the school without major adjustments.	5.00	HA	4.50	HA
Average Weighted Mean		5.00	HA	4.50	HA
Overall Weighted Mean		4.71	HA	4.81	HA

Table 7 presents the level of acceptability of the InterACTS system as evaluated by IT experts and administrators based on ISO 9126 software quality standards. The results show that both groups give consistently high ratings across all criteria.

For Functionality, IT experts give an Average Weighted Mean of 4.54, while administrators give 5.00, both interpreted as Highly Acceptable. This indicates that the system effectively supports attendance tracking, accurately identifies individuals through face recognition, maintains organized attendance records, and ensures data security.

For Reliability, IT experts give an Average Weighted Mean of 4.50 and administrators give 5.00, both interpreted as Highly Acceptable. The findings indicate that the system continues to operate effectively even when minor errors occur and can quickly recover from interruptions or failures.

Regarding Usability, IT experts give an Average Weighted Mean of 4.92, while administrators give 4.83, both interpreted as Highly Acceptable. This suggests that the system interface is understandable, easy to learn, and efficient to operate for attendance monitoring and management.

For Efficiency, IT experts give an Average Weighted Mean of 4.50, while administrators give 4.75, both interpreted as Highly Acceptable. This indicates that the system processes attendance data quickly and utilizes hardware resources efficiently during operation.

In terms of Maintainability, IT experts give an Average Weighted Mean of 4.81, while administrators give 4.75, both interpreted as Highly Acceptable. These results

show that the system is easy to analyze, modify, maintain, and test without affecting its stability and performance.

Lastly, for Portability, IT experts give an Average Weighted Mean of 5.00, while administrators give 4.50, both interpreted as Highly Acceptable. This indicates that the system can be easily installed and deployed across different school environments with minimal adjustments.

Overall, the system obtains an Overall Weighted Mean of 4.71 from IT experts and 4.81 from administrators, both interpreted as Highly Acceptable. These findings confirm that the InterACTS system is effective, reliable, and suitable for academic implementation.

Table 8
Level of Acceptability of InterACTS as Perceived by the Faculty and Students

Criteria	Sub-Characteristic	Faculty	Students		
		Rate	AR	Rate	AR
Functionality	1. The InterACTS system provides features that effectively support attendance tracking in hallway, monitoring, and event modes.	5.00	HA	5.00	HA
	2. The system accurately records and displays attendance data using face recognition technology.	5.00	HA	4.79	HA
	3. The InterACTS system supports efficient monitoring of attendance during special events and activities.	5.00	HA	4.70	HA
	4. The system provides complete attendance monitoring through reports, history tracking, and real-time notifications.	5.00	HA	4.79	HA
Average Weighted Mean		5.00	HA	4.82	HA
Reliability	5. The system consistently produces reliable attendance records during continuous use.	5.00	HA	4.90	HA
	6. The system maintains stable performance even when handling multiple users at the same time.	4.67	HA	4.75	HA
Average Weighted Mean		4.84	HA	4.83	HA
Usability	7. The system allows users to quickly understand how to use each mode (hallway, monitoring, events).	5.00	HA	4.68	HA
	8. The layout and design of the system help users perform tasks without confusion.	4.67	HA	4.82	HA
	9. The system minimizes the need for assistance when being used by first-time users.	5.00	HA	4.78	HA
	10. The system reduces manual effort in recording attendance	5.00	HA	4.89	HA
	Average Weighted Mean	4.92	HA	4.79	HA
Efficiency	11. The system processes face recognition and attendance logging without noticeable delay.	5.00	HA	4.74	HA
	12. The system can handle large numbers of users efficiently during peak hours (e.g., events or class transitions).	4.67	HA	4.67	HA
Average Weighted Mean		4.84	HA	4.71	HA
Maintainability	13. The system allows administrators to easily manage and update user data	5.00	HA	4.94	HA
	14. The system supports smooth implementation of improvements and additional features.	4.67	HA	4.84	HA
	15. The system supports smooth implementation of improvements and additional features.	4.67	HA	4.75	HA
	16. The system allows easy monitoring and troubleshooting of system issues.	5.00	HA	4.72	HA
Average Weighted Mean		4.84	HA	4.81	HA
Portability	17. The system can operate effectively across different locations within the school (hallways, classrooms, event areas).	4.67	HA	4.87	HA
	18. The system can be accessed and operated in different areas of the school without requiring major setup adjustments.	5.00	HA	4.79	HA
Average Weighted Mean		4.84	HA	4.83	HA
Overall Weighted Mean		4.88	HA	4.80	HA

Table 8 presents the level of acceptability of the InterACTS system as perceived by faculty and student respondents based on ISO software quality standards. The results show that the system obtains consistently high ratings across all indicators.

In terms of Functionality, the system is rated 5.00 by faculty and 4.82 by students, indicating that it is highly effective in supporting attendance tracking and providing accurate and comprehensive monitoring features.

For Reliability, both groups give similarly high ratings, with 4.84 for faculty and 4.83 for students, reflecting the system's stable and dependable performance even under continuous use.

Regarding Usability, the system obtains 4.92 from faculty and 4.79 from students, suggesting that it is user-friendly and easy to operate.

In terms of Efficiency, ratings of 4.84 and 4.71 indicate that the system processes attendance data effectively with minimal delay.

For Maintainability, both groups give high scores of 4.84 and 4.81, showing that the system is easy to manage and update.

Lastly, for Portability, both faculty and students provide almost identical ratings of 4.84 and 4.83, indicating that the system can be deployed across different environments with ease.

Overall, the system obtains an Overall Weighted Mean of 4.88 from faculty and 4.80 from students, both interpreted as Highly Acceptable. These results confirm that the InterACTS system demonstrates strong performance across all ISO software quality standards, making it suitable for implementation in academic institutions

CHAPTER 5

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary of findings, conclusions, and recommendations of the study entitled InterACTS. It highlights the overall results based on the objectives of the research and evaluates the effectiveness of the developed system. The findings show that the interACTS system is able to achieve its intended purpose of improving interaction, monitoring, and management of records in a more efficient and organized manner. Based on the results of system testing and user evaluation, the system is found to be functional, user-friendly, and reliable. The features implemented in the system address the identified problems encountered in the existing process. The conclusions drawn from the study confirm that the proposed system provides a practical solution and contributes positively to the workflow of the users. Furthermore, this chapter also presents recommendations for future researchers and system developers. Suggested improvements include system enhancement, feature expansion, and integration of additional technologies to further improve performance and usability.

Summary

The InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management is an innovative platform designed to modernize attendance recording and student monitoring through facial recognition technology and web-based information management. The system integrates three major functional modes: Attendance Mode, Events Mode, and Monitoring Mode, providing a comprehensive solution for daily class operations and school activities.

The Attendance Mode enables automatic recording of student presence using facial recognition, eliminating manual logging and minimizing errors such as proxy attendance. The Events Mode allows the institution to manage and monitor attendance during school programs, seminars, and other activities outside regular classes. The Monitoring Mode provides real-time tracking of student records and generates reports that can be accessed by teachers and administrators through the web system.

Email notification via Gmail is also integrated into the system, which automatically informs parents or guardians about the attendance status of their children, strengthening communication between school and home. The system was developed using the Software Development Life Cycle (SDLC) and evaluated based on functionality, usability, reliability, and performance. Results showed that the system is highly acceptable and effective in improving attendance management and student monitoring.

Conclusion

1. The InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management is successfully designed and developed to modernize traditional attendance recording in educational institutions, providing a centralized platform for students, teachers, and administrators, and enabling efficient management, monitoring, and communication of attendance and school-related activities. The system effectively operates through three main functional modes Hallway Attendance Mode, Events Mode, and Monitoring Mode with each mode utilizing distinct logic and algorithms to ensure accurate facial recognition, real-time attendance tracking,

and efficient data processing. Email notification (via Gmail) services were also integrated to provide timely updates to parents regarding student attendance and school-related activities. All proposed features presented during the title proposal are successfully implemented and integrated into the system in accordance with the study objectives.

2. Based on the evaluation conducted using ISO software quality standards, the system achieves a high level of acceptability across all respondent groups, including students, teachers, IT experts, and administrators, with an overall weighted mean ranging from 4.71 to 4.88. These results indicate strong performance in terms of functionality, reliability, usability, performance, security, maintainability, and compatibility, showing that the system is reliable, user-friendly, and effective for school management operations.

3. The implementation plan and development timeline provided a structured and systematic framework for the successful deployment of the system, covering system setup, testing, training, and full integration. Overall, the system prove to be highly effective, reliable, and suitable for full adoption in educational institutions, delivering a technology-driven solution that improves accuracy, minimizes manual workload, and significantly enhances the efficiency of attendance and monitoring processes.

Recommendation

The following recommendations are provided to further enhance the functionality, reliability, and security of the InterACTS system and to support its continuous improvement and effective implementation in educational institutions.

1. Facial recognition training may be conducted during the enrollment stage under controlled and consistent lighting conditions. Facial images may be properly captured using multiple angles such as front, left, right, upward, and smiling positions to ensure a complete and high-quality dataset. This may improve the accuracy and effectiveness of the system during the learning or training phase.
2. Further improvement of the facial recognition module may help the actual system deployment, particularly in handling real-world conditions such as varying lighting and different facial angles during live attendance scanning. This ensures that the system remains accurate and reliable even when used in hallway, event, and monitoring environments.
3. Parents or guardians may encourage to create and maintain an active Gmail account to ensure the successful and timely delivery of email notifications regarding student attendance updates.
4. The system may require a stable and reliable internet connection to maintain its real-time functionality, especially for facial recognition, attendance logging, and notification services.
5. The system may be expanded through integration with additional devices or kiosks to enable smoother and more accessible attendance recording across different locations such as hallways, classrooms, and event areas.

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Appendix A

Team Member Role in the Development of the Thesis Project

Project Manager:	Dilean James B. Vito
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System Analyst:	John Denver A. Esperanzate Jhon Webster P. Fortuna Dilean James B Vito
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Programmer:	Jhon Webster P. Fortuna Dilean James B. Vito
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Database Designer:	John Mark S. Manangkil Jhon Webster P. Fortuna
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Q/A Tester:	JC Jeric M. Rodelas Jhon Webster P. Fortuna Dilean James B Vito
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Documentary:	Dilean James B. Vito Maricon S. Dela Cruz
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Appendix B
Implementation Plan

Table 3

Implementation Plan Table for InterACTS

STRATEGY	ACTIVITIES	PERSONAL INVOLVE	DURATION
Provide a letter for the user of the InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management.	System user for Implementation	Developers, Admin, students, parents and the IT Experts.	One (1) day
Coordination and assessment with the Admin, Students, Parents and IT experts for the Development of InterACTS: A Web-Based Multi-Mode Face Recognition Attendance and Monitoring System for Educational Management.	System Implementation	Developers, Admin, students, parents and the IT Experts	Five (5) days

Appendix C

Evaluation Form for Administrator

InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management.

Name: _____ Position: _____ Date: _____

Instructions: Please read each statement carefully and check the number that best represents your evaluation of the interACTS system using the provided scale.

Point of Scale	Range of Weighted Mean	Adjectival Rating
5	4.21 – 5.00	Highly Acceptable
4	3.41 – 4.20	Very Acceptable
3	2.61 – 3.40	Acceptable
2	1.81 – 2.60	Not Acceptable
1	1.00 – 1.80	Highly Not Acceptable

	Sub-Characteristics	Ratings				
		1	2	3	4	5
Functionality	Sustainability – The system’s features effectively support attendance tracking across hallway, monitoring, and event modes.					
	Accuracy – The system accurately identifies individuals and records attendance using face recognition.					
	Compliance – The system ensures that attendance data is properly recorded and managed in an organized and systematic manner.					
	Security – The system is secure by allowing access to its administration, whereas the function of confidentiality of the data.					
Reliability	Fault Tolerance – The system continues to operate and record attendance even when minor errors occur during scanning.					

	Recoverability – The system can quickly restore its operation and data after system interruptions or failures.					
Usability	Understandability – The system interface is clear and easy to understand across all modes (hallway, monitoring, events).					
	Learnability – Users can easily learn how to use the system with minimal guidance.					
	Operability – Users and administrators can efficiently operate the system for attendance tracking and monitoring.					
Efficiency	Time Behavior – The system quickly processes face recognition and attendance recording with minimal delay.					
	Resource Behavior – The system efficiently utilizes hardware resources such as CPU, memory, and network during operation.					
Maintainability	Analyzability – The system allows easy detection and identification of errors during operation.					
	Changeability – The system can be modified or updated easily to improve features or performance.					
	Stability – The system maintains consistent performance even after updates or modifications.					
	Testability – The system can be easily tested to ensure that updates and changes function correctly.					
Portability	Installability – The system can be easily installed and set up in different monitoring locations (e.g., hallways, entrances, event areas).					
	Conformance The system can be deployed and used in various environments within the school without major adjustments.					

Appendix D

Evaluation Form for IT Experts

InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management.

Name: _____ Position: _____ Date: _____

Instructions: Please read each statement carefully and check the number that best represents your evaluation of the interACTS system using the provided scale.

Point of Scale	Range of Weighted Mean	Adjectival Rating
5	4.21 – 5.00	Highly Acceptable
4	3.41 – 4.20	Very Acceptable
3	2.61 – 3.40	Acceptable
2	1.81 – 2.60	Not Acceptable
1	1.00 – 1.80	Highly Not Acceptable

Criteria	Sub-Characteristics	Ratings				
		1	2	3	4	5
Functionality	Sustainability – The system’s features effectively support attendance tracking across hallway, monitoring, and event modes.					
	Accuracy – The system accurately identifies individuals and records attendance using face recognition.					
	Compliance – The system ensures that attendance data is properly recorded and managed in an organized and systematic manner.					
	Security – The system is secure by allowing access to its administration, whereas the function of confidentiality of the data.					
Reliability	Fault Tolerance – The system continues to operate and record attendance even when minor errors occur during scanning.					

	Recoverability – The system can quickly restore its operation and data after system interruptions or failures.					
Usability	Understandability – The system interface is clear and easy to understand across all modes (hallway, monitoring, events).					
	Learnability – Users can easily learn how to use the system with minimal guidance.					
	Operability – Users and administrators can efficiently operate the system for attendance tracking and monitoring.					
Efficiency	Time Behavior – The system quickly processes face recognition and attendance recording with minimal delay.					
	Resource Behavior – The system efficiently utilizes hardware resources such as CPU, memory, and network during operation.					
Maintainability	Analyzability – The system allows easy detection and identification of errors during operation.					
	Changeability – The system can be modified or updated easily to improve features or performance.					
	Stability – The system maintains consistent performance even after updates or modifications.					
	Testability – The system can be easily tested to ensure that updates and changes function correctly.					
Portability	Installability – The system can be easily installed and set up in different monitoring locations (e.g., hallways, entrances, event areas).					
	Conformance The system can be deployed and used in various environments within the school without major adjustments.					

Appendix E

Questionnaire/Evaluation Form for Students

Evaluation Form for IT Experts InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management.

Name: _____ Position: _____ Date: _____

Instructions: Please read each statement carefully and check the number that best represents your evaluation of the interACTS system using the provided scale.

Point of Scale	Range of Weighted Mean	Adjectival Rating
5	4.21 – 5.00	Highly Acceptable
4	3.41 – 4.20	Very Acceptable
3	2.61 – 3.40	Acceptable
2	1.81 – 2.60	Not Acceptable
1	1.00 – 1.80	Highly Not Acceptable

Part I - Functionality		Ratings				
No.	Statement	5	4	3	2	1
1	The InterACTS system provides features that effectively support attendance tracking in hallway, monitoring, and event modes.					
2	The system accurately records and displays attendance data using face recognition technology.					
4	The InterACTS system supports efficient monitoring of attendance during special events and activities.					
5.	The system provides complete attendance monitoring through reports, history tracking, and real-time notifications.					
Part II - Reliability						
6	The system consistently produces reliable attendance records during continuous use.					

7	The system maintains stable performance even when handling multiple users at the same time.					
Part III - Usability						
8	The system allows users to quickly understand how to use each mode (hallway, monitoring, events).					
9	The layout and design of the system help users perform tasks without confusion.					
10	The system minimizes the need for assistance when being used by first-time users.					
11.	The system reduces manual effort in recording attendance.					
Part IV - Efficiency						
12	The system processes face recognition and attendance logging without noticeable delay.					
13	The system can handle large numbers of users efficiently during peak hours (e.g., events or class transitions).					
Part V - Maintainability						
14	The system allows administrators to easily manage and update user data.					
15	The system supports smooth implementation of improvements and additional features.					
16	The system maintains performance quality after updates or system changes.					
17	The system allows easy monitoring and troubleshooting of system issues.					
Part VI - Portability						
18	The system can operate effectively across different locations within the school (hallways, classrooms, event areas).					
19	The system can be accessed and operated in different areas of the school without requiring major setup adjustments.					

Appendix F

Questionnaire/Evaluation Form for Faculty

InterACTS: A Web-Based Multi-mode Face Recognition Attendance and Monitoring System for Educational Management.

Name: _____ Course: _____ Date: _____

Instructions: Please read each statement carefully and check the number that best represents your evaluation of the interACTS system using the provided scale.

Point of Scale	Range of Weighted Mean	Adjectival Rating
5	4.21 – 5.00	Highly Acceptable
4	3.41 – 4.20	Very Acceptable
3	2.61 – 3.40	Acceptable
2	1.81 – 2.60	Not Acceptable
1	1.00 – 1.80	Highly Not Acceptable

Part I - Functionality		Ratings				
No.	Statement	5	4	3	2	1
1	The InterACTS system provides features that effectively support attendance tracking in hallway, monitoring, and event modes.					
2	The system accurately records and displays attendance data using face recognition technology.					
4	The InterACTS system supports efficient monitoring of attendance during special events and activities.					
5.	The system provides complete attendance monitoring through reports, history tracking, and real-time notifications.					
Part II - Reliability						
6	The system consistently produces reliable attendance records during continuous use.					

7	The system maintains stable performance even when handling multiple users at the same time.					
Part III - Usability						
8	The system allows users to quickly understand how to use each mode (hallway, monitoring, events).					
9	The layout and design of the system help users perform tasks without confusion.					
10	The system minimizes the need for assistance when being used by first-time users.					
11.	The system reduces manual effort in recording attendance.					
Part IV - Efficiency						
12	The system processes face recognition and attendance logging without noticeable delay.					
13	The system can handle large numbers of users efficiently during peak hours (e.g., events or class transitions).					
Part V - Maintainability						
14	The system allows administrators to easily manage and update user data.					
15	The system supports smooth implementation of improvements and additional features.					
16	The system maintains performance quality after updates or system changes.					
17	The system allows easy monitoring and troubleshooting of system issues.					
Part VI - Portability						
18	The system can operate effectively across different locations within the school (hallways, classrooms, event areas).					
19	The system can be accessed and operated in different areas of the school without requiring major setup adjustments.					

Appendix G

Picture with Students during System Evaluation

The researchers tested the Face Recognition and Monitoring System to make sure it works well and reliably in different situations, such as when the lighting changes, when faces are seen from different angles, or when many people are detected at the same time.



Appendix H

Picture with IT Expert during System Evaluation

The researchers conducted an evaluation involving IT experts to assess the Face Recognition and Monitoring System. This evaluation aimed to determine the level of acceptability of the system in terms of performance, functionality, reliability, usability, efficiency, maintainability and portability to obtain expert feedback for system enhancement.



Appendix I

Picture with the Faculty during System Evaluation

The researchers carried out an evaluation with faculty members to examine the Face Recognition and Monitoring System. The purpose of this assessment was to gauge the system's acceptability regarding its functionality, usability, and effectiveness in handling attendance and monitoring tasks, while also collecting feedback to guide future improvements.



Appendix J

Picture with the Administrator during System Evaluation

The researchers performed a system assessment together with the administrator to evaluate the Face Recognition and Monitoring System. This assessment was conducted to measure the system's level of acceptability based on its functionality, usability, and efficiency in handling attendance and monitoring operations. It also provided a means to collect the administrator's insights and suggestions for the continued improvement and refinement of the system.



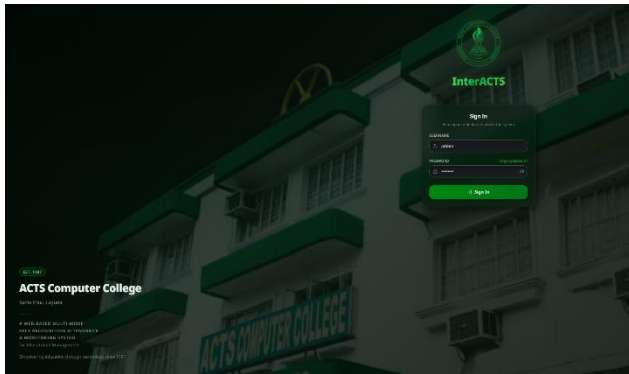
Appendix K

User's Guide

InterACTS Manual's Guide (Admin's Side)

How to Access InterACTS System?

1. The admin is instructed to connect their device to an internet connection
2. Open any browser, the developers suggest using google chrome as the default browser.
3. Access <https://interacts-neon.vercel.app/>

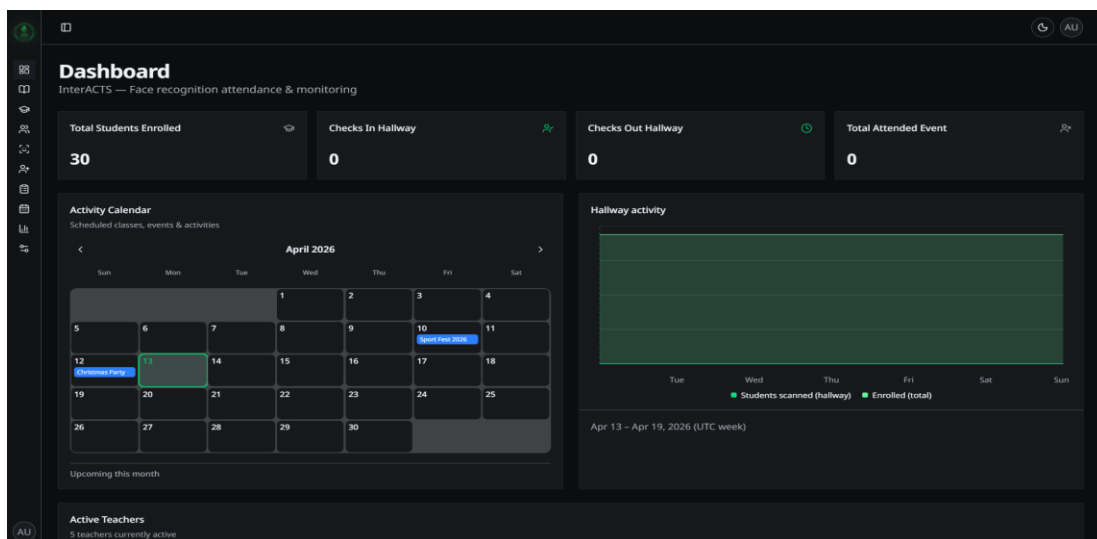


Signing into Account

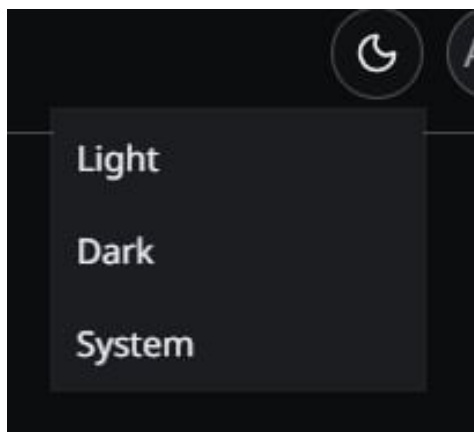
By clicking the Sign in button, please enter the provided username and password. It will be redirected to the Dashboard once logged in.

The picture above shows the login page for the system

Admin Dashboard



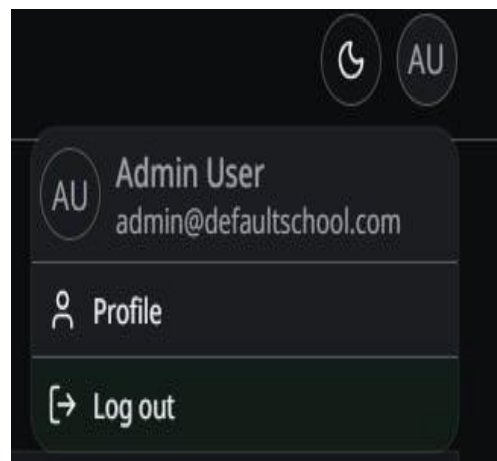
The **Admin Dashboard** serves as the main interface for monitoring system activities and data within InterACTS. Upon accessing this section, administrators can view key information such as the total number of enrolled students, hallway check-in and check-out records, and the number of events attended. The dashboard also provides a calendar for tracking scheduled activities and displays a list of active teachers currently registered in the system. By regularly using this feature, administrators can efficiently oversee student participation and monitor overall system usage in real time.



Display Mode

The **Display Mode** feature allows users to switch between light and dark themes in the system. It provides an option to change the interface appearance to either light mode or dark mode depending on user preference.

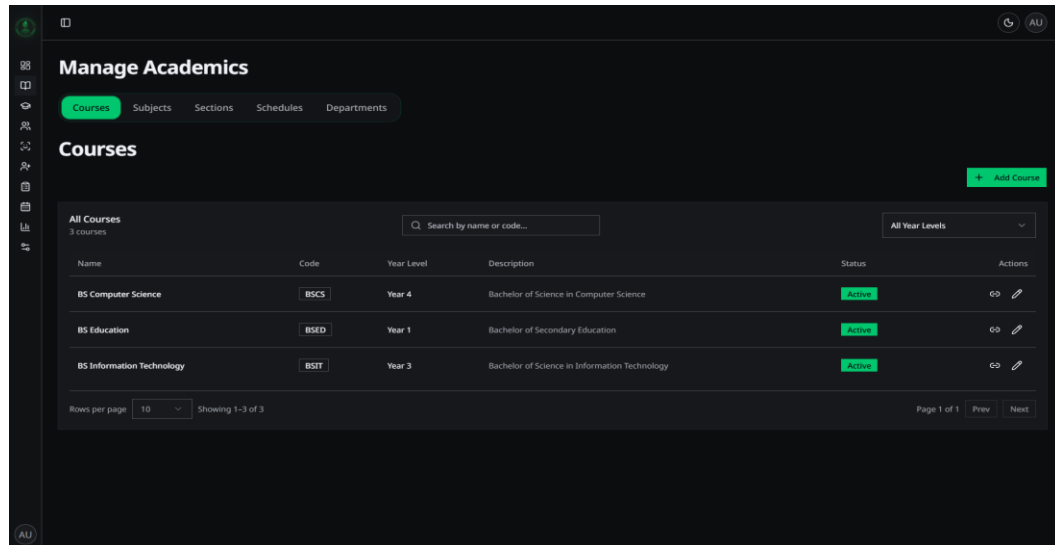
In the **Admin Dashboard**, a dropdown menu located on the right side provides access to the Admin Profile section. This allows the administrator to view and update personal information, including account details and credentials. It also enables changes to profile information within the system



Personal information (Admin's View)

The screenshot displays a web application interface for an Admin's profile. On the left is a dark sidebar with a vertical list of icons. The main content area has a dark background. At the top, it says 'Profile' and 'Manage your personal information and account settings.' Below this is a circular profile picture placeholder with the letters 'AU' and a green camera icon. The form contains several input fields: 'Username' (with 'admin' entered), 'Role' (with 'Admin' entered), 'First Name' (with 'Admin' entered), 'Last Name' (with 'User' entered), 'Email' (with 'admin@defaultschool.com' entered), 'Phone Number', 'Birthday' (with a date picker showing 'dd/mm/yyyy'), and 'Address'. A green 'Save Profile' button is located below the address field. Below the form is a 'Change Password' section with fields for 'Current Password', 'New Password', and 'Confirm New Password'. At the bottom left of the main area is a small circular icon with 'AU'.

The **Personal Information** section allows users to view and update their account details, including username, first name, last name, email, phone number, birthday, and address. It also enables users to save profile changes and update their password within the system. In addition, this section ensures that user information remains accurate and up to date, supporting proper account management and secure system access.

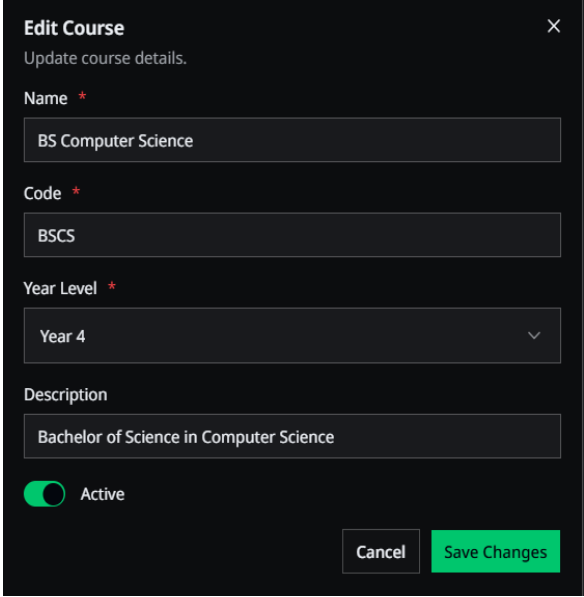


Manage Academics Courses (Admin's View)

The **Manage Academic Courses** section displays all created courses within the system. It allows the administrator to view and manage the list of available academic courses in an organized format.

The **Add Course** button allows the administrator to add a new academic course in the system. It enables the input of course details such as course name, course code, and year level.

The **Edit Course** feature allows the administrator to update existing academic course details in the system. It enables modification of information such as course name, course code, and year level.



Edit Course ✕
Update course details.

Name *
BS Computer Science

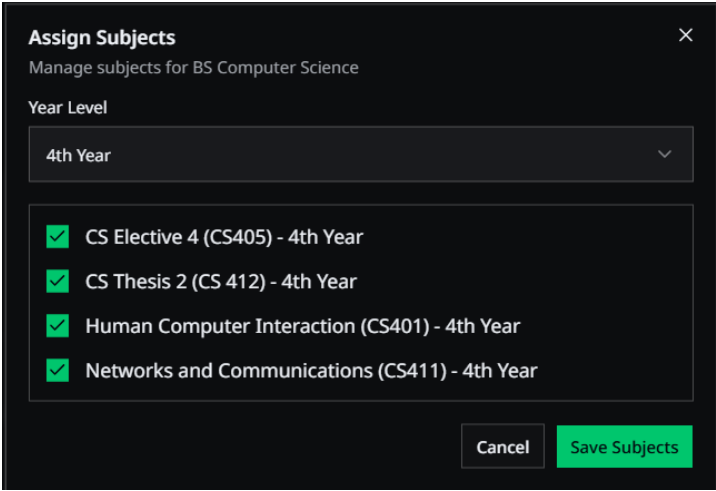
Code *
BSCS

Year Level *
Year 4 ▾

Description
Bachelor of Science in Computer Science

☒ Active

Cancel Save Changes



Assign Subjects ✕
Manage subjects for BS Computer Science

Year Level
4th Year ▾

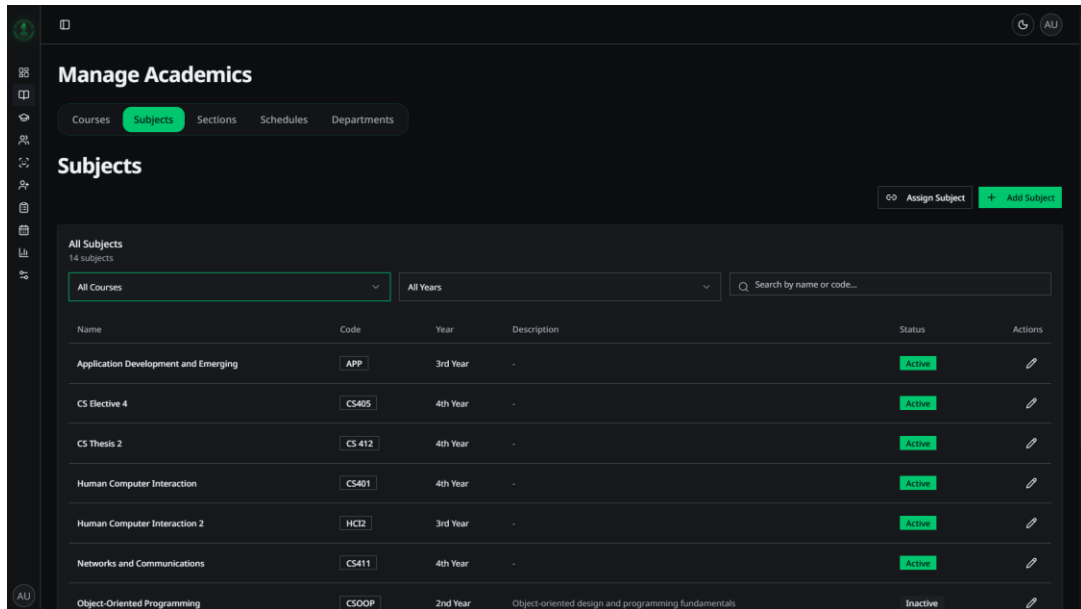
- ☒ CS Elective 4 (CS405) - 4th Year
- ☒ CS Thesis 2 (CS 412) - 4th Year
- ☒ Human Computer Interaction (CS401) - 4th Year
- ☒ Networks and Communications (CS411) - 4th Year

Cancel Save Subjects

The **Assign Subjects** section allows the administrator to view all created subjects and assign them to a specific year level. It enables selection of subjects and linking them to the

appropriate year level within the system.

Manage Academics Subjects (Admin's View)



The **Manage Academic** Subjects section allows the administrator to view all created subjects in the system. It displays subjects across all courses and year levels in an organized list.

The 'Add Subject' modal form contains the following fields and controls:

- Name ***: Text input with placeholder 'e.g. Mathematics'
- Code ***: Text input with placeholder 'e.g. MATH101'
- Description**: Text input with placeholder 'Optional description'
- Year ***: Dropdown menu with '1st Year' selected
- Active**: Toggle switch (currently turned on)
- Buttons**: 'Cancel' and 'Create Subject'

The **Add Subject** button allows the administrator to add a new subject in the system. It enables the input of subject details such as subject name, subject code, and year level.

The **Edit Subject** button allows the administrator to add a new subject in the system. It enables the input of subject details such as subject name, subject code, and year level.

✕

Edit Subject

Update subject details.

Name *

Application Development and Emerging

Code *

APP

Description

Optional description

Year *

3rd Year

Active

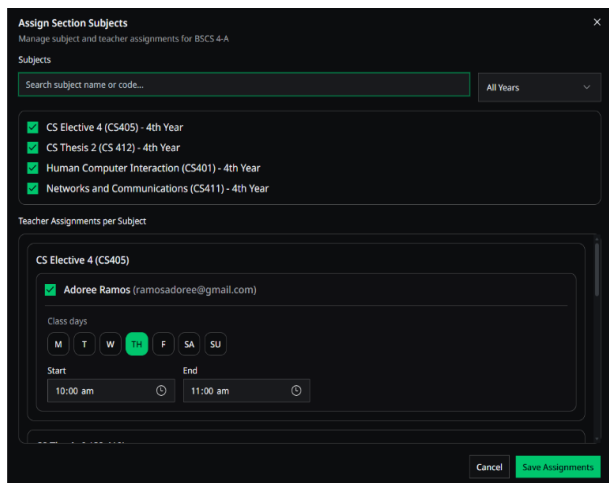
Cancel

Save Changes

Manage Academics Sections (Admin's View)

[illegible]

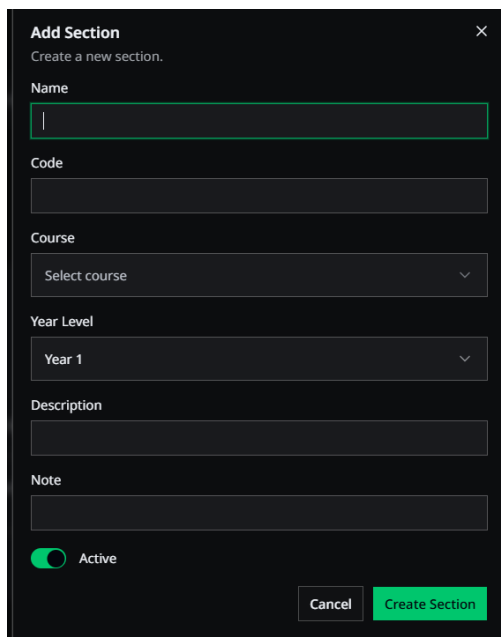
The **Manage Academic Section** allows the administrator to view all created sections across different year levels. It also provides a search function to quickly locate a specific section within the system.



The screenshot shows a dialog box titled "Assign Section Subjects" with a subtitle "Manage subject and teacher assignments for BSCS 4-A". It features a search bar for "Search subject name or code..." and a dropdown for "All Years". Below these, a list of subjects is shown with checkboxes: "CS Elective 4 (CS405) - 4th Year", "CS Thesis 2 (CS 412) - 4th Year", "Human Computer Interaction (CS401) - 4th Year", and "Networks and Communications (CS411) - 4th Year". A section titled "Teacher Assignments per Subject" is expanded for "CS Elective 4 (CS405)", showing a teacher "Adoree Ramos (ramosadoree@gmail.com)" with a class schedule of "Th" (Thursday) from "10:00 am" to "11:00 am". The dialog includes "Cancel" and "Save Assignments" buttons.

The **Assign Section Subjects** feature allows the administrator to assign subjects to specific sections across all year levels. It also enables selection of subjects individually and viewing of already assigned teachers for each subject. In addition, the

administrator can set the day and time schedule for each assigned subject.



The screenshot shows a dialog box titled "Add Section" with a subtitle "Create a new section.". It contains input fields for "Name", "Code", "Course" (with a "Select course" dropdown), "Year Level" (with a "Year 1" dropdown), "Description", and "Note". There is also a toggle switch for "Active" which is currently turned on. The dialog includes "Cancel" and "Create Section" buttons.

The **Add Section** feature allows the administrator to create a new section in the system. It enables the input of section details such as section name, subject code, year level, and course assignment.

The **Edit Section** feature allows the administrator to update existing section details in the system. It enables modification of information such as section name, subject code, year level, and course assignment.

Edit Section

Update section details.

Name

BSCS 4-A

Code

CS4A

Course

BS Computer Science - Year 4

Year Level

Year 4

Description

CS4A

Note

☒ Active

Cancel

Save Changes

Manage Academics Schedules (Admin's View)

Schedules

+ Add Schedule

Filters

Filter both tables by course, teacher, and subject.

Course

All courses

Teacher

All teachers

Subject

All subjects

Section class schedules

Active assignments from section subject teachers (7 rows).

Course	Section	Subject	Teacher	Days	Time	Action
BS Information Technology	BSIT 3-A IT3A	Application Development and Emerging APP	Jerome Aguilar	Wed	16:00-17:30	Edit Remove
BS Information Technology	BSIT 3-A IT3A	Statistics with Computer Application STATS	Maricris Amolaza	Wed	14:30-16:00	Edit Remove
BS Computer Science	BSCS 4-A CS4A	CS Elective 4 CS405	Adoree A. Ramos	Thu	10:00-11:00	Edit Remove
BS Computer Science	BSCS 4-A CS4A	CS Thesis 2 CS 412	Adoree A. Ramos	Thu	11:00-12:00	Edit Remove
BS Computer Science	BSCS 4-A CS4A	Human Computer Interaction CS401	Randie Ian A. Villatuya	Fri	08:00-09:00	Edit Remove
BS Computer Science	BSCS 4-A CS4A	Networks and Communications CS411	Randie Ian A. Villatuya	Fri	09:00-10:30	Edit Remove
BS Information Technology	BSIT 3-A IT3A	Human Computer Interaction 2 HCI2	Randie Ian A. Villatuya	Wed	13:00-14:30	Edit Remove

Rows per page: 10

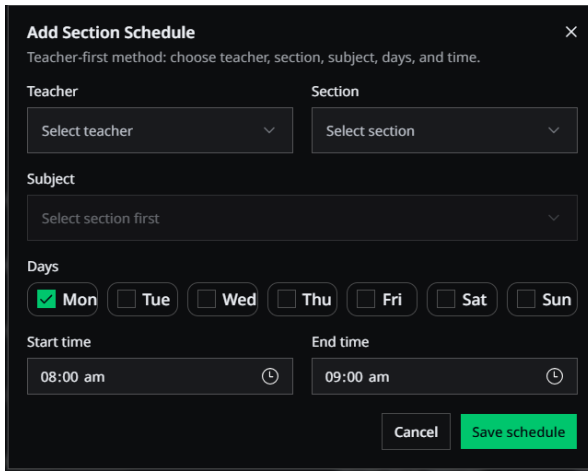
Showing 1-7 of 7

Page 1 of 1

Prev

Next

The **Manage Academic Schedules** section allows the administrator to view all created schedules in the system. It displays assigned teachers, subjects, days, and time schedules in an organized list. It also provides dropdown selections for courses, teachers, and subjects to manage and filter schedule entries efficiently.



Add Section Schedule ✕

Teacher-first method: choose teacher, section, subject, days, and time.

Teacher

Section

Subject

Days

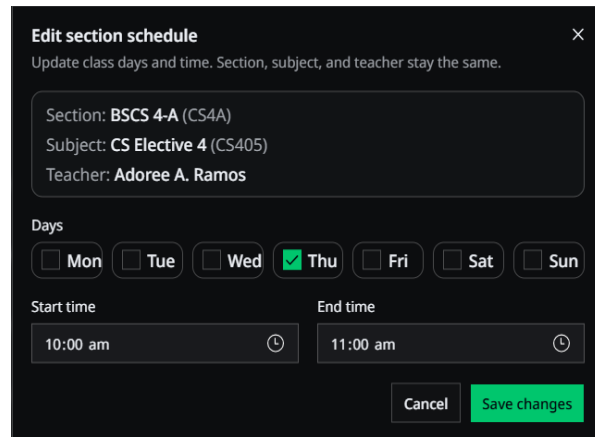
☒ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri ☐ Sat ☐ Sun

Start time ⌚

End time ⌚

The **Add Section Schedule** feature allows the administrator to create a class schedule by assigning a teacher, section, and subject. It also enables the selection of the day, start time, and end time of the class.

The **Edit Section Schedule** feature allows the administrator to modify existing class schedules by selecting a teacher, section, and subject. It enables updates to the schedule details, specifically the day and time, including the start time and end time of the class.



Edit section schedule ✕

Update class days and time. Section, subject, and teacher stay the same.

Section: **BSCS 4-A (CS4A)**

Subject: **CS Elective 4 (CS405)**

Teacher: **Adoree A. Ramos**

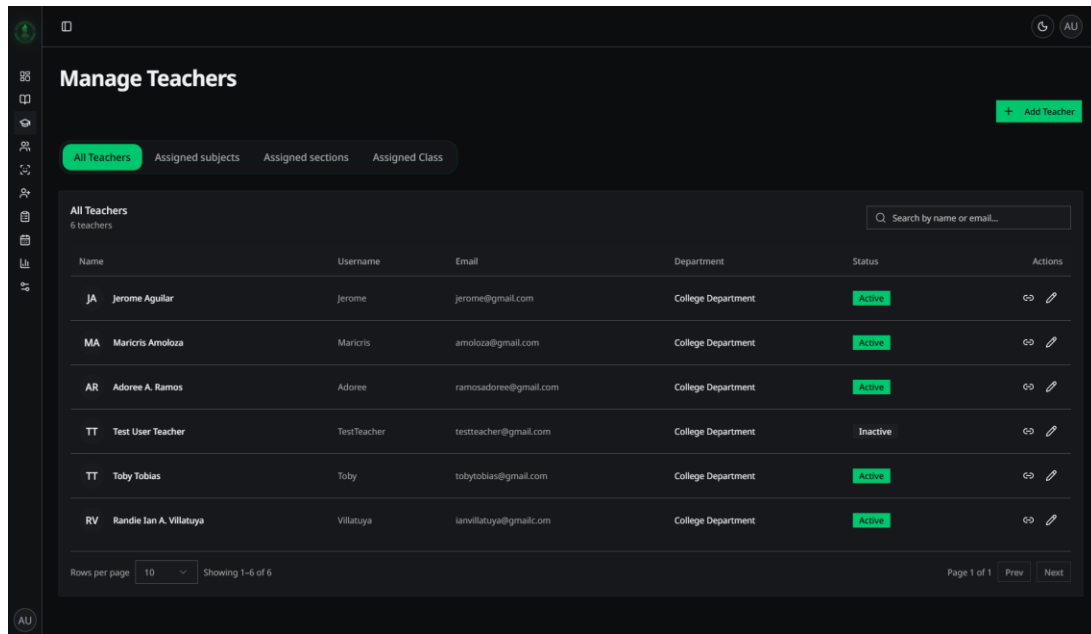
Days

☐ Mon ☐ Tue ☐ Wed ☒ Thu ☐ Fri ☐ Sat ☐ Sun

Start time ⌚

End time ⌚

Manage Teacher's (Admin's View)



The **Manage Teachers** section allows the administrator to view all registered teachers in the system. It displays the complete list of added teachers and provides a search function to easily locate a specific teacher

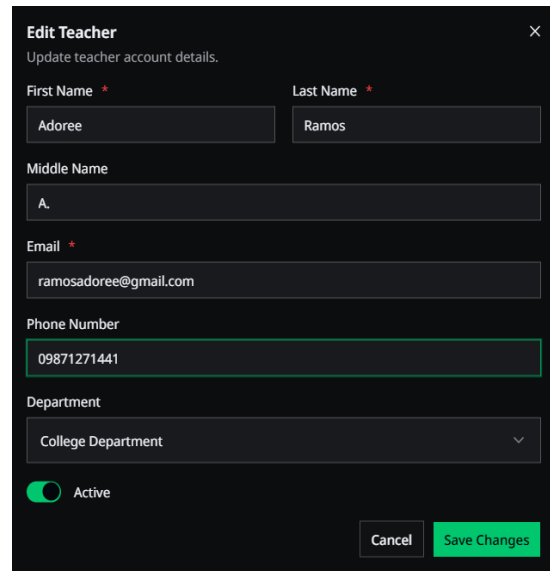
The screenshot shows the 'Add Teacher' form, which is a modal window titled 'Add Teacher' with a close button 'X'. The subtitle is 'Create a new teacher account.' The form contains the following fields:

- First Name * (text input)
- Last Name * (text input)
- Middle Name (text input)
- Username * (text input)
- Password * (password input)
- Email * (text input)
- Phone Number (text input)
- Department (dropdown menu with 'Select department' as the placeholder)

At the bottom right, there are two buttons: 'Cancel' and 'Create Teacher'.

The **Add Teacher** feature allows the administrator to register a new teacher in the system. It enables the input of teacher details such as first name, last name, middle name, username, password, email, phone number, and

The **Edit Teacher** feature allows the administrator to update existing teacher information in the system. It enables modification of details such as first name, last name, middle name, username, password, email, phone number, and department.



Edit Teacher
Update teacher account details.

First Name * Last Name *

Adoree Ramos

Middle Name

A.

Email *

ramosadoree@gmail.com

Phone Number

09871271441

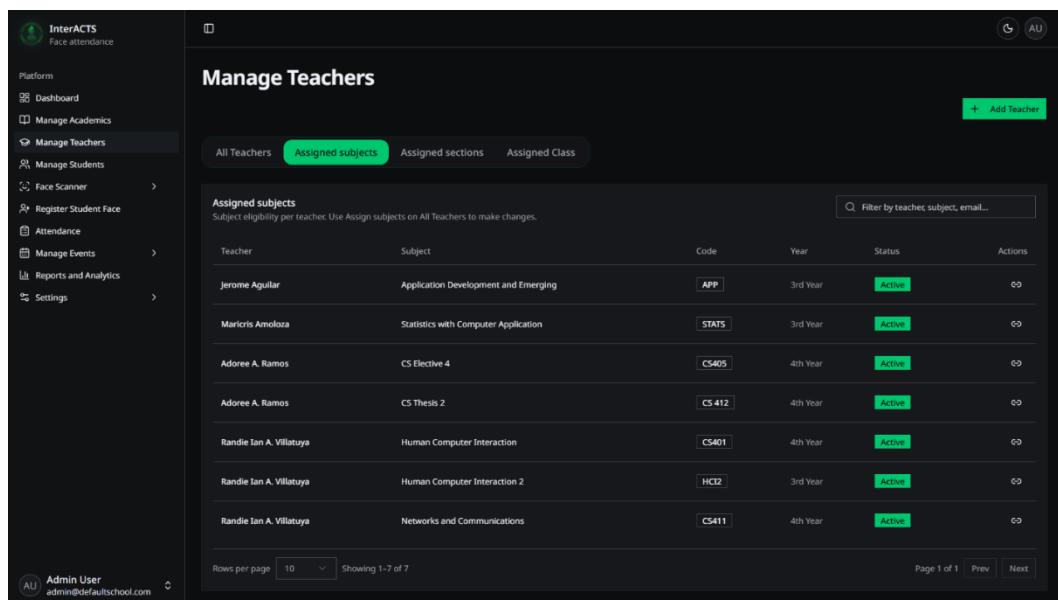
Department

College Department

☒ Active

Cancel Save Changes

Manage Teacher's Assigned Subjects (Admin's View)



InterACTS
Face attendance

Platform

- Dashboard
- Manage Academics
- Manage Teachers
- Manage Students
- Face Scanner
- Register Student Face
- Attendance
- Manage Events
- Reports and Analytics
- Settings

Manage Teachers

+ Add Teacher

All Teachers Assigned subjects Assigned sections Assigned Class

Assigned subjects
Subject eligibility per teacher. Use Assign subjects on All Teachers to make changes.

Filter by teacher, subject, email...

Teacher	Subject	Code	Year	Status	Actions
Jerome Aguilar	Application Development and Emerging	APP	3rd Year	Active	GO
Maricris Amoloza	Statistics with Computer Application	STATS	3rd Year	Active	GO
Adoree A. Ramos	CS Elective 4	CS405	4th Year	Active	GO
Adoree A. Ramos	CS Thesis 2	CS 412	4th Year	Active	GO
Randie Ian A. Villatuya	Human Computer Interaction	CS401	4th Year	Active	GO
Randie Ian A. Villatuya	Human Computer Interaction 2	HCI2	3rd Year	Active	GO
Randie Ian A. Villatuya	Networks and Communications	CS411	4th Year	Active	GO

Rows per page: 10 Showing 1-7 of 7 Page 1 of 1 Prev Next

Admin User
admin@defaultschool.com

The **Teacher's Assigned Subjects** section allows the administrator to view all subjects assigned to teachers in the system. It displays the list of assigned teachers and provides a search function to easily locate specific records.

Assign Subjects ×

Manage subjects for Jerome Aguilar

All Years ▼

- ☒ Application Development and Emerging (APP) - 3rd Year
- ☐ CS Elective 4 (CS405) - 4th Year
- ☐ CS Thesis 2 (CS 412) - 4th Year
- ☐ Human Computer Interaction (CS401) - 4th Year
- ☐ Human Computer Interaction 2 (HCI2) - 3rd Year
- ☐ Networks and Communications (CS411) - 4th Year
- ☐ Readings in Philippine History (GERPH) - 1st Year
- ☐ Science, Technology and Society (GESTS) - 2nd Year
- ☐ Statistics with Computer Application (STATS) - 3rd Year
- ☐ Understanding the Self (GEUTS) - 1st Year

Cancel Save Subjects

The **Teacher's Assigned Subjects** section allows the administrator to view all subjects assigned to teachers in the system. It displays the list of assigned teachers and provides a search function to easily locate specific records.

Manage Teacher's Assigned Section (Admin's View)

InterACTS
Face attendance

Platform

- Dashboard
- Manage Academics
- Manage Teachers**
- Manage Students
- Face Scanner
- Register Student Face
- Attendance
- Manage Events
- Reports and Analytics
- Settings

Admin User
admin@defaultschool.com

Manage Teachers + Add Teacher

All Teachers Assigned subjects **Assigned sections** Assigned Class

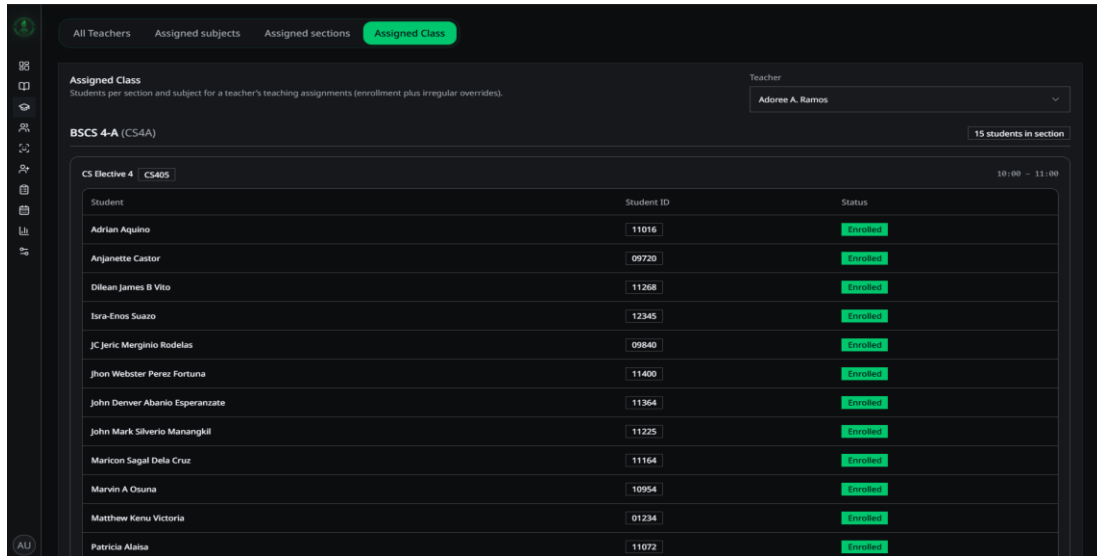
Filter by teacher, section, subject...

Teacher	Section	Subject	Days	Time	Status
Jerome Aguilar	BST 3-A (BT3A) - Year 3	Application Development and Emerging (APP)	Wed	16:00 - 17:30	Active
Maricris Amoloza	BST 3-A (BT3A) - Year 3	Statistics with Computer Application (STATS)	Wed	14:30 - 16:00	Active
Adoree A. Ramos	BSCS 4-A (CS4A) - Year 4	CS Elective 4 (CS405)	Thu	10:00 - 11:00	Active
Adoree A. Ramos	BSCS 4-A (CS4A) - Year 4	CS Thesis 2 (CS 412)	Thu	11:00 - 12:00	Active
Randle Ian A. Villatuya	BSCS 4-A (CS4A) - Year 4	Human Computer Interaction (CS401)	Fri	08:00 - 09:00	Active
Randle Ian A. Villatuya	BSCS 4-A (CS4A) - Year 4	Networks and Communications (CS411)	Fri	09:00 - 10:30	Active
Randle Ian A. Villatuya	BST 3-A (BT3A) - Year 3	Human Computer Interaction 2 (HCI2)	Wed	13:00 - 14:30	Active

Rows per page: 10 Showing 1-7 of 7 Page 1 of 1 Prev Next

The **Teacher's Assigned Section** allows the administrator to view all sections assigned to each teacher in the system. It displays the corresponding schedule, including the day and time for each assigned section.

Manage Teacher's Assigned Class (Admin's View)



Assigned Class
Students per section and subject for a teacher's teaching assignments (enrollment plus irregular overrides).

Teacher: **Adornee A. Ramos**

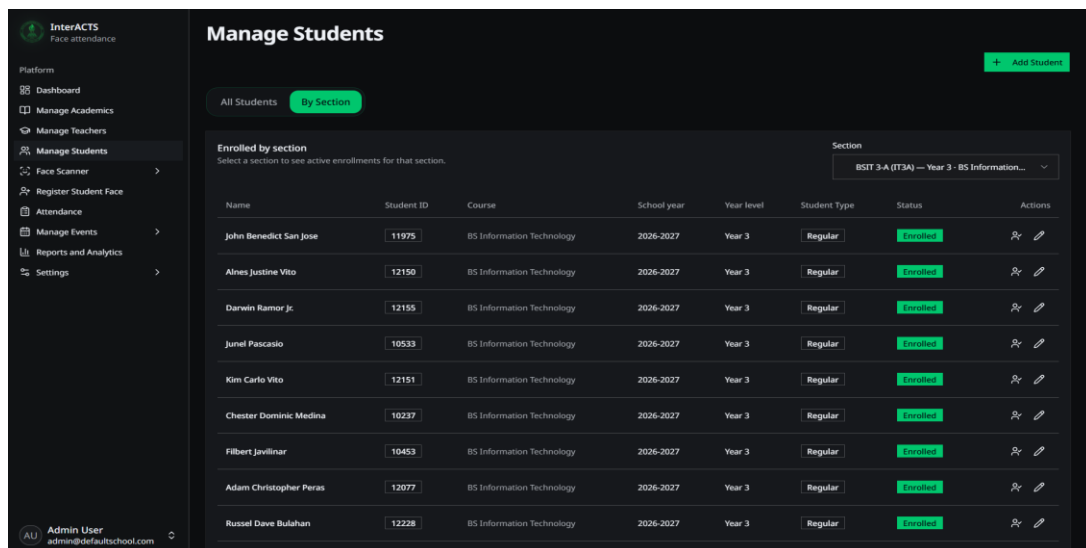
BSCS 4-A (CS4A) 15 students in section

CS Elective 4 CS405 10:00 - 11:00

Student	Student ID	Status
Adrian Aquino	11016	Enrolled
Anjanette Castor	09720	Enrolled
Dilean James B Vito	11208	Enrolled
Isra-Enos Suazo	12345	Enrolled
JC Jeric Merginio Rodelas	09840	Enrolled
Jhon Webster Perez Fortuna	11400	Enrolled
John Denver Abanilo Esperanzate	11364	Enrolled
John Mark Silverio Manangkil	11225	Enrolled
Maricon Sagal Dela Cruz	11164	Enrolled
Marvin A Osuna	10954	Enrolled
Matthew Kenu Victoria	01234	Enrolled
Patricia Alaisa	11072	Enrolled

The **Teacher's Assigned Class** section allows the administrator to view all classes assigned to teachers in the system. It displays the list of students under each class, including their status such as enrolled or pending. It also provides a dropdown selection to choose and view a specific teacher's assigned class.

Manage Student's (Admin's View)



Manage Students

All Students **By Section** [+ Add Student](#)

Enrolled by section
Select a section to see active enrollments for that section.

Section: **BSIT 3-A (IT3A) - Year 3 - BS Information...**

Name	Student ID	Course	School year	Year level	Student Type	Status	Actions
John Benedict San Jose	11975	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎
Alnei Justine Vito	12150	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎
Darwin Ramor Jr.	12155	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎
Junel Pascasio	10533	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎
Kim Carlo Vito	12151	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎
Chester Dominic Medina	10237	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎
Filbert Jawlinar	10453	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎
Adam Christopher Peras	12077	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎
Russel Dave Bulahan	12228	BS Information Technology	2026-2027	Year 3	Regular	Enrolled	✕ ✎

The **Manage Students** section allows the administrator to view all students in the system. It displays student details such as enrollment status, year level, classification (regular or irregular), course, and student ID.

Add Student

A student user account will be created automatically. The default password is the student ID plus the last name with spaces removed, all lowercase (e.g. 00152 + Dela Cruz → 00152delacruz).

First Name *
 Last Name *
 Middle Name
 Student ID Number *
 Year Level *
 Year 1
 Email *
 Contact Number *
 Birthday
 dd/mm/yyyy
 Enrolled Date
 14/04/2026
 Address
 Guardian Information
 Guardian Contact *
 Guardian Email *
 Cancel Create Student

The **Add Student** feature allows the administrator to register a new student in the system. It enables the input of student details such as first name, last name, middle name, student ID, year level, email, contact number, birthday, enrolled date, address, guardian contact, and guardian email.

The **Edit Student** feature allows the administrator to update existing student information in the system. It enables modification of details such as first name, last name, middle name, student ID, year level, email, contact number, birthday, enrolled date, address, guardian contact, and guardian email. It also allows updating of the student's status, such as enrolled, pending, dropped, or graduated.

Edit Student

Updating this student will also sync the linked user account.

First Name *
 John Benedict
 Last Name *
 San Jose
 Middle Name
 Student ID Number *
 11975
 Year Level *
 Year 3
 Email *
 johnbenedictsanjose001@gmail.com
 Contact Number *
 123
 Birthday
 10/04/2026
 Enrolled Date
 10/04/2026
 Address
 Guardian Information
 Guardian Contact *
 123
 Guardian Email *
 johnbenedictsanjose001@gmail.com
 Status
 Enrolled
 Active
 Cancel Save Changes

The **Edit Enrollment** feature allows the administrator to update student enrollment details in the system. It enables modification of information such as course, section, school year, and year level. It also allows updating of the student's enrollment status, including enrolled, dropped, or graduated, as well as classification as regular or irregular

Edit enrollment
View or update enrollment for John Benedict San Jose

Course *
BS Information Technology

Section
BSIT 3-A (IT3A) - Year 3

School Year *
2026-2027

Year Level *
Year 3

Enrolled Date
10/04/2026

Status
Enrolled

Student Type *
Regular

Remarks

☒ Set as active enrollment

Cancel Update enrollment

Irregular Subject Plan
Add or remove subject schedule overrides for John Benedict San Jose.

Active Overrides
No active overrides.

Add Override

Assignment type
ADD

Base Section Assignment (for REMOVE/REPLACE)
None

Reuse Existing Assignment (optional)
None

Subject
None

Teacher
None

Effective From
dd/mm/yyyy

Effective To
dd/mm/yyyy

Days of Week
Mon Tue Wed Thu Fri Sat Sun

Start Time
--:--

End Time
--:--

Remarks

Add override

The **Manage Irregular Plan** feature allows the administrator to create and manage irregular or override class assignments in the system. It enables selection of assignment types such as add, move, replace sections, or reuse existing

assignments. The administrator can assign subjects and teachers, set the effective date range, specify days of the week, and define the start and end time of classes. It also displays the list of assigned subjects for reference.

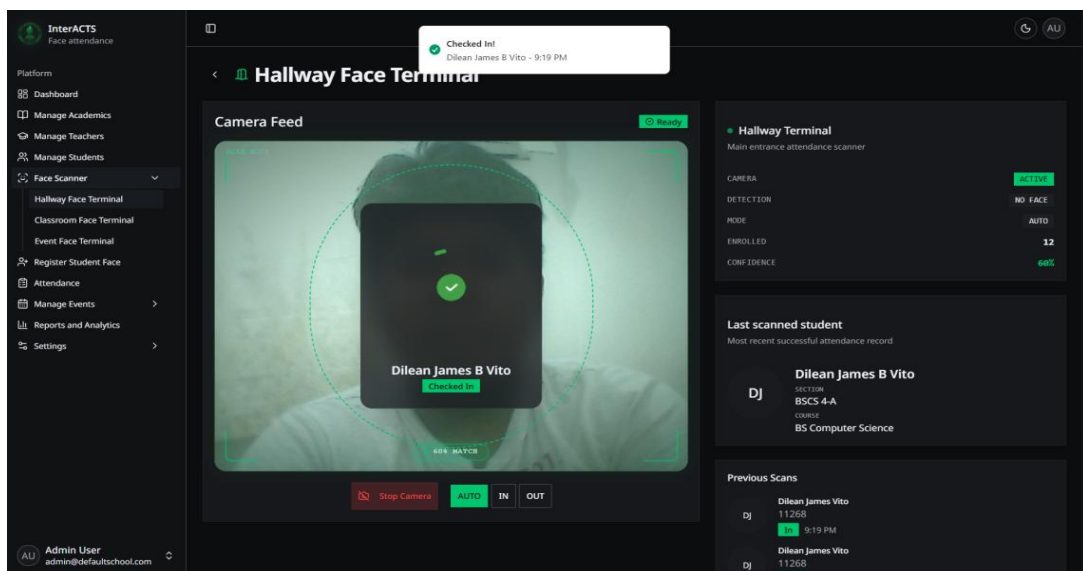
Register Student Face (Admin's View)

The **Register Student Face** feature allows the administrator to register a student's face in the system. It enables selection of the student's name and section before capturing facial data. The registration requires multiple angles including front, left, right, upward view, and a smiling expression to ensure accurate face recognition.

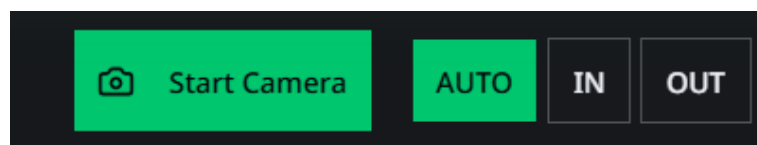
Student Name	Student ID	Latest enrolled	Credential refs	Actions
test number 3 33333	@ Student ID: d884b778a-4122-b968-88d201c10b28	Latest enrolled: 4/14/2026	1 credential PRIMARY	Re-register Delete
Kristopher Alonuevo 12255	@ Student ID: f9d295a0-8f05-49b1-9ba5-56aee57a1f3	Latest enrolled: 4/10/2026	1 credential PRIMARY	Re-register Delete
Adam Christopher Peras 12077	@ Student ID: b2094308-7b02-4c54-bcc9-2a50397deedc	Latest enrolled: 4/10/2026	1 credential PRIMARY	Re-register Delete
Filbert Javlinar 10453	@ Student ID: 75d6072e-d996-400d-a5f9-26f1e41821ef	Latest enrolled: 4/10/2026	1 credential PRIMARY	Re-register Delete
Kim Carlo Vito 12151	@ Student ID: c9261a18-0030-4b0f-9795-86d78aa490c	Latest enrolled: 4/10/2026	1 credential PRIMARY	Re-register Delete

The **Re-Register Student Face** feature allows the administrator to update or replace an existing student's facial data in the system. It enables deletion of previous face records and re-registration using the same required angles, including front, left, right, upward view, and smiling expression.

Hallway Face Terminal (Admin's View)



The **Hallway Face Terminal** serves as the attendance scanning point for students



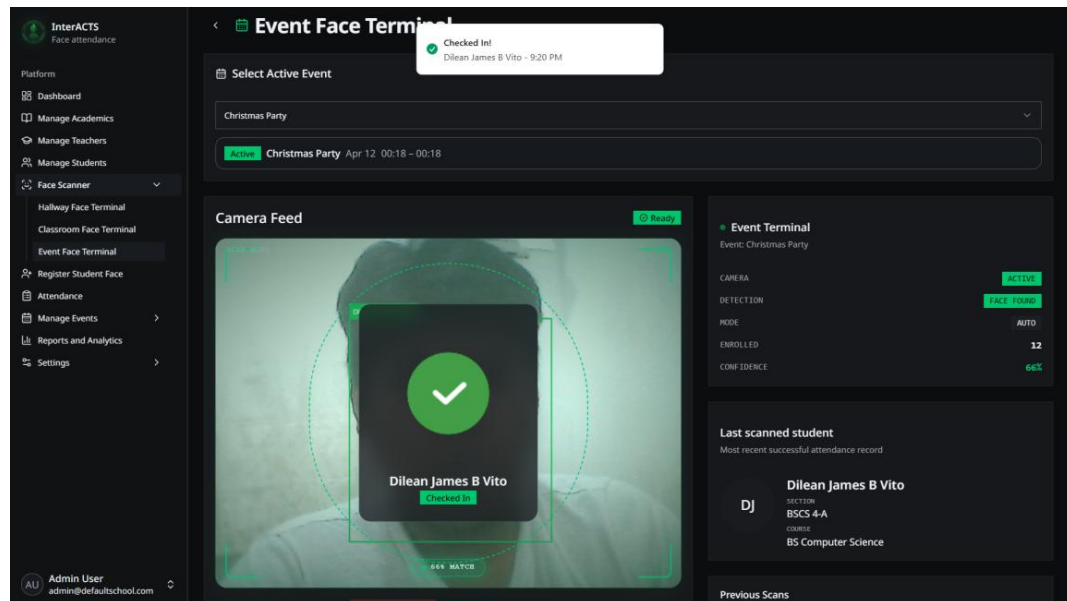
using facial recognition. It allows students to check in

and check out within the hallway after face registration. The system automatically records each attendance activity and sends a notification to the student's parents or guardians via email upon check-out.

The **Face Terminal camera** feature allows the system to capture student attendance using facial recognition. It includes automatic check-in and check-out

functions when a face is detected. It also provides manual check-in and manual check-out buttons that can be used when needed to record attendance.

Event's Face Terminal (Admin's View)

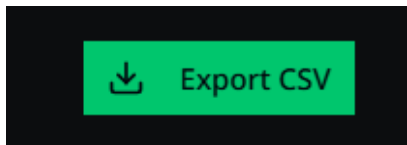


The **Event's Face Terminal** is used for recording student attendance during events using facial recognition, and it only activates when there is a scheduled event on the selected date. It requires the student's payment receipt to be approved before allowing access to the scanning feature, ensuring that only verified participants can check in. Once verified, the system records attendance in real time for proper monitoring of event participation.

Attendance Log (Admin's View)

Student	Type	Source Type	Date	Time	Confidence
Dilean James Vito	Check Out	WEB_PORTAL	Apr 13, 2025	9:22:06 PM	56%
Dilean James Vito	Check In	WEB_PORTAL	Apr 13, 2025	9:21:49 PM	72%
JC Jaric Rodelas	Check Out	MANUAL	Apr 13, 2025	10:30:00 AM	0%
Adrian Aquino	Check Out	MANUAL	Apr 13, 2025	10:30:00 AM	0%
Anjanette Castor	Check Out	MANUAL	Apr 13, 2025	10:30:00 AM	0%
Dilean James Vito	Check Out	MANUAL	Apr 13, 2025	10:30:00 AM	0%
Jhon Webster Fortuna	Check Out	MANUAL	Apr 13, 2025	10:30:00 AM	0%
John Mark Manangkil	Check Out	MANUAL	Apr 13, 2025	10:30:00 AM	0%
John Denver Esperanzate	Check Out	MANUAL	Apr 13, 2025	10:30:00 AM	0%

The **Attendance Log** displays all recorded attendance activities in the system, including face recognition logs from the hallway and events, monitoring records, and manual check-ins and check-outs. It allows the administrator to track whether students have checked in or checked out and also provides a date filter to view attendance records for a selected date only.



The **Attendance Log Export** feature allows the administrator to export all recorded attendance data in the system. It includes check-in and check-out records from face recognition, manual attendance, and event monitoring, along with student details such as name, course, and time records. This feature also supports exporting filtered data based on selected dates for easier reporting and documentation.

Event Management (Admin's View)

The screenshot displays the 'Events' section of an admin dashboard. It features a sidebar with navigation icons, a top header with a user profile 'AU', and a main content area. The 'Events' section includes a '+ Add Event' button, tabs for 'Event Management' (selected) and 'Event Attendance', and a table of 'All Events'. Below the events table is a 'Payments by event' section with a filter dropdown set to 'All Events' and a table of payment receipts.

Name	Event Date	Start	End	Time	Description	Status	Actions
Sport Fest 2026	Apr 10, 2026	Apr 10, 2026	Apr 10, 2026	11:17 PM – 5:17 PM	-	Active	
Christmas Party	Apr 12, 2026	Apr 12, 2026	Apr 12, 2026	12:18 AM – 12:18 AM	rower	Active	
ACQUAINTANCE PARTY	Apr 15, 2026	Apr 14, 2026	Apr 23, 2026	7:07 AM – 5:10 PM	-	Active	

Receipt	Student	Event	Amount	Purpose	Status	Submitted
	Dilean James Vito	ACQUAINTANCE PARTY	0.00	Payment and Proof	Pending	Apr 14, 2026

The **Event Management** section allows the administrator to view and manage event details such as event date, start date, end date, event start time, and event end time. It also displays uploaded event payment receipts submitted by students for approval, ensuring that only verified payments are marked as approved within the system.

Event Management Attendance (Admin's View)

The screenshot shows the 'Events' section of the system. At the top, there's a sidebar with navigation icons and a main header with 'Events' and a '+ Add Event' button. Below the header, there are tabs for 'Event Management' and 'Event Attendance', with 'Event Attendance' being the active tab. The main content area is titled 'Event Attendance Viewer' and contains a sub-header 'Current active event student logs grouped per section.' Below this, there are date pickers for 'From' and 'To' (both set to 14/04/2026) and a 'Refresh' button. A section titled 'ACQUAINTANCE PARTY' is shown with an 'Active window: 2026-04-14 to 2026-04-23'. Below this, there are filters for 'Sections 0', 'Students 0', 'Logs 0', 'IN 0', and 'OUT 0'. The main area displays a message: 'No event scan logs found for selected range.'

The **Event Management Attendance** section allows the administrator to view all student attendance records for specific events. It displays the check-in and check-out logs of students during the event, providing an organized record of participation for monitoring purposes.

The 'Add Event' form is a modal window with a close button (X) in the top right corner. It contains the following fields and controls:

- Name ***: A text input field with a placeholder 'e.g. Foundation Day'.
- Description**: A text input field with a placeholder 'Optional description'.
- Start Date ***: A date picker field with a placeholder 'dd/mm/yyyy'.
- End Date ***: A date picker field with a placeholder 'dd/mm/yyyy'.
- Event Date ***: A date picker field with a placeholder 'dd/mm/yyyy'.
- Primary date used for calendar display**: A label below the Event Date field.
- Start Time ***: A time picker field with a placeholder '--:--'.
- End Time ***: A time picker field with a placeholder '--:--'.
- Active**: A toggle switch that is currently turned on.
- Buttons**: 'Cancel' and 'Create Event' buttons at the bottom right.

The Add Event feature allows the administrator to create a new event in the system. It enables input of event details such as event name, description, start date, end date, event date, start time, and end time. It also allows setting the event status as active or inactive to control event availability within the system.

✕

Edit Event

Update event details.

Name *

Sport Fest 2026

Description

Optional description

Start Date *

10/04/2026

📅

End Date *

10/04/2026

📅

Event Date *

10/04/2026

📅

Primary date used for calendar display

Start Time *

11:17 pm

🕒

End Time *

05:17 pm

🕒

☒ Active

Cancel

Save Changes

The **Edit Event** feature allows the administrator to update existing event details in the system. It enables modification of information such as event name, description, start date, end date, event date, start time, end time, and event status.

Event's Payment (Admin's View)

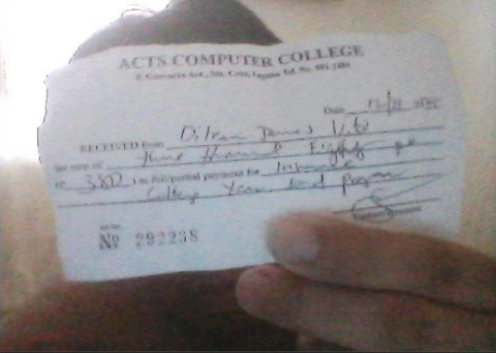
[illegible]

The **Event's Payment** section allows the administrator to view all student payments for specific events. It displays details such as event name, payment amount,

and payment date, and enables the administrator to review submitted payments for verification and approval.

Review payment

Amount and purpose are read-only. Update status and remarks for the student.



Amount

0.00

Purpose

Payment and Proof

Status

Pending

Pending

Approved

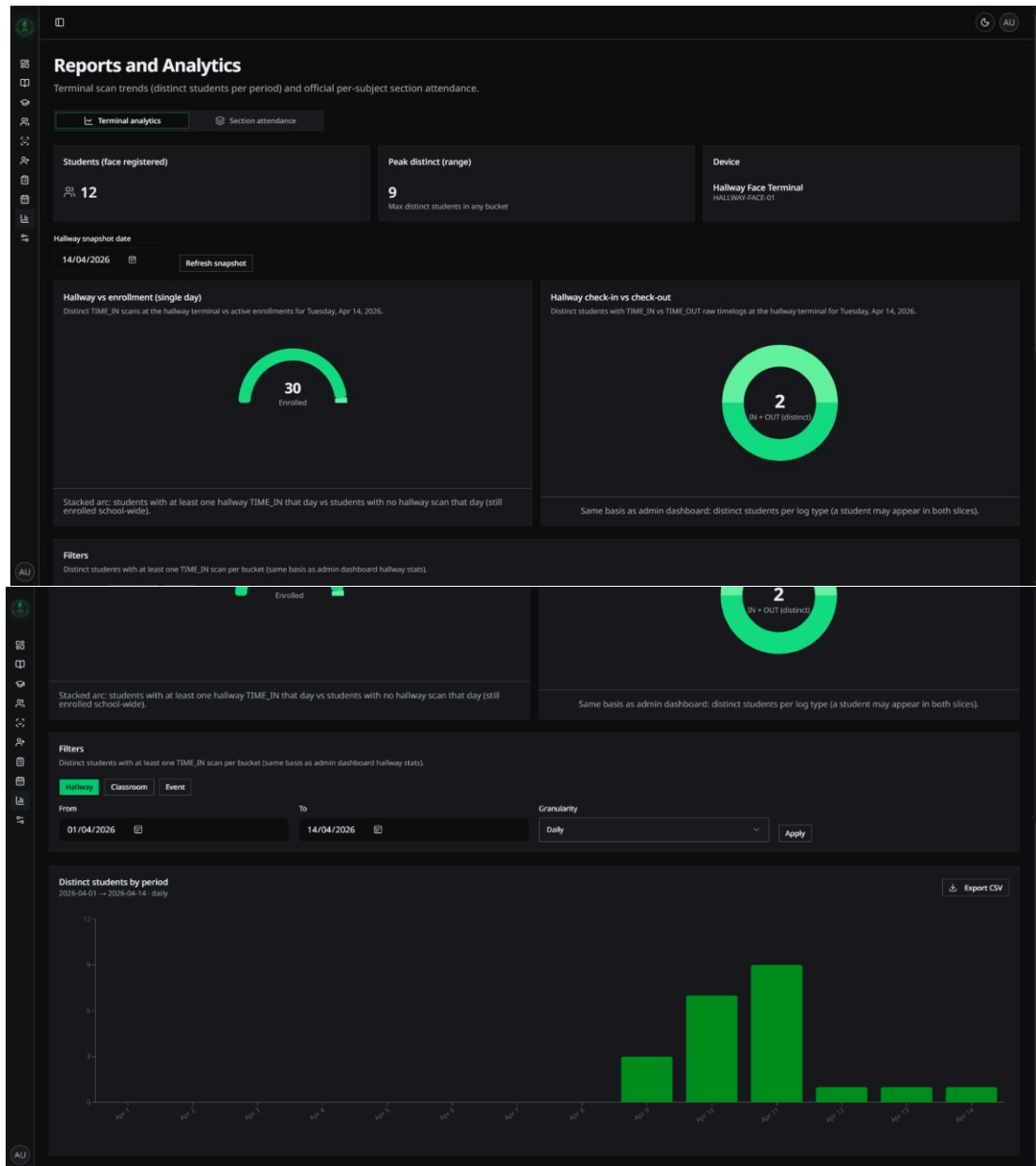
Rejected

Cancel

Save

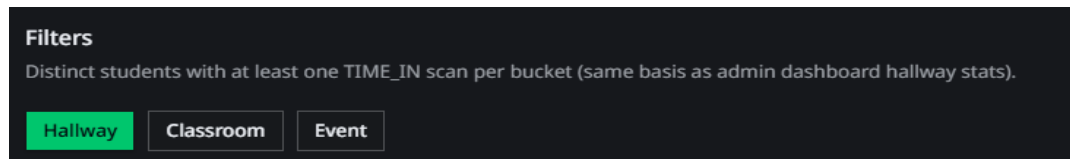
The **Review Event Payment** feature allows the administrator to view the uploaded receipt image submitted by students for event payments. It enables the administrator to verify the payment and set its status as approved, rejected, or pending, along with adding a description or remarks for further details

Report and Analytics (Admin's View)



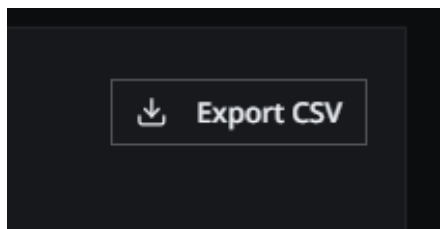
The **Reports and Analytics** section allow the administrator to view overall system analytics, including registered student faces, training data, peak distance range, and device usage. It also displays comparisons such as hallway check-ins versus enrolled

students and check-ins versus check-outs. In addition, the section provides distinct student reports by period, with options to filter data by daily, weekly, or monthly views



using date selection.

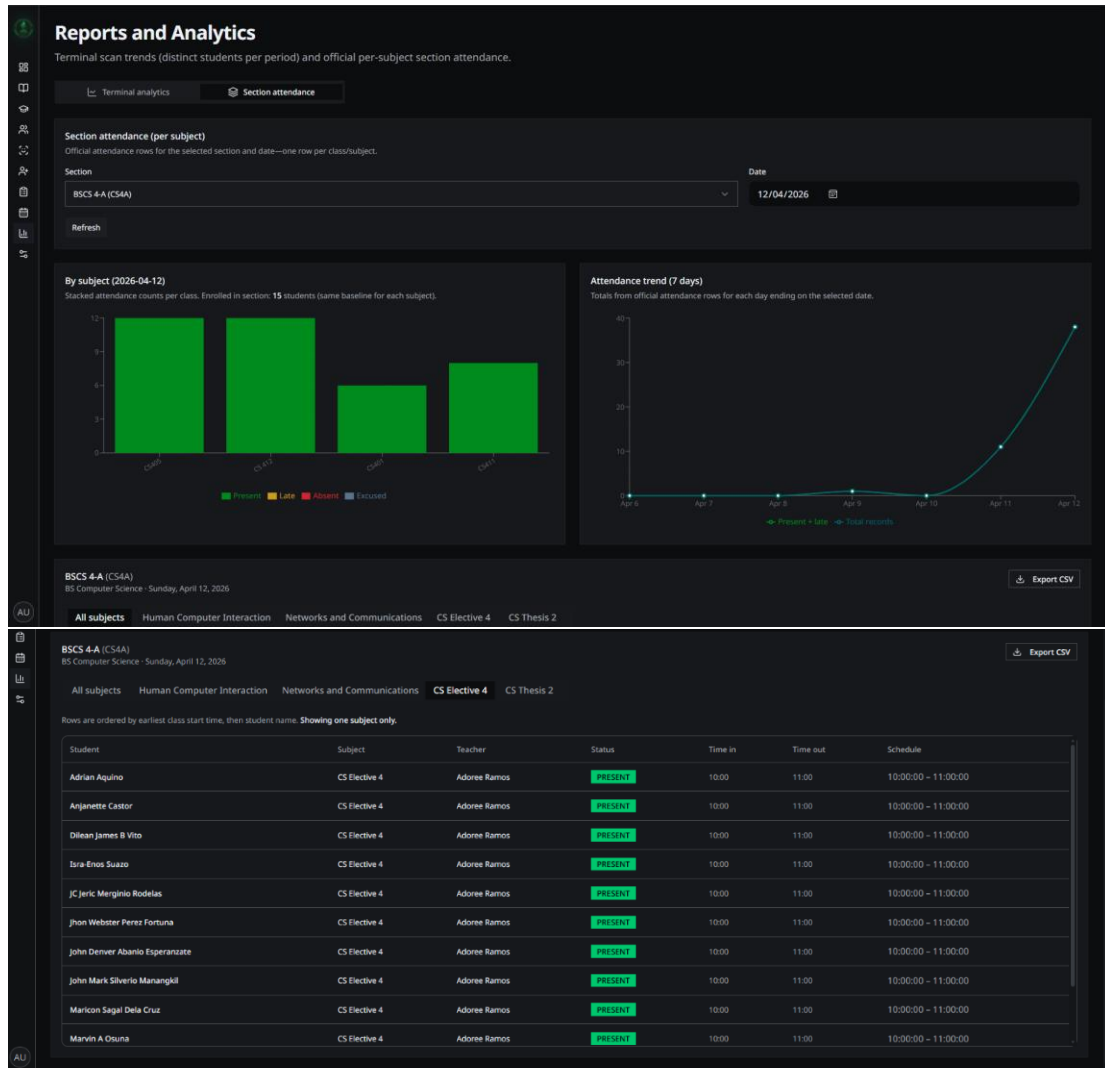
The system also includes filters to view analytics based on selected areas such as hallway, classroom, or events.



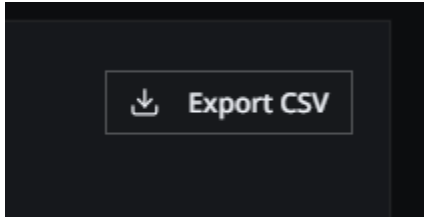
The **Analytics Export** feature allows the administrator to export detailed system analytics and attendance records. It includes time logs such as hallway check-in and check-out, classroom attendance per subject, and event attendance records.

Section	Course	Student full name	Date	Time check-in	Time check-out
BSCS 4-A	BS Computer Science	Adrian Aquino	09/04/2026	8:00:00	9:00:00
BSIT 3-A	BS Information Technology	Adam Christopher Peras	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Alnes Justine Vito	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Chester Dominic Medina	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Darwin Ramor Jr.	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Dehn Mark Oliveros	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Filbert Javilinar	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Honey Grace Uriarte	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Jack Daniel Rana	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	John Benedict San Jose	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	John Rey Bahan	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Junel Pascasio	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Kim Carlo Vito	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Kristopher AA+onuevo	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Renz Ivan Rongavilla	10/04/2026	13:00:00	14:30:00
BSIT 3-A	BS Information Technology	Russel Dave Bulahan	10/04/2026	13:00:00	14:30:00
BSCS 4-A	BS Computer Science	Adrian Aquino	11/04/2026	8:00:00	9:00:00
BSCS 4-A	BS Computer Science	Anjanette Castor	11/04/2026	8:00:00	9:00:00
BSCS 4-A	BS Computer Science	Dilean James B Vito	11/04/2026	8:00:00	12:13:17
BSCS 4-A	BS Computer Science	Isra-Enos Suazo	11/04/2026	8:00:00	9:00:00
BSCS 4-A	BS Computer Science	JC Jeric Merginio Rodelas	11/04/2026	8:00:00	12:13:06
BSCS 4-A	BS Computer Science	Jhon Webster Perez Fortuna	11/04/2026	8:00:00	9:00:00
BSCS 4-A	BS Computer Science	John Denver Abanio Esperanza	11/04/2026	8:00:00	12:12:22
BSCS 4-A	BS Computer Science	John Mark Silverio Manangkil	11/04/2026	8:00:00	12:12:04
BSCS 4-A	BS Computer Science	Mark Anthony Dela Cruz	11/04/2026	8:00:00	12:12:12

Report and Analytics Section Attendance (Admin's View)



The **Report and Analytics Section Attendance** allow the administrator to view attendance analytics of students per section and subject. It displays detailed records including student status such as present, along with time in, time out, and scheduled class information for accurate monitoring and evaluation.

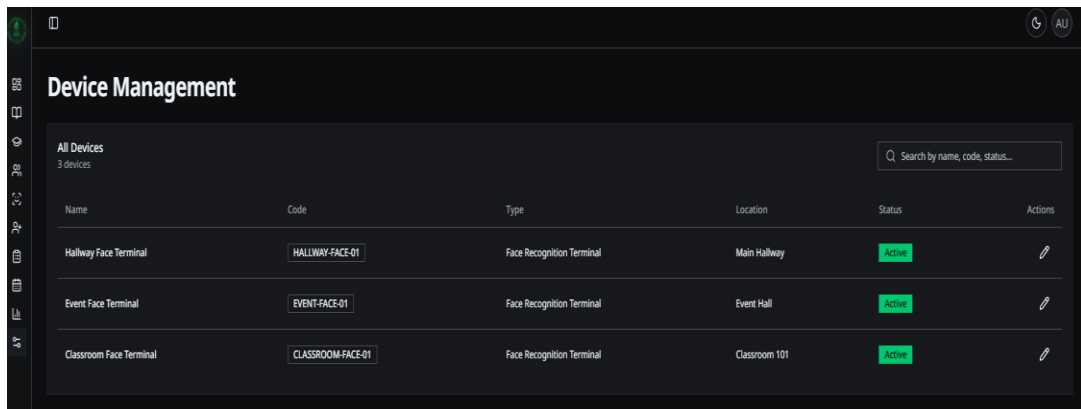


The Report and Analytics Section

Attendance allow the administrator to view attendance analytics of students per section and subject. It displays detailed records including student status such as present, along with time in, time out, and scheduled class information. It also provides an option to export the data in CSV format for reporting and documentation purposes.

Student	Subject	Teacher	Status	Time in	Time out	Class window
Adrian Aquino	CS Elective 4	Adoree Ramos	PRESENT	9:57	11:00	10:00:00-11:00:00
Anjanette Castor	CS Elective 4	Adoree Ramos	PRESENT	10:00	11:00	10:00:00-11:00:00
Dilean James B Vito	CS Elective 4	Adoree Ramos	PRESENT	9:58	11:00	10:00:00-11:00:00
Isra-Enos Suazo	CS Elective 4	Adoree Ramos	PRESENT	9:58	11:00	10:00:00-11:00:00
JC Jeric Merginio Rodelas	CS Elective 4	Adoree Ramos	PRESENT	10:00	11:00	10:00:00-11:00:00
Jhon Webster Perez Fortuna	CS Elective 4	Adoree Ramos	PRESENT	9:59	11:00	10:00:00-11:00:00
John Denver Abanio Esperanzate	CS Elective 4	Adoree Ramos	PRESENT	10:00	11:00	10:00:00-11:00:00
John Mark Silverio Manangkil	CS Elective 4	Adoree Ramos	PRESENT	10:00	11:00	10:00:00-11:00:00
Maricon Sagal Dela Cruz	CS Elective 4	Adoree Ramos	PRESENT	9:59	11:00	10:00:00-11:00:00
Marvin A Osuna	CS Elective 4	Adoree Ramos	PRESENT	10:00	11:00	10:00:00-11:00:00
Matthew Kenu Victoria	CS Elective 4	Adoree Ramos	PRESENT	10:00	11:00	10:00:00-11:00:00
Patricia Alaisa	CS Elective 4	Adoree Ramos	PRESENT	10:00	11:00	10:00:00-11:00:00

Settings Device Management (Admin's View)



Edit Timelog Device
Update the selected timelog device details.

Name	Code
Hallway Face Terminal	HALLWAY-FACE-01
Device Type	Department
Face Recognition Terminal	Senior High School Department
Serial Number	MAC Address
5H-HALLWAY-FACE-01	98-1A2B-3C4D-5E
IP Address	Location Name
192.168.1.101	Main Hallway
Latitude	Longitude
14.59991234	120.58421256
Timezone	Status
Asia/Makila	Active
Installed Date	
09/04/2026	
Address	
Main Building Hallway	
Settings (JSON object)	
{ "mode": "FACE" }	

The **Settings Device Management** section allows the administrator to view all face recognition terminals in the system. It enables management of device status, including setting devices as active, inactive, under maintenance, or offline, as well as editing device information when necessary.

Settings Attendance Policy (Admin's View)

Attendance policy
School-wide defaults for late / absent rules, how early students may scan before class start, and how long after class end a check-out scan is accepted. Teachers can override per class.

School default
Applies when a class has no per-assignment policy. Used by the classroom face terminal for eligibility and status.

On-time grace (minutes after start)	Late window (minutes after start)
10	30
Early arrival allowance (minutes before start)	Late check-out grace (minutes after class end)
0	20

Face check-out is allowed until this many minutes after the scheduled end time (also used when resolving back-to-back classes).

Absent after late window
If off, scans stay "late" instead of absent after the late window.

☒

[Save school policy](#)

The **Settings Attendance Policy** allows the administrator to update attendance parameters in the system. It enables modification of settings such as on-time grace period, late window, late check-out grace, and early arrival allowance.

Settings Email Logs (Admin's View)

Attendance date	Student	Email type	Guardian email	Hallway in	Hallway out	Subject	Status	Sent at	Actions
2026-04-09	Jack Daniel Rana	Event Scan In	ranajack369@gmail.com	—	—	Event scan notification — Jack Daniel R...	SENT	4/10/2026, 3:08:09 PM	ⓘ
2026-04-10	Dilean James B Vito	Hallway Checkout	dileanjames29@gmail.com	4/10/2026, 7:16:19 AM	4/10/2026, 11:25:51 AM	Attendance summary — Dilean James B...	SENT	4/10/2026, 11:25:58 AM	ⓘ

The **Settings Email Logs** section allows the administrator to view all sent and failed email notifications in the system. It provides details of student attendance notifications sent to parents or guardians, including check-in and check-out records, as well as subject-based attendance information for monitoring and review.

Settings Users (Admin's View)

Name	Username	Role	Email	Status	Actions
Randie Ian A. Villatuya	Villatuya	Teacher	ianvillatuya@gmail.com	Active	ⓘ
Jack Daniel Rana	11976	Student	ranajack369@gmail.com	Active	ⓘ
Deho Mark Oliveros	10752	Student	oliverosdehohmark@gmail.com	Active	ⓘ
John Denver Abanilo Espenansate	11364	Student	denver@gmail.com	Active	ⓘ
Maricon Sagal Dela Cruz	11164	Student	maricon@gmail.com	Active	ⓘ
Patricia Alaka	11072	Student	patricia@gmail.com	Active	ⓘ
Ira-Eros Suarez	12345	Student	eros@gmail.com	Active	ⓘ
Adrian Aquino	11016	Student	adrian@gmail.com	Active	ⓘ
Jerome Aguilar	Jerome	Teacher	jerome@gmail.com	Active	ⓘ
Aines Justine Vito	12150	Student	ainesvito@gmail.com	Active	ⓘ
Junel Pascasio	10533	Student	junelmontepascasio0917@gmail.com	Active	ⓘ

The **Settings Email Logs** section allows the administrator to view all users in the system, including both students and teachers. It displays user-related email log information for monitoring and reference purpose.

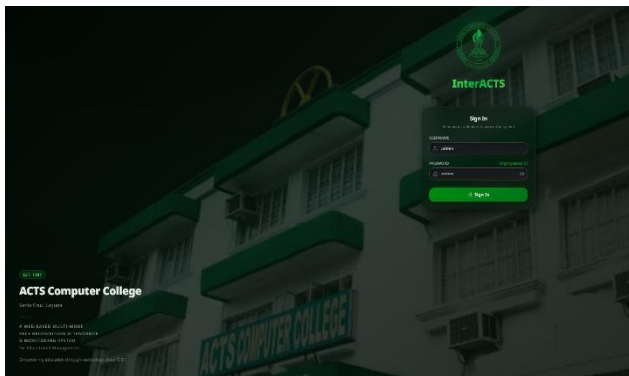
Appendix L

User's Guide

InterACTS Manual's Guide (Teacher's View)

How to Access InterACTS System?

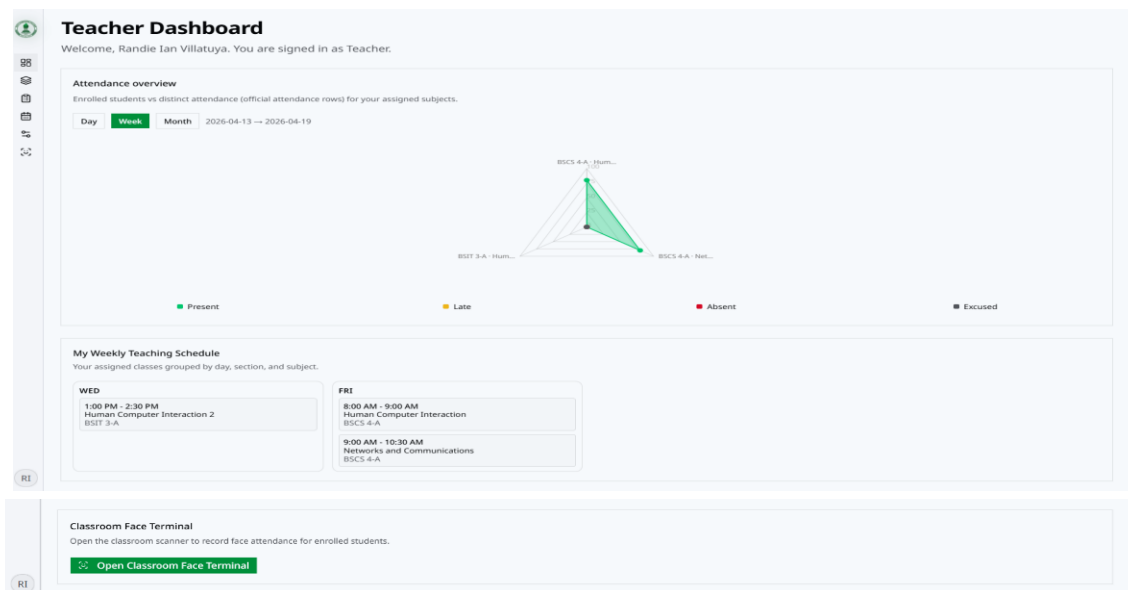
1. The Teacher's is instructed to connect their device to an internet connection
2. Open any browser, the developers suggest using google chrome as the default browser.
3. Access <https://interacts-neon.vercel.app/>



Signing into Account

By clicking the Sign in button, please enter the provided username and password. It will be redirected to the Dashboard once logged in.

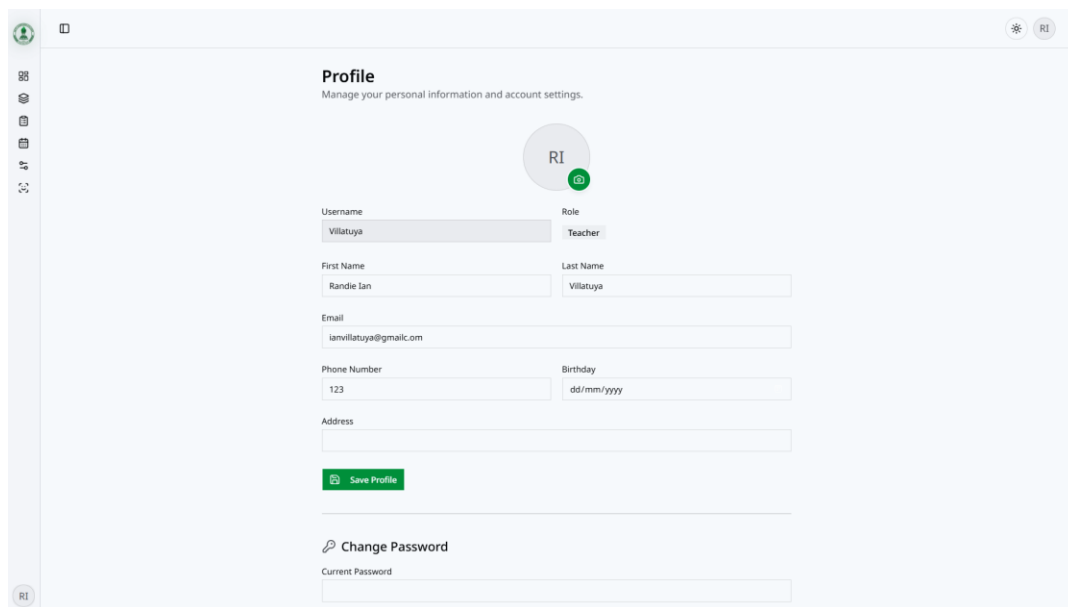
The picture above shows the login page for the system



Teacher's Dashboard

The **Teacher Dashboard** allows teachers to view their assigned classes and related analytics. It displays student attendance summaries such as present, late, absent, and excused, with options to filter data by daily, weekly, or monthly views. It also shows the teacher's weekly class schedule and provides quick access to the classroom face terminal for attendance monitoring.

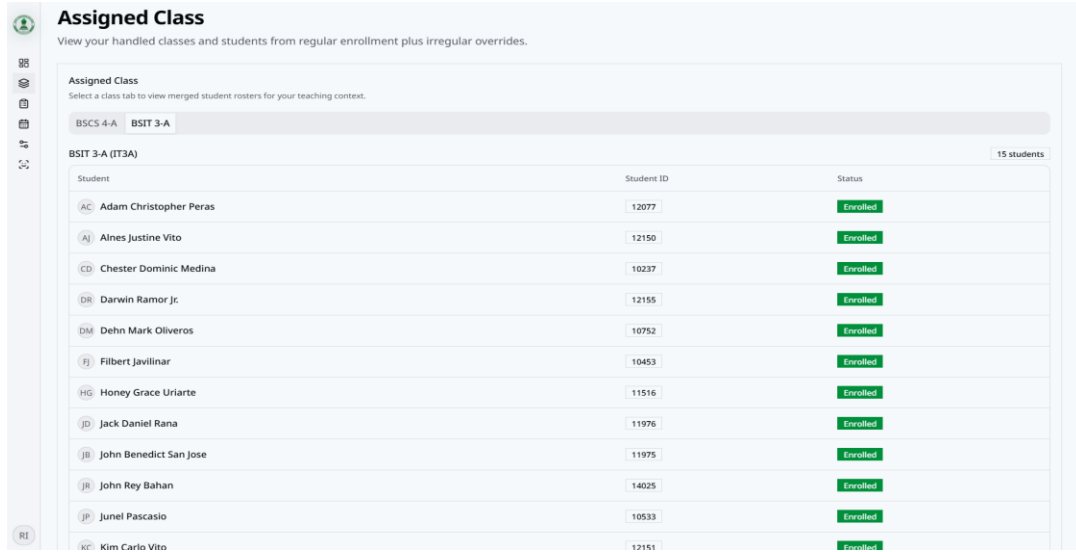
Teacher's Information



The screenshot shows a web application interface for a teacher's profile. On the left is a vertical sidebar with icons for home, classes, analytics, and settings. The main content area is titled 'Profile' with the subtitle 'Manage your personal information and account settings.' Below this is a circular profile picture placeholder with the initials 'RI' and a green camera icon. The form contains several input fields: 'Username' (filled with 'Villatuya'), 'Role' (filled with 'Teacher'), 'First Name' (filled with 'Randle Ian'), 'Last Name' (filled with 'Villatuya'), 'Email' (filled with 'ianvillatuya@gmail.com'), 'Phone Number' (filled with '123'), 'Birthday' (placeholder 'dd/mm/yyyy'), and 'Address' (empty). A green 'Save Profile' button is located below the address field. At the bottom, there is a 'Change Password' section with a lock icon and a 'Current Password' input field.

The **Teacher's Information** section allows teachers to view and update their personal account details within the system. It enables modification of information such as username, last name, email, phone number, birthday, and address to ensure that their profile information remains accurate and up to date. It also provides the option to change the account password for security and account protection purposes, ensuring safe access to the system and maintaining the integrity of user accounts.

Assigned Class (Teacher's View)



The screenshot shows the 'Assigned Class' section of a teacher's interface. It includes a sidebar with navigation icons, a main header with the title 'Assigned Class' and a subtitle 'View your handled classes and students from regular enrollment plus irregular overrides.' Below this is a tabbed interface with 'BSCS 4-A' and 'BSIT 3-A'. The 'BSIT 3-A' tab is selected, showing a list of 15 students. Each student row includes a small circular icon with a letter, the student's name, their Student ID, and their enrollment status (Enrolled or Not Enrolled).

Student	Student ID	Status
AC Adam Christopher Peras	12077	Enrolled
AJ Aines Justine Vito	12150	Enrolled
CD Chester Dominic Medina	10237	Enrolled
DR Darwin Ramor Jr.	12155	Enrolled
DM Dehn Mark Oliveros	10752	Enrolled
FJ Filbert Javillinar	10453	Enrolled
HG Honey Grace Uriarte	11516	Enrolled
JD Jack Daniel Rana	11976	Enrolled
JB John Benedict San Jose	11975	Enrolled
JR John Rey Bahan	14025	Enrolled
JP Junel Pascasio	10533	Enrolled
KC Kim Carlo Vito	12161	Enrolled

The **Assigned Class section** allows teachers to view all students under their assigned classes. It displays the list of students including their enrollment status, showing whether they are enrolled or not enrolled within the teacher's handled classes.

Attendance of Students (Teacher's View)

Attendance

Class attendance

From: 14/04/2026 To: 14/04/2026 Status filter: All statuses

Summary: Total records 5 Present 5 Late 0 Absent 0 Excused 0

BSCS ... Human Computer Intera... BSCS ... Networks and Communi... BSIT ... Human Computer Intera...

Search by name or student ID...

Student	Student ID	Status	Time in	Time out	First scan	Last scan	Logs
AA Adrian Aquino	11016	PRESENT	8:00 AM	9:00 AM	8:00 AM	9:00 AM	Mark Absent Logs
AC Anjanette Castor	09720	PRESENT	8:00 AM	9:00 AM	8:00 AM	9:00 AM	Mark Absent Logs
DJ Dilean James B Vito	11268	PRESENT	8:00 AM	9:00 AM	8:00 AM	9:00 AM	Mark Absent Logs
IS Isra-Enos Suazo	12345	PRESENT	8:00 AM	9:00 AM	8:00 AM	9:00 AM	Mark Absent Logs
J JC Jeric Merginio Rodelas	09840	PRESENT	8:00 AM	9:00 AM	8:00 AM	9:00 AM	Mark Absent Logs
JW Jhon Webster Perez Fortuna	11400	ABSENT	—	—	—	—	Mark Present Logs
JD John Denver Abanio Esperanzate	11364	ABSENT	—	—	—	—	Mark Present Logs
JM John Mark Silverio Manangkil	11225	ABSENT	—	—	—	—	Mark Present Logs

The **Attendance of Students** section allows teachers to view student attendance per subject, including time in and time out records. It also provides attendance management buttons that can be used to mark students as present during online classes when automatic scanning is not applicable.

Summary: Total records 5 Present 5 Late 0 Absent 0 Excused 0

BSCS ... Human Computer Intera... BSCS ... Networks and Communi... BSIT ... Human Computer Intera...

Search by name or student ID...

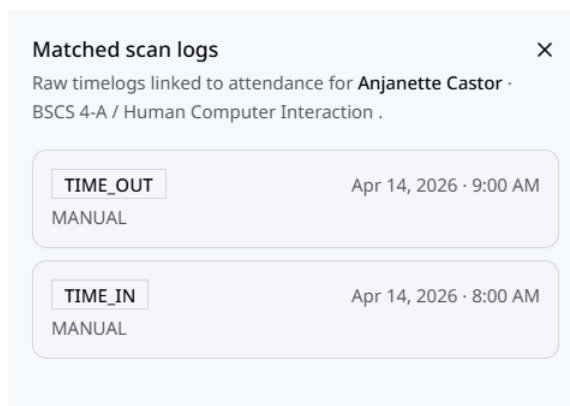
AA Adrian Aquino Present Logs	AC Anjanette Castor Present Logs	DJ Dilean James B Vito Present Logs	IS Isra-Enos Suazo Present Logs	J JC Jeric Merginio Rode... Present Logs	JW Jhon Webster Perez F... Absent Logs	JD John Denver Abanio E... Absent Logs
JM John Mark Silverio Ma... Absent Logs	MS Maricon Sagal Dela Cr... Absent Logs	MA Marvin A Osuna Absent Logs				

Rows per page: 10 Showing 1-10 of 15 Page 1 of 2 Prev Next

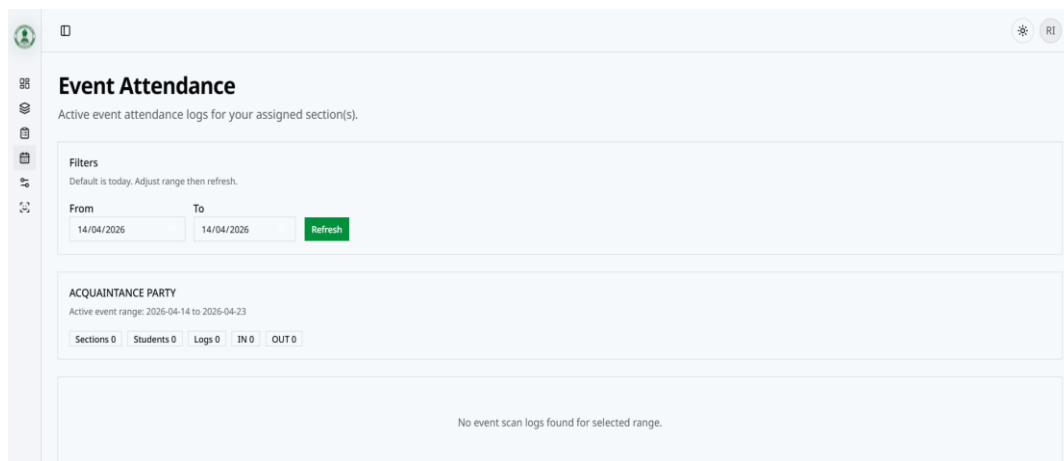


The **Attendance View Mode** allows teachers to display student attendance records in either list or grid format. It provides flexibility in viewing student information based on preference, making it easier to organize and review attendance data.

The **Matched Scan Logs** feature allows teachers to view raw attendance scan records of students. It displays detailed logs including the date, time in, and time out of each student, providing an accurate record of individual attendance activities.



Event's Attendance (Teacher's View)



The **Event's Attendance** section allows teachers to view attendance records of students under their handled classes who participated in events. It enables filtering by

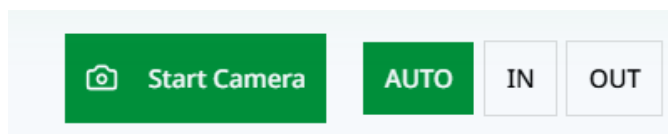
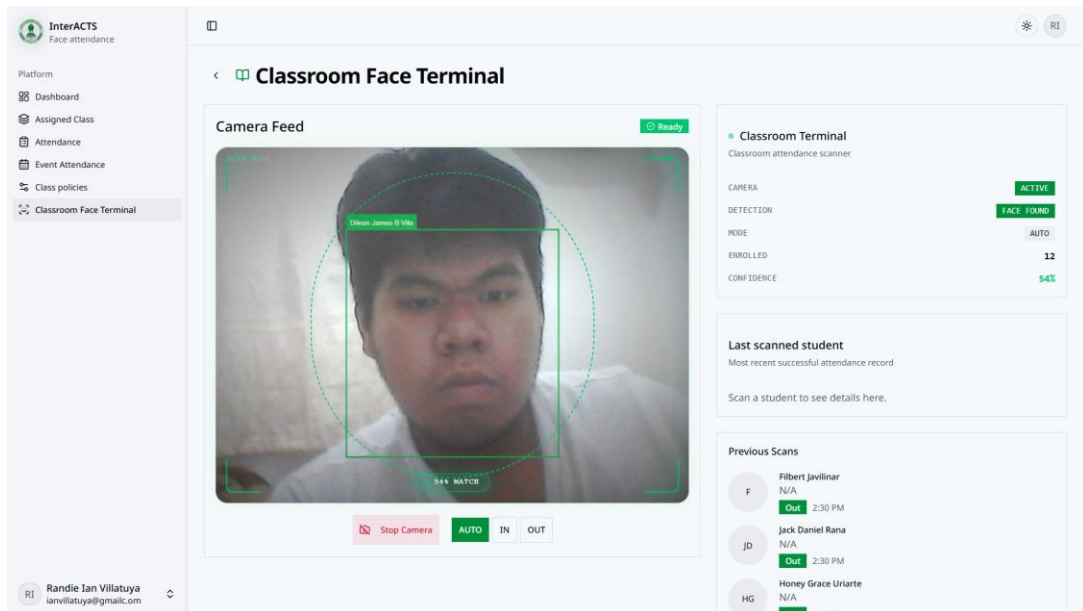
date using a calendar to easily check specific event attendance records and monitor student participation.

Class attendance policies (Teacher's View)

The screenshot shows a web interface for managing class attendance policies. At the top, there's a header with a user profile icon and a 'RI' button. Below the header, the title 'Class attendance policies' is displayed, followed by a subtitle: 'Set rules for your assigned sections and subjects. Scans outside the allowed window are blocked at the classroom terminal.' The main content area is titled 'Per-class overrides' and includes a note: 'Values below apply to this class only. School defaults are shown for reference.' Below this, there's a text box showing 'School defaults: grace 10m · late window 30m · early allowance 0m · late check-out 20m'. A section for '3SCS 4-A · Human Computer Interaction 3CS 4-A · Networks and Communicator SIT 3-A · Human Computer Interaction' is highlighted. Underneath, a 'Custom override' section contains four input fields: 'On-time grace (minutes)' with a value of 10, 'Late window (minutes)' with a value of 20, 'Early arrival allowance (minutes)' with a value of 10, and 'Late check-out grace (minutes after class end)' with a value of 20. A note below these fields states: 'Allows face check-out for this many minutes after the scheduled end; also used when the next class starts immediately after.' At the bottom of this section, there's a toggle switch for 'Absent after late window' which is currently turned on. A green 'Save for this class' button is located at the bottom left of the form.

The **Class Attendance Policies** section allows teachers to view and manage attendance rules within their classes. It enables adjustment of settings such as on-time grace period, late window, late check-out grace, and early arrival allowance to ensure accurate and flexible attendance recording for class activities.

Classroom Face Terminal (Teacher's View)



The Classroom Face Terminal allows teachers to

record student attendance per subject using facial recognition. It includes automatic check-in and check-out through the camera system, enabling real-time attendance tracking for classroom activities.

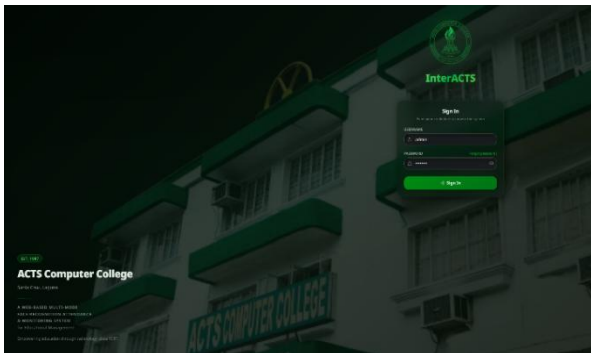
Appendix M

User's Guide

InterACTS Manual's Guide (Student's View)

How to Access InterACTS System?

1. The student's is instructed to connect their device to an internet connection
2. Open any browser, the developers suggest using google chrome as the default browser.
3. Access <https://interacts-neon.vercel.app/>

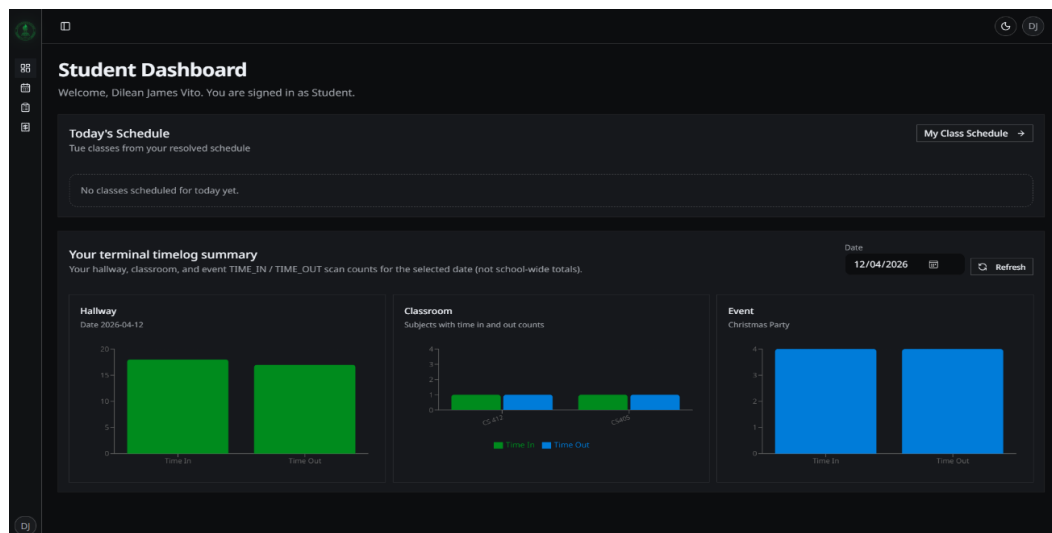


Signing into Account

By clicking the Sign in button, please enter the provided username and password. It will be redirected to the Dashboard once logged in.

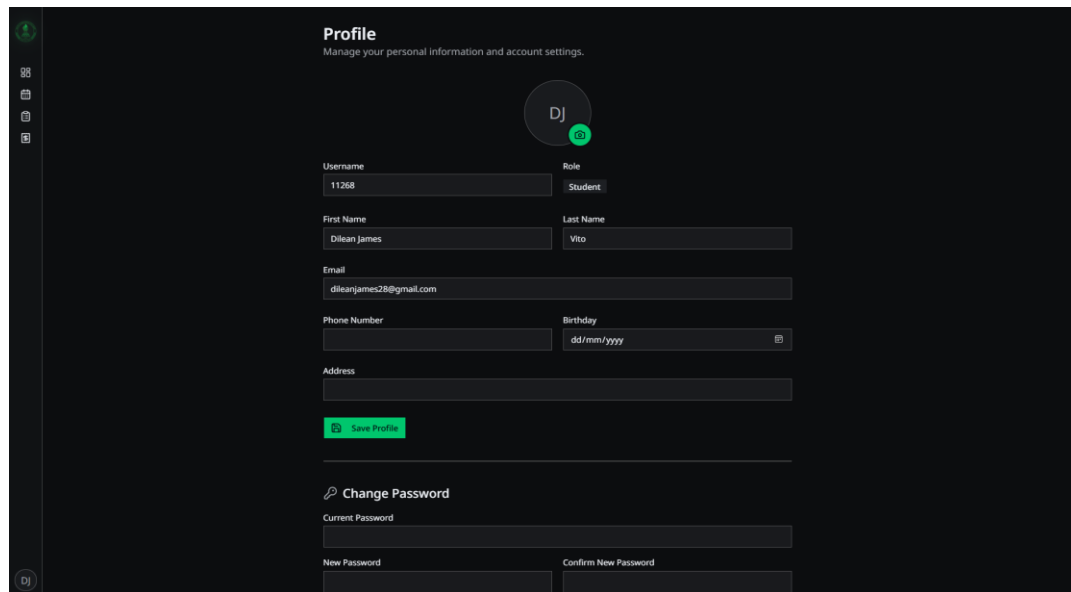
The picture above shows the login page for the system

Student's Dashboard



The **Student's Dashboard** allows students to view their personal attendance analytics for hallway, classroom, and event activities. It also displays their schedule for the current day and provides a shortcut to access their full weekly class schedule for better time management and organization.

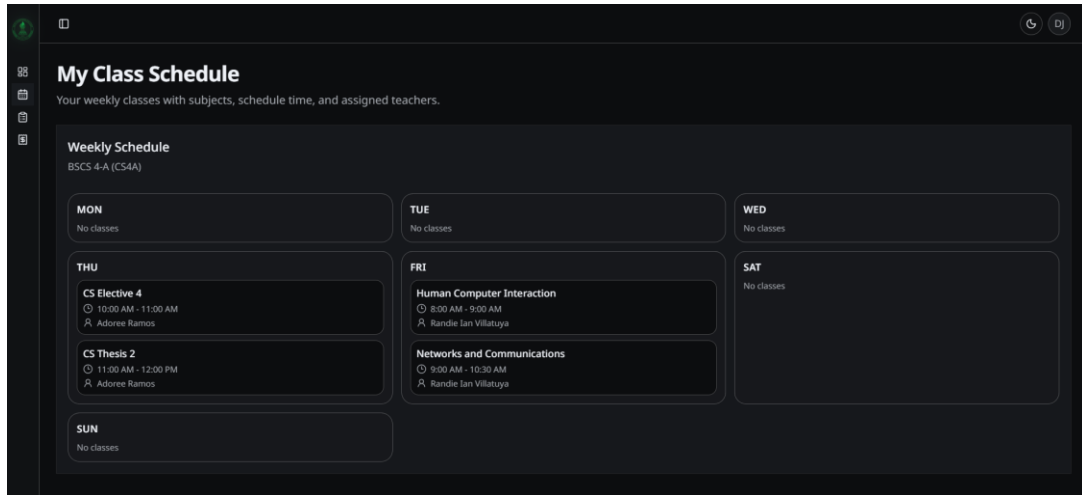
Student's Personal Information



The screenshot shows a web interface for a student's profile. On the left is a dark sidebar with icons for home, calendar, and profile. The main content area is titled 'Profile' with the subtitle 'Manage your personal information and account settings.' Below this is a circular profile picture placeholder with the initials 'DJ' and a green camera icon. The form contains several input fields: Username (11268), Role (Student), First Name (Dilean James), Last Name (Vito), Email (dileanjames28@gmail.com), Phone Number, Birthday (dd/mm/yyyy), and Address. A green 'Save Profile' button is located below the address field. Below the profile section is a 'Change Password' section with fields for Current Password, New Password, and Confirm New Password.

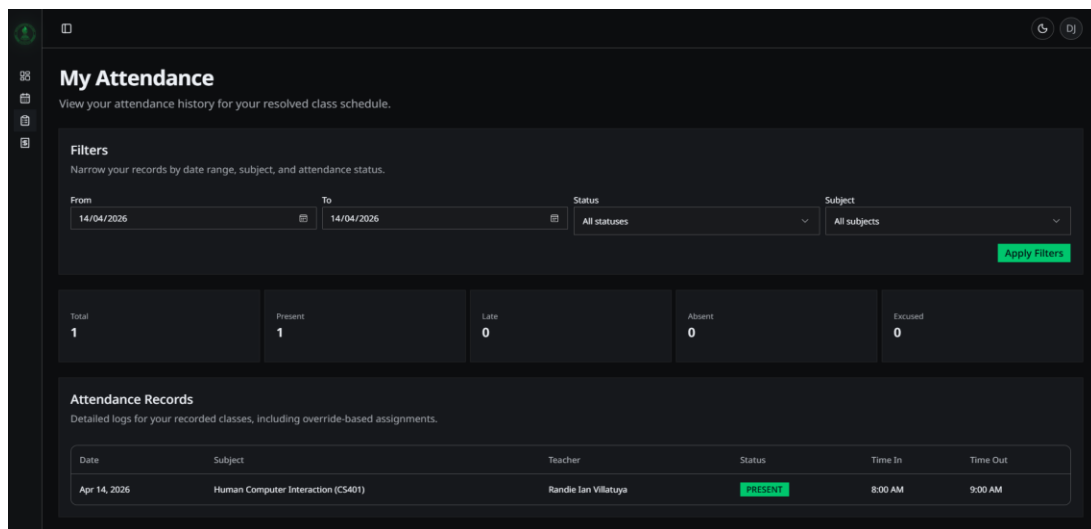
The **Student's Personal Information** section allows students to view and update their personal account details within the system. It enables modification of information such as username, first name, last name, email, phone number, birthday, and address. It also allows students to change their account password for security and account protection purposes.

Class Schedule's (Student's View)



The **Class Schedule section** allows students to view their complete weekly class schedule. It displays all assigned subjects along with their corresponding days and time slots, helping students easily track and manage their academic timetable.

Student's Attendance (Student's View)



The **Student's Attendance section** allows students to view their personal attendance records per subject. It displays their attendance status including time in, time out, and overall attendance summary, helping students monitor their class participation

Submit Payment (Student's View)

Submit payment

Capture a photo of your receipt for an active event. An administrator will review and verify it.

New submission

Use your camera to take a clear photo of your receipt.

Event

ACQUAINTANCE PARTY

Purpose

Proof of Payment

Receipt photo

Recapture

Photo ready

Your submissions

Status and any notes from administration.

Receipt	Event	Purpose	Status	Remarks	Submitted
	ACQUAINTANCE PARTY	Proof of Payment	Pending	—	Apr 14, 2026

The **Submit Payment** feature allows students to submit event payment receipts using a real-time camera capture. It enables the selection of the event and purpose before submission. Students can also track the status of their submission, including pending, approved, or rejected, as reviewed by the administrator.

Appendix N

Sample Code

```
import {Navigate } from "react-router-dom"

import SchoolLogo from "@assets/SchoolLogo.png"
import SchoolBuilding from "@assets/SchoolBuilding.png"
import { LoginForm } from "@components/login-form"
import { getDashboardPath, useAuth } from "@contexts/auth-context"

export function LoginPage() {
  const { user, isLoading } = useAuth()

  if (isLoading) {
    return (
      <div className="flex min-h-dvh items-center justify-center bg-background">
        <div
          className="size-8 animate-spin rounded-full border-2 border-primary border-t-transparent"
          aria-hidden
        />
      </div>
    )
  }

  if (user) {
    return <Navigate to={getDashboardPath(user.role?.name)} replace />
  }
}
```

```

return (
  <div className="relative flex min-h-dvh w-full overflow-hidden">
    {/ * — Background: school building image with dark green overlay — */}
    <div className="absolute inset-0 z-0">
      <img
        src={SchoolBuilding}
        alt="ACTS Computer College"
        className="h-full w-full object-cover object-center"
      />
      {/ * Multi-layer dark green gradient overlay */}
      <div
        className="absolute inset-0"
        style={{
          background:
            "linear-gradient(135deg, oklch(0.12 0.03 155 / 84%) 0%, oklch(0.16 0.025
160 / 80%) 50%, oklch(0.14 0.028 148 / 86%) 100%)",
          }}
      />
      {/ * Subtle noise/texture overlay for depth */}
      <div
        className="absolute inset-0 opacity-20"
        style={{
          backgroundImage:
            "radial-gradient(ellipse at 20% 50%, oklch(0.62 0.19 145 / 12%) 0%,
transparent 60%), radial-gradient(ellipse at 80% 20%, oklch(0.50 0.16 148 / 10%) 0%,
transparent 50%)",
          }}
      />
    </div>
  )

```

```

    { /* — Desktop: Left panel with building info — */ }

    <div className="relative z-10 hidden lg:flex lg:w-1/2 lg:flex-col lg:items-start
lg:justify-end lg:p-12">
      <div className="max-w-md">
        <div
          className="mb-4 inline-block rounded-full px-3 py-1 text-xs font-semibold
tracking-widest uppercase"
          style={{
            background: "oklch(0.62 0.19 145 / 15%)",
            border: "1px solid oklch(0.62 0.19 145 / 30%)",
            color: "oklch(0.75 0.18 145)",
          }}
        >
          Est. 1987
        </div>
        <h2
          className="mb-3 text-4xl font-black leading-tight tracking-tight"
          style={{ color: "oklch(0.93 0.018 150)" }}
        >
          ACTS Computer College
        </h2>
        <p
          className="text-base leading-relaxed"
          style={{ color: "oklch(0.68 0.022 150)" }}
        >
          Santa Cruz, Laguna
        </p>
      </div>

```

```

        className="mt-6 h-px w-16"
        style={{
            background:
                "linear-gradient(90deg, oklch(0.62 0.19 145 / 70%), transparent)",
        }}
    />
    /* System description — desktop left panel */
    <p
        className="mt-5 text-sm font-semibold uppercase leading-relaxed tracking-
widest"
        style={{ color: "oklch(0.62 0.025 150)" }}
    >
        A Web-Based Multi-Mode
    <br />
        Face Recognition Attendance
    <br />
        & Monitoring System
    <br />
    <span
        className="font-medium normal-case tracking-wide"
        style={{ color: "oklch(0.50 0.020 150)" }}
    >
        for Educational Management
    </span>
</p>
<p
    className="mt-4 text-sm leading-relaxed"
    style={{ color: "oklch(0.55 0.02 150)" }}

```

```

    >
    Empowering education through technology since 1987.
  </p>
</div>
</div>

```

```

    { /* — Main content: vertically centered on mobile; top-center on desktop — */ }

    <div className="relative z-10 flex w-full flex-col items-center justify-center px-3
py-8 sm:px-4 sm:py-10 lg:w-1/2 lg:justify-start lg:px-12 lg:pt-12 lg:pb-10">
      <div className="w-full max-w-sm">
        { /* — Logo + App Identity — */ }

        <div className="mb-6 flex flex-col items-center text-center sm:mb-8">
          { /* School logo — large, no circular frame */ }

          <img
            src={SchoolLogo}
            alt="ACTS Computer College Logo"
            className="mb-4 h-20 w-20 object-contain sm:mb-5 sm:h-32 sm:w-32 lg:h-
40 lg:w-40"
            style={{
              filter:
                "drop-shadow(0 4px 20px oklch(0 0 0 / 45%)) drop-shadow(0 0 14px
oklch(0.62 0.19 145 / 35%))",
            }}
          />

          { /* App name */ }

          <h1
            className="mb-1 text-3xl font-black tracking-tight md:text-4xl"
            style={{

```

```

background:
  "linear-gradient(135deg, oklch(0.80 0.20 145), oklch(0.65 0.19 148),
oklch(0.72 0.18 142))",
  WebkitBackgroundClip: "text",
  WebkitTextFillColor: "transparent",
  backgroundClip: "text",
  filter: "drop-shadow(0 0 8px oklch(0.62 0.19 145 / 30%))",
  }}

```

```
>
```

```
InterACTS
```

```
</h1>
```

```
{/* Decorative divider */}
```

```
<div className="mt-4 flex items-center gap-2">
```

```
<div
```

```
  className="h-px w-12"
```

```
  style={{
```

```
    background:
```

```
      "linear-gradient(90deg, transparent, oklch(0.62 0.19 145 / 40%))",
```

```
  }}

```

```
/>
```

```
<div
```

```
  className="h-1 w-1 rounded-full"
```

```
  style={{ background: "oklch(0.62 0.19 145 / 50%)" }}
```

```
/>
```

```
<div
```

```
  className="h-px w-12"
```

```
  style={{
```

```

        background:
            "linear-gradient(90deg, oklch(0.62 0.19 145 / 40%), transparent)",
        }}
    />
</div>
</div>

{ /* — Login Card (glass + neumorphic) — */}
<div
    className="rounded-2xl p-4 sm:p-6"
    style={{
        background: "oklch(0.20 0.025 155 / 50%)",
        backdropFilter: "blur(20px) saturate(1.6)",
        WebkitBackdropFilter: "blur(20px) saturate(1.6)",
        border: "1px solid oklch(1 0 0 / 9%)",
        boxShadow:
            "8px 8px 20px oklch(0 0 0 / 50%), -5px -5px 14px oklch(0.24 0.032 155 / 16%), inset 0 1px 0 oklch(1 0 0 / 6%)",
    }}
>
    <LoginForm />
</div>

{ /* — Footer info (mobile only — desktop shows in left panel) — */}
<div className="mt-6 flex flex-col items-center gap-1 text-center lg:hidden">
    <p
        className="text-xs font-semibold tracking-wide"
        style={{ color: "oklch(0.55 0.02 150)" }}
    >

```

```

    >
      ACTS Computer College
    </p>
    <p
      className="text-xs"
      style={{ color: "oklch(0.42 0.016 150)" }}
    >
      Santa Cruz, Laguna &nbsp;·&nbsp; Est. 1987
    </p>
  </div>
</div>
</div>
</div>
)
}

```

```

import { Link } from "react-router-dom"
import { useCallback, useEffect, useMemo, useState } from "react"
import { ArrowRight, Clock3, RefreshCcw, UserRound } from "lucide-react"
import { format } from "date-fns"
import { Bar, BarChart, CartesianGrid, Legend, ResponsiveContainer, Tooltip, XAxis, YAxis } from "recharts"
import { Card, CardContent, CardDescription, CardHeader, CardTitle } from "@components/ui/card"
import { Button } from "@components/ui/button"
import { MainLayout } from "@components/layout/MainLayout"
import { useAuth } from "@contexts/auth-context"
import {

```

```

dashboardService,
type StudentScheduleItem,
type StudentTerminalTimelogSummaryResponse,
type StudentTodaySchedule
} from "@services/dashboard.service"
import { getApiErrorMessage } from "@services/api"

function toDisplayTime(value: string | null): string {
  if (!value) return "TBA"
  const [h, m] = value.split(":")
  const hours = Number(h)
  const minutes = Number(m)
  if (Number.isNaN(hours) || Number.isNaN(minutes)) return value

  const period = hours >= 12 ? "PM" : "AM"
  const hour12 = hours % 12 || 12
  return `${hour12}:${String(minutes).padStart(2, "0")} ${period}`
}

function toMinutes(value: string | null): number {
  if (!value) return Number.MAX_SAFE_INTEGER
  const [h, m] = value.split(":")
  const hours = Number(h)
  const minutes = Number(m)
  if (Number.isNaN(hours) || Number.isNaN(minutes)) return
  Number.MAX_SAFE_INTEGER
  return (hours * 60) + minutes
}

```

```

function getScheduleStatus(item: StudentScheduleItem): "ongoing" | "upcoming" |
"ended" | "tba" {
    const start = toMinutes(item.startTime)
    const end = toMinutes(item.endTime)

    if (start === Number.MAX_SAFE_INTEGER || end ===
Number.MAX_SAFE_INTEGER) return "tba"

    const now = new Date()
    const currentMinutes = (now.getHours() * 60) + now.getMinutes()
    if (currentMinutes >= start && currentMinutes <= end) return "ongoing"
    if (currentMinutes < start) return "upcoming"
    return "ended"
}

```

```

function statusClass(status: ReturnType<typeof getScheduleStatus>): string {
    if (status === "ongoing") return "bg-emerald-100 text-emerald-700"
    if (status === "upcoming") return "bg-blue-100 text-blue-700"
    if (status === "ended") return "bg-slate-200 text-slate-700"
    return "bg-amber-100 text-amber-700"
}

```

```

function currentDateInputValue(): string {
    return format(new Date(), "yyyy-MM-dd")
}

```

```

function toInOutBars(point: { time_in_scans: number; time_out_scans: number } |
null) {
    if (!point) return []
}

```

```

return [
  { name: "Time In", value: point.time_in_scans },
  { name: "Time Out", value: point.time_out_scans }
]
}

export default function StudentDashboard() {
  const { user } = useAuth()
  const displayName = user ? `${user.firstName} ${user.lastName}`.trim() : "Student"
  const [schedule, setSchedule] = useState<StudentTodaySchedule | null>(null)
  const [loading, setLoading] = useState(true)
  const [error, setError] = useState<string | null>(null)
  const [terminalDate, setTerminalDate] = useState(currentDateInputValue)
  const [terminalSummary, setTerminalSummary] =
    useState<StudentTerminalTimelogSummaryResponse | null>(null)
  const [terminalLoading, setTerminalLoading] = useState(true)
  const [terminalError, setTerminalError] = useState<string | null>(null)

  useEffect(() => {
    let isMounted = true

    const loadTodaySchedule = async () => {
      setLoading(true)
      setError(null)
      try {
        const data = await dashboardService.getStudentTodaySchedule()
        if (!isMounted) return
        setSchedule(data)
      }
    }
  })
}

```

```

    } catch (loadError) {
      if (!isMounted) return
      setError(getApiErrorMessage(loadError, "Failed to load your class schedule for
today."))
    } finally {
      if (isMounted) {
        setLoading(false)
      }
    }
  }
}

```

```

loadTodaySchedule()
return () => {
  isMounted = false
}
}, [])

```

```

const loadTerminalSummary = useCallback(async () => {
  setTerminalLoading(true)
  setTerminalError(null)
  try {
    const data = await dashboardService.getStudentTerminalTimelogSummary({ date:
terminalDate })
    setTerminalSummary(data)
  } catch (loadError) {
    setTerminalError(getApiErrorMessage(loadError, "Failed to load terminal timelog
summary."))
    setTerminalSummary(null)
  } finally {

```

```

    setTerminalLoading(false)
  }
}, [terminalDate])

useEffect(() => {
  void loadTerminalSummary()
}, [loadTerminalSummary])

const sortedItems = useMemo(
  () => [...(schedule?.today.items ?? []).sort((a, b) => toMinutes(a.startTime) -
toMinutes(b.startTime)),
  [schedule]
)

const hallwayPoint = terminalSummary?.hallway.series?.[0] ?? null
const eventPoint = terminalSummary?.event.series?.[0] ?? null
const hallwayInOutData = useMemo(() => toInOutBars(hallwayPoint),
[hallwayPoint])
const eventInOutData = useMemo(() => toInOutBars(eventPoint), [eventPoint])
const classroomChartData = useMemo(
  () =>
    (terminalSummary?.classroom.series ?? []).map((entry) => ({
      name: (entry.subject_code || entry.subject_name || "Unmatched").slice(0, 14),
      fullName: entry.subject_name || entry.subject_code || "Unmatched",
      timeIn: entry.time_in_scans,
      timeOut: entry.time_out_scans
    })),
  [terminalSummary]
)

```

```

return (
  <MainLayout>
    <div className="space-y-4 sm:space-y-6">
      <div className="space-y-1 sm:space-y-2">
        <h1 className="text-2xl font-bold tracking-tight sm:text-3xl">Student
Dashboard</h1>
        <p className="text-sm text-muted-foreground sm:text-base">
          Welcome, {displayName}. You are signed in as Student.
        </p>
      </div>

      <Card>
        <CardHeader className="pb-3">
          <div className="flex flex-col gap-2 sm:flex-row sm:items-start sm:justify-
between sm:gap-4">
            <div>
              <CardTitle className="text-base sm:text-lg">Today's Schedule</CardTitle>
              <CardDescription className="text-xs sm:text-sm">
                {schedule ? `${schedule.today.day} classes from your resolved schedule` :
                "Your classes and teachers for today."}
              </CardDescription>
            </div>

            <Button asChild variant="outline" size="sm" className="self-start text-xs
sm:text-sm">
              <Link to="/my-class-schedule">
                My Class Schedule
                <ArrowRight className="ml-2 h-4 w-4" />
              </Link>
            </Button>
          </div>
        </CardHeader>
      </Card>
    </div>
  </MainLayout>
)

```

```

    </Button>
  </div>
</CardHeader>
<CardContent className="space-y-2.5 sm:space-y-3">
  {loading ? (
    <p className="text-sm text-muted-foreground">Loading today's
classes...</p>
  ) : error ? (
    <p className="text-sm text-destructive">{error}</p>
  ) : sortedItems.length === 0 ? (
    <div className="rounded-lg border border-dashed p-3 text-xs text-muted-
foreground sm:p-4 sm:text-sm">
      No classes scheduled for today yet.
    </div>
  ) : (
    <div className="grid gap-3 md:grid-cols-2">
      {sortedItems.map((item) => {
        const status = getScheduleStatus(item)

        return (
          <div
            key={item.assignmentId}
            className="rounded-xl border bg-linear-to-br from-background to-
muted/30 p-3 shadow-sm sm:p-4"
          >
            <div className="mb-1.5 flex items-start justify-between gap-2 sm:mb-
2">
              <p className="text-sm font-semibold sm:text-
base">{item.subjectName || "Subject"}</p>

```

```

        <span className={`rounded-full px-2 py-0.5 text-[11px] font-semibold
capitalize sm:py-1 sm:text-xs ${statusClass(status)} `}>
            {status}
        </span>
    </div>

    <div className="space-y-1 text-xs sm:space-y-1.5 sm:text-sm">
        <p className="flex items-center gap-1.5 text-muted-foreground
sm:gap-2">
            <Clock3 className="h-3.5 w-3.5 sm:h-4 sm:w-4" />
            {toDisplayTime(item.startTime)} - {toDisplayTime(item.endTime)}
        </p>
        <p className="flex items-center gap-1.5 text-muted-foreground
sm:gap-2">
            <UserRound className="h-3.5 w-3.5 sm:h-4 sm:w-4" />
            {item.teacherName || "Teacher TBA"}
        </p>
        <p className="text-xs text-muted-foreground">
            {item.sectionName || "Section"} {item.sectionCode ? `
($ {item.sectionCode})` : ""}
        </p>
    </div>
</div>
    )
    }}
</div>
    )
</CardContent>
</Card>

```

```

<Card>
  <CardHeader className="pb-3">
    <div className="flex flex-col gap-2 sm:flex-row sm:items-end sm:justify-between">
      <div>
        <CardTitle className="text-base sm:text-lg">Your terminal timelog summary</CardTitle>
        <CardDescription className="text-xs sm:text-sm">
          Your hallway, classroom, and event TIME_IN / TIME_OUT scan counts for the selected date (not school-wide totals).
        </CardDescription>
      </div>
      <div className="flex flex-col gap-2 sm:flex-row sm:items-end">
        <div className="space-y-1">
          <label htmlFor="student-terminal-date" className="text-xs text-muted-foreground">
            Date
          </label>
          <input
            id="student-terminal-date"
            type="date"
            className="flex h-9 w-full min-w-44 rounded-md border border-input bg-background px-3 py-2 text-xs sm:w-auto sm:text-sm"
            value={terminalDate}
            onChange={(event) => setTerminalDate(event.target.value)}
          />
        </div>
        <Button
          type="button"
          variant="outline"

```

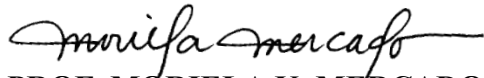
```
size="sm"  
className="w-full sm:w-auto"  
onClick={() => void loadTerminalSummary()}  
disabled={terminalLoading}  
>  
<RefreshCcw className="mr-2 h-3.5 w-3.5" />
```

Certificate of Grammarian

GRAMMARIAN'S CERTIFICATE

This is to certify that the undersigned has reviewed and went through all the pages of the **manuscript** developed by **Dela Cruz Maricon S. Esperanzate John Denver A., Fortuna Jhon Webster P. Manangkil John Mark S. Rodelas JC Jeric M. and Vito Dilean James B.** It is also to certify that the content is aligned with the set of structural rules that governs the composition of sentences, phrases, and words in the English language.

Signed:


PROF. MORIELA K. MERCADO

Grammarian

Date Signed:

May 9, 2026

Curriculum Vitae



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San Juan Santa Cruz, Laguna

PERSONAL INFORMATION

DATE OF BIRTH:	JANUARY 23, 2003
PLACE OF BIRTH:	ESCALANTE CITY, NEGROS OCCIDENTAL
SEX:	FEMALE
HEIGHT:	5'6
AGE:	22
RELIGION:	CATHOLIC
CIVIL STATUS:	SINGLE
NATIONALITY	FILIPINO
FATHER'S NAME:	REMEGIO N. DELA CRUZ
MOTHER'S NAME:	JULIET S. DELA CRUZ

EDUCATION

TERTIARY-CURRENTLY IN FOURTH YEAR
BACHELOR OF SCIENCE IN COMPUTER SCIENCE
ACTS COMPUTER COLLEGE

SENIOR HIGH SCHOOL
OLD POBLACION NATIONAL HIGH SCHOOL
COMPLETED IN 2022

SECONDARY (JUNIOR HIGH SCHOOL)
OLD POBLACION NATIONAL HIGH SCHOOL
COMPLETED IN 2020

ELEMENTARY
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COMPLETED IN 2016



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PLACE OF BIRTH:	VALENZUELA CITY, VALENZUELA
SEX:	MALE
HEIGHT:	5'6
AGE:	23
RELIGION:	CATHOLIC
CIVIL STATUS:	SINGLE
NATIONALITY	FILIPINO
FATHER'S NAME:	CALIXTRO ESPERANZATE
MOTHER'S NAME:	JONALYN ESPERANZATE

EDUCATION

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CHARIS CHRISTIAN SCHOOL
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LAWANG BATO NATIONAL HIGH SCHOOL
COMPLETED IN 2018

ELEMENTARY
LAWANG BATO ELEMENTARY SCHOOL
COMPLETED IN 2014



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PLACE OF BIRTH:	CALAUAN, LAGUNA
SEX:	MALE
HEIGHT:	5'6
AGE:	22
RELIGION:	CATHOLIC
CIVIL STATUS:	SINGLE
NATIONALITY	FILIPINO
FATHER'S NAME:	MICHAEL C. FORTUNA
MOTHER'S NAME:	EDEN P. FORTUNA

EDUCATION

TERTIARY-CURRENTLY IN FOURTH YEAR
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SENIOR HIGH SCHOOL
DAYAP NATIONAL INTEGRATED HIGH SCHOOL
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SECONDARY (JUNIOR HIGH SCHOOL)
MARANATHA CHRISTIAN ACADEMY BAY CHAPTER
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ELEMENTARY
DAYAP ELEMENTARY SCHOOL MAIN
COMPLETED IN 2016



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DATE OF BIRTH:	APRIL 21, 2004
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SEX:	MALE
HEIGHT:	5'4
AGE:	21
RELIGION:	CATHOLIC
CIVIL STATUS:	SINGLE
NATIONALITY	FILIPINO
FATHER'S NAME:	ARIEL MANANGKIL
MOTHER'S NAME:	DARLINA MANANGKIL

EDUCATION

TERTIARY-CURRENTLY IN FOURTH YEAR

BACHELOR OF SCIENCE IN COMPUTER SCIENCE
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PILA SENIOR HIGH SCHOOL
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ELEMENTARY

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PLACE OF BIRTH:	PILA, LAGUNA
SEX:	MALE
HEIGHT:	5'5
AGE:	22
RELIGION:	ROMAN CATHOLIC
CIVIL STATUS:	SINGLE
NATIONALITY	FILIPINO
FATHER'S NAME:	JOEL V. RODELAS
MOTHER'S NAME:	MARITES M. RODELAS

EDUCATION

TERTIARY-CURRENTLY IN FOURTH YEAR
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PLACE OF BIRTH:	SANTA CRUZ, LAGUNA
SEX:	MALE
HEIGHT:	5'
AGE:	22
RELIGION:	CATHOLIC
CIVIL STATUS:	SINGLE
NATIONALITY	FILIPINO
FATHER'S NAME:	ALNES VITO
MOTHER'S NAME:	CANDELARIA BALANQUIT PAMBUHAN

EDUCATION

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STI SANTA CRUZ LAGUNA
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PERDO GUEVARA MEMORIAL NATIONAL HIGH SCHOOL
COMPLETED IN 2020

ELEMENTARY

UNSON ELEMENTARY SCHOOL
COMPLETED IN 2016